

# Service Manual

## LG-H422

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# 1. INTRODUCTION

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## 1.1 Purpose

This manual provides the information necessary to repair, calibration, description and download the features of this model.

## 1.2 Regulatory Information

### A. Security

This material is prohibited to share and release to unauthorized person, in accordance with the regulations, LG Electronics, Civil / criminal responsibility in accordance with the relevant provisions violate.

### B. Precautions for repair

- In case of Disassembly or Assembly to repair product, be careful of a product failure caused by RF signals and Static electricity.
- When using Magnetic tool for the Phone's SVC repair, you should check affect the Electric parts according to effect of Magnet.
- When fastening the screw, be careful not to damage the head of screw and even product.

### C. Attention

**Boards, which contain Electrostatic Sensitive Device (ESD), are indicated by the  sign.**

**Following information is ESD handling:**

- Service personal should ground themselves by using a wrist, strap when exchange system board.
- When repair are made to a system board, they should spread the floor with anti-static mat which is also grounded.
- Use a suitable, grounded soldering iron.
- Keep sensitive parts in these protective packages until these are used.
- When returning system board or parts like EEPROM to the Factory, use the protective package as described.

## 2. PERFORMANCE

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### 2.1 Band Specification

| Support Band | TX Freq (MHz) | RX Freq (MHz) |
|--------------|---------------|---------------|
| WCDMA(FDD1)  | 1920 – 1980   | 2110 – 2170   |
| WCDMA(FDD5)  | 824 – 849     | 869 – 894     |
| WCDMA(FDD8)  | 880 – 915     | 925 – 960     |
| EGSM         | 880 – 915     | 925 – 960     |
| GSM850       | 824 – 849     | 869 – 894     |
| DCS1800      | 1710 – 1785   | 1805 – 1880   |
| PCS1900      | 1850 – 1910   | 1930 – 1990   |

## 2. PERFORMANCE

### 2.2 HW Features

| Item               |                        | Feature                                                                                                                                                                | Comment                                                                           |
|--------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Form Factor        |                        | DOP type                                                                                                                                                               |                                                                                   |
| Battery            |                        | 1) Capacity<br>Standard : Li-Ion, 2100mAh                                                                                                                              |                                                                                   |
|                    |                        | 2) Packing Type : Changeable                                                                                                                                           |                                                                                   |
| Size               |                        | Standard : 133.25mm x 66.12mm x 9.95mm                                                                                                                                 | mm                                                                                |
| Weight             |                        | 120.4g                                                                                                                                                                 | With Battery                                                                      |
| Volume             |                        | 62cc                                                                                                                                                                   | Without Battery                                                                   |
| PCB                |                        | 10 STG Layers , 0.8t                                                                                                                                                   |                                                                                   |
| Charging time      |                        | 4.2hrs                                                                                                                                                                 | @ Power Off / 2100mAh                                                             |
| RX sensitivity     |                        | WCDMA(FDD1) : -106.7 dBm<br>WCDMA(FDD5) : -104.7 dBm<br>WCDMA(FDD8) : -103.7 dBm<br>EGSM : -105 dBm<br>GSM850 : -105 dBm<br>DCS 1800 : -105 dBm<br>PCS 1900 : -105 dBm |                                                                                   |
| TX output power    | WCDMA/<br>GSM/<br>GPRS | WCDMA : 23dBm/3.84MHz, +2/-2dBm<br>EGSM : 33 dBm<br>GSM850 : 33 dBm<br>DCS 1800 : 30.7 dBm<br>PCS 1900 : 30 dBm                                                        | Class3(WCDMA)<br>Class4 (EGSM)<br>Class4 (GSM850)<br>Class1 (DCS)<br>Class1 (PCS) |
| GPRS compatibility |                        | GPRS Class 12                                                                                                                                                          |                                                                                   |
| EDGE compatibility |                        | EDGE Class 12 Downlink(Rx) Only                                                                                                                                        |                                                                                   |
| SIM card type      |                        | Plug-In SIM<br>3V /1.8V                                                                                                                                                |                                                                                   |

## 2. PERFORMANCE

|                            |                                                                            |                                                                                             |
|----------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Display                    | Main LCD (4.7", (1280H) X (720V))                                          |                                                                                             |
| Built-in Camera            | 8M AF / 1.0M VT                                                            |                                                                                             |
| Status Indicator           | No                                                                         |                                                                                             |
| Keypad                     | Function Key : 4<br>Back Key : 3                                           | Function Key: Home, Back, menu, S<br>IM-switch<br><br>Back Key : Volume up, down, power key |
| ANT                        | Main : Internal Fixed Type                                                 |                                                                                             |
| System connector           | 5 Pin                                                                      | micro USB                                                                                   |
| Ear Phone Jack             | 3.5Phi, 4 Pole, Stereo                                                     |                                                                                             |
| PC synchronization         | Yes                                                                        |                                                                                             |
| Memory                     | NAND Flash : 8GByte<br>SDRAM : 1GByte LPDDR3                               |                                                                                             |
| Speech coding              | FR, EFR, HR, AMR                                                           |                                                                                             |
| Data & Fax                 | Built in Data & Fax support                                                |                                                                                             |
| Vibrator                   | Built in Vibrator                                                          |                                                                                             |
| Bluetooth                  | V4.1                                                                       |                                                                                             |
| MIDI(for Buzzer Function)  | 128 Poly                                                                   |                                                                                             |
| Music Player               | MP3,AAC,(e)AAC+,WMA,RA,WAV                                                 |                                                                                             |
| Video Player               | Playback : 30fps 1080p (H.264/263, MPEG 4/2, WMV9, VC1, VP8, DivX, XVID)   |                                                                                             |
|                            | Recording : 30fps 1080p (H.264/263, MPEG4/8)<br>30fps 720p (Wi-Fi display) |                                                                                             |
| Camcorder                  | MPEG4/8                                                                    |                                                                                             |
| Voice Recording            | Yes                                                                        |                                                                                             |
| Speaker Phone mode Support | Yes                                                                        |                                                                                             |
| Travel Adapter             | Yes                                                                        | In-box item                                                                                 |
| CDROM                      | No                                                                         |                                                                                             |
| Stereo Headset             | Yes                                                                        |                                                                                             |
| Data Cable                 | Yes                                                                        | In-box item                                                                                 |
| T-Flash(External Memory)   | Yes                                                                        |                                                                                             |

## 2. PERFORMANCE

### 2.3 RSSI Display

| Antenna BAR | Specification |            | Unit |
|-------------|---------------|------------|------|
|             | WCDMA         | GSM        |      |
| 5→4         | -89 ±2dBm     | -91 ±2dBm  | dBm  |
| 4→3         | -94 ±2dBm     | -96 ±2dBm  |      |
| 3→2         | -100 ±2dBm    | -99 ±2dBm  |      |
| 2→1         | -104 ±2dBm    | -103 ±2dBm |      |
| 1→0         | -110 ±2dBm    | -105 ±2dBm |      |

### 2.4 Current Consumption

|                                         | Specification                |                                                                     |
|-----------------------------------------|------------------------------|---------------------------------------------------------------------|
|                                         | GSM                          | WCDMA                                                               |
| Stand by current<br>(Sleep & Idle Avg.) | Under 7mA @ P.P 5            | Under 7mA @ DRX 7                                                   |
| Talk current                            | Under 280 mA<br>@GSM Level 5 | Under 550mA<br>@ Max Power Avg.<br>Under 230 mA<br>@ Tx -10dBm Avg. |

## 2. PERFORMANCE

### 2.5 Battery bar

| Battery Bar      | Specification | Battery Bar        | Specification |
|------------------|---------------|--------------------|---------------|
| Bar 20(Full)     | 98%이상         | Bar 9 -> Bar 8     | 43% -> 42%    |
| Bar 20 -> Bar 19 | 98% -> 97%    | Bar 8 -> Bar 7     | 38% -> 37%    |
| Bar 19 -> Bar 18 | 93% -> 92%    | Bar 7 -> Bar 6     | 33% -> 32%    |
| Bar 18 -> Bar 17 | 88% -> 87%    | Bar 6 -> Bar 5     | 28% -> 27%    |
| Bar 17 -> Bar 16 | 83% -> 82%    | Bar 5 -> Bar 4     | 23% -> 22%    |
| Bar 16 -> Bar 15 | 78% -> 77%    | Bar 4 -> Bar 3     | 16% -> 15%    |
| Bar 15 -> Bar 14 | 73% -> 72%    | Bar 3 -> Bar 2     | 13% -> 12%    |
| Bar 14 -> Bar 13 | 68% -> 67%    | Bar 2 -> Bar 1     | 8% -> 7%      |
| Bar 13 -> Bar 12 | 63% -> 62%    | Bar 1 -> Bar 0     | 3% -> 2%      |
| Bar 12 -> Bar 11 | 58% -> 57%    | Power off          | 1%이하          |
| Bar 11 -> Bar 10 | 53% -> 52%    | Low battery pop-up | 15% , 5%, 0%  |
| Bar 10 -> Bar 9  | 48% -> 47%    |                    |               |

## 2. PERFORMANCE

### 2.6 SW Specification

| Item                  | Feature                                                                                                      | Comment                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| RSSI                  | 0 ~ 5 Levels                                                                                                 |                                                                                          |
| Battery Charging      | 0 ~ 20 Levels                                                                                                |                                                                                          |
| Key Volume            | 0 ~ 7 Level                                                                                                  |                                                                                          |
| Audio Volume          | 0 ~ 15 Level                                                                                                 |                                                                                          |
| Time / Date Display   | Yes                                                                                                          |                                                                                          |
| Multi-Language        | Yes                                                                                                          | depending on build language                                                              |
| Quick Access Mode     | Phone / Messaging / Browser/ Applications                                                                    | Phone / Contact / Messaging / Applications                                               |
| PC Sync               | Yes                                                                                                          |                                                                                          |
| Speed Dial            | Yes                                                                                                          | Voice mail center -> 1 key                                                               |
| Profile               | Yes                                                                                                          | not same with feature phone setting                                                      |
| CLIP / CLIR           | Yes                                                                                                          |                                                                                          |
| Phone Book            | Name / Number / Email / Groups / Postal addresses / Organizations / IM / Note / Nickname / Website / Event / | There is no limitation on the number of items.<br>It depends on available memory amount. |
| Last Dial Number      | Yes                                                                                                          |                                                                                          |
| Last Received Number  | Yes                                                                                                          |                                                                                          |
| Last Missed Number    | Yes                                                                                                          |                                                                                          |
| Search by Number/Name | Yes                                                                                                          |                                                                                          |
| Group                 | Yes                                                                                                          | There is no limitation on the number of items.<br>It depends on available memory amount. |
| Fixed Dial Number     | Yes                                                                                                          |                                                                                          |
| Own Number            | Yes                                                                                                          | My Profile (add/edit/delete are supported)                                               |
| Voice Memo            | Yes                                                                                                          | Support voice recorder                                                                   |
| Network Selection     | Automatic                                                                                                    |                                                                                          |
| Mute                  | Yes                                                                                                          |                                                                                          |
| Call Divert           | Yes                                                                                                          |                                                                                          |
| Call Barring          | Yes                                                                                                          |                                                                                          |
| Call Duration         | Yes                                                                                                          |                                                                                          |
| SMS (EMS)             | There is no limitation on the number of items.<br>It depends on available memory amount.                     | EMS does not support.                                                                    |
| SMS Over GPRS         | Yes                                                                                                          |                                                                                          |

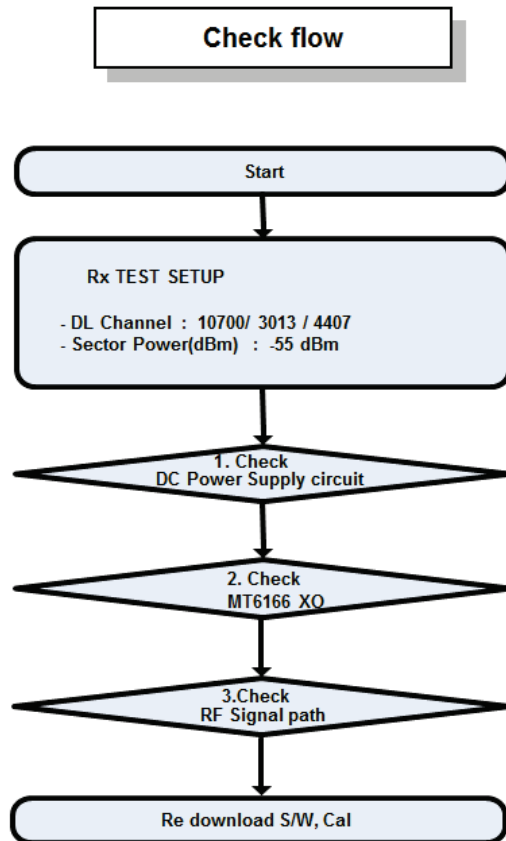
## 2. PERFORMANCE

|                                    |                     |                                                                                                                                                                                                                                                                                                                       |
|------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MMS MPEG4<br>Send / Receive / Save | Yes                 | <ul style="list-style-type: none"> <li>➤ <b>Send / Receive</b> : Yes</li> <li>➤ <b>Save</b> : depends on content type<br/>Support video content type list               <ol style="list-style-type: none"> <li>1. video/mp4</li> <li>2. video/h263</li> <li>3. video/3gpp2</li> <li>video/3gpp</li> </ol> </li> </ul> |
| Long Message                       | MAX 2000 characters | The standard of Open vender                                                                                                                                                                                                                                                                                           |
| Cell Broadcast                     | Yes                 |                                                                                                                                                                                                                                                                                                                       |
| Download                           | Over the Web        |                                                                                                                                                                                                                                                                                                                       |
| Calendar                           | Yes                 |                                                                                                                                                                                                                                                                                                                       |
| Memo                               | Yes                 | There is no limitation on the number of items.<br>It depends on available memory amount.                                                                                                                                                                                                                              |
| World Clock                        | Yes                 |                                                                                                                                                                                                                                                                                                                       |
| Stop Watch                         | Yes                 |                                                                                                                                                                                                                                                                                                                       |
| Wall Paper                         | Yes                 |                                                                                                                                                                                                                                                                                                                       |
| Download Melody / Wallpaper        | Yes                 | Over web browser                                                                                                                                                                                                                                                                                                      |
| SIM Toolkit                        | Yes                 |                                                                                                                                                                                                                                                                                                                       |
| MMS                                | Yes                 | Google MMS Client                                                                                                                                                                                                                                                                                                     |
| EONS                               | Yes                 |                                                                                                                                                                                                                                                                                                                       |
| CPHS                               | Yes                 | V4.2                                                                                                                                                                                                                                                                                                                  |
| Camera                             | Yes                 | 8M AF /<br>Digital Zoom : x4                                                                                                                                                                                                                                                                                          |
| Bluetooth                          | Yes                 | Ver. 4.1                                                                                                                                                                                                                                                                                                              |
| FM radio                           | Yes                 |                                                                                                                                                                                                                                                                                                                       |
| GPRS                               | Yes                 | Class 12                                                                                                                                                                                                                                                                                                              |
| EDGE                               | Yes                 | Class 12(Rx only)                                                                                                                                                                                                                                                                                                     |
| Hold / Retrieve                    | Yes                 |                                                                                                                                                                                                                                                                                                                       |
| Conference Call                    | Yes                 | Max. 6                                                                                                                                                                                                                                                                                                                |
| DTMF                               | Yes                 |                                                                                                                                                                                                                                                                                                                       |
| AMR                                | Yes                 |                                                                                                                                                                                                                                                                                                                       |
| IM                                 | Yes                 | Google Hangout                                                                                                                                                                                                                                                                                                        |
| Email                              | Yes                 |                                                                                                                                                                                                                                                                                                                       |

## 3.1 WCDMA Rx Part

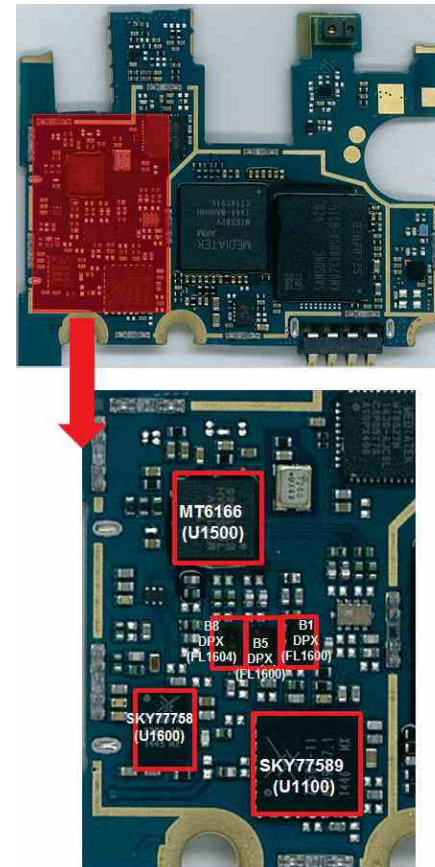
### WCDMA B1/B5/B8 Rx

#### Checking Flow



#### Image

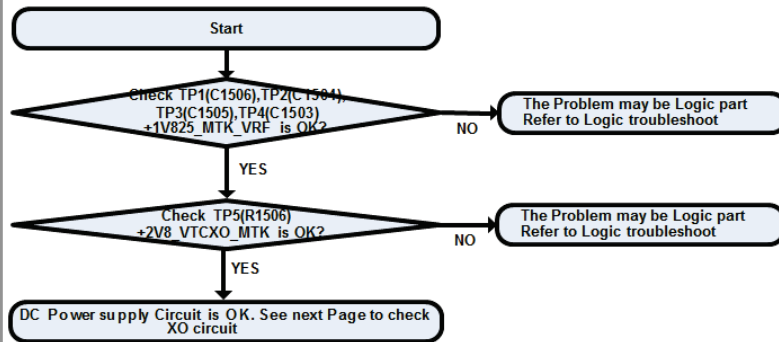
Main  
Top



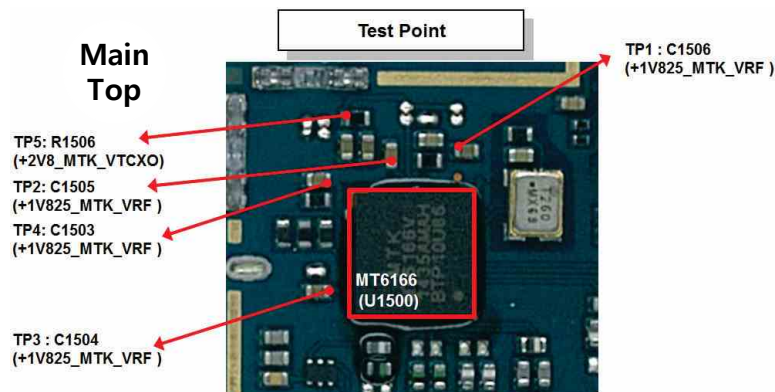
## 3.1 WCDMA Rx Part

### Checking DC Power supply circuit

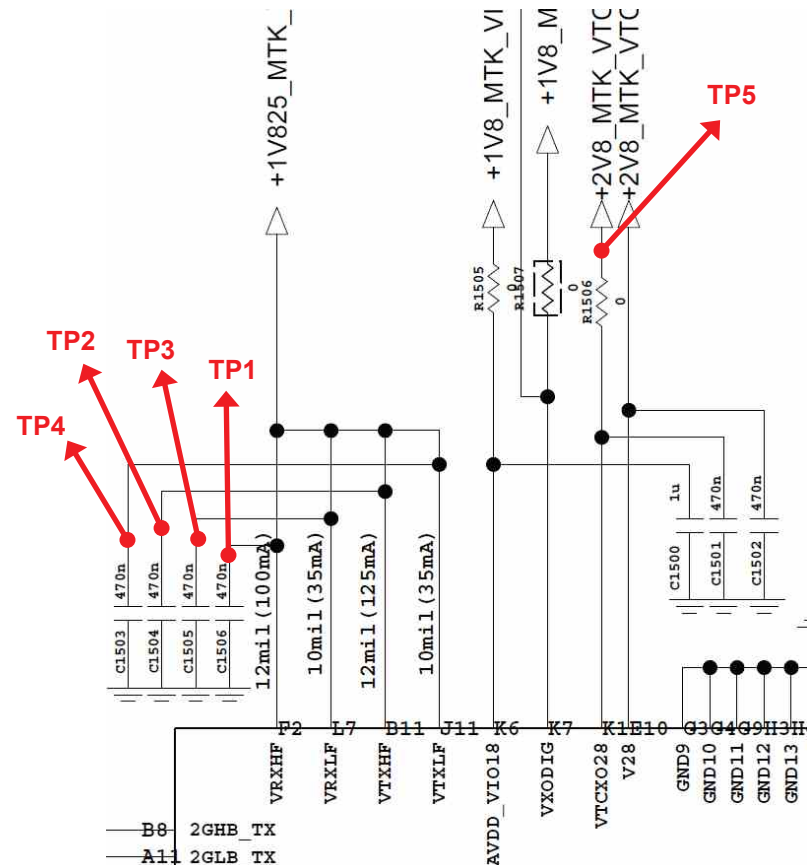
#### Checking Flow



#### Image



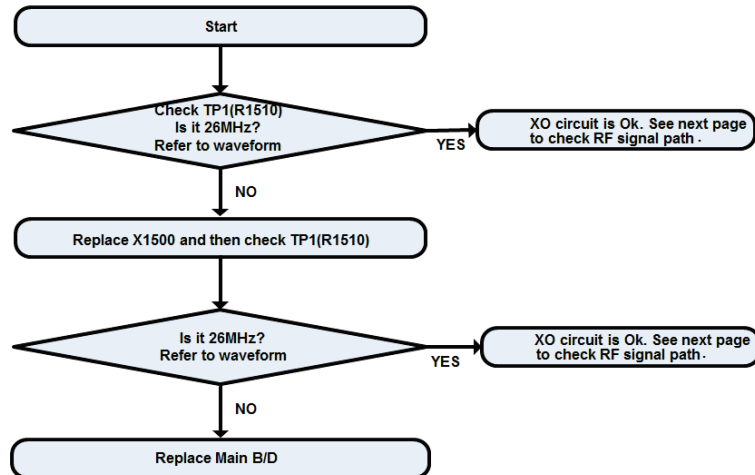
#### Circuit Diagram



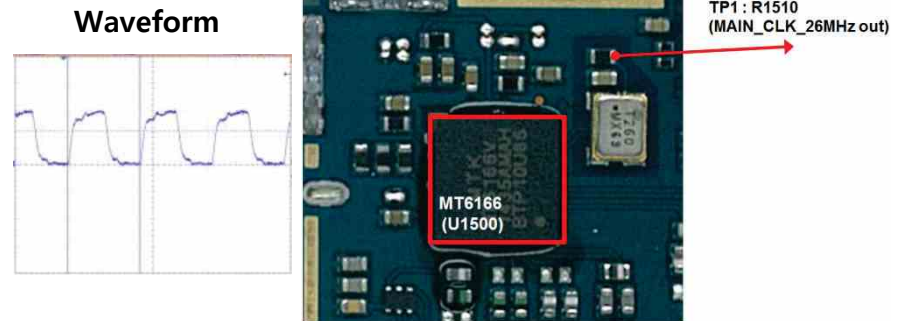
## 3.1 WCDMA Rx Part

### Checking MT6166 XO circuit

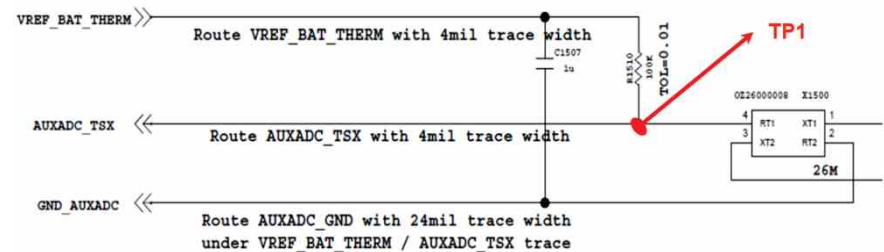
#### Checking Flow



#### Image



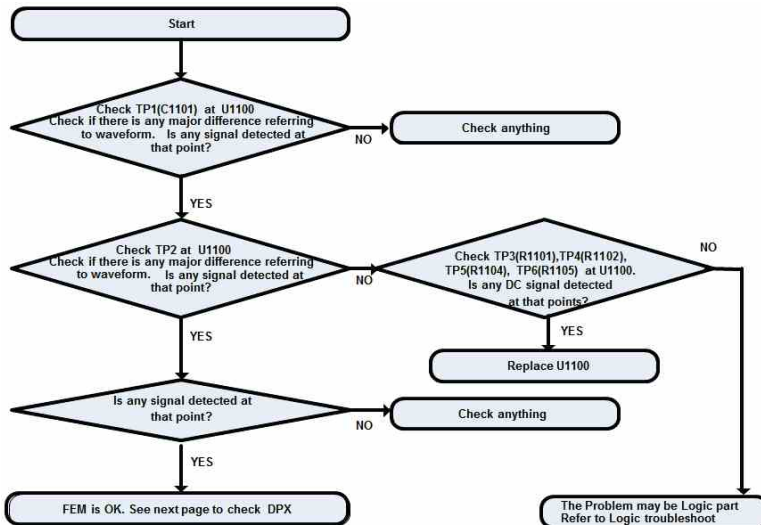
#### Circuit Diagram



## 3.1 WCDMA Rx Part

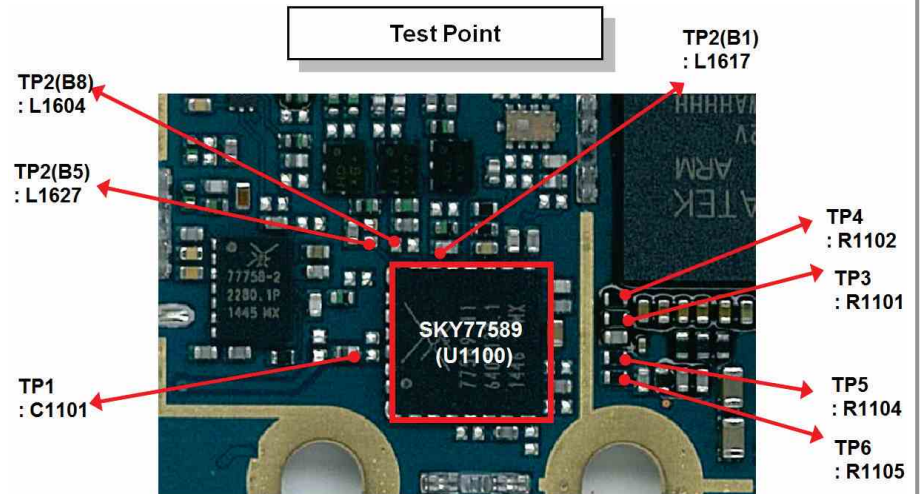
### Checking RF signal path (FEM)

#### Checking Flow



#### Image

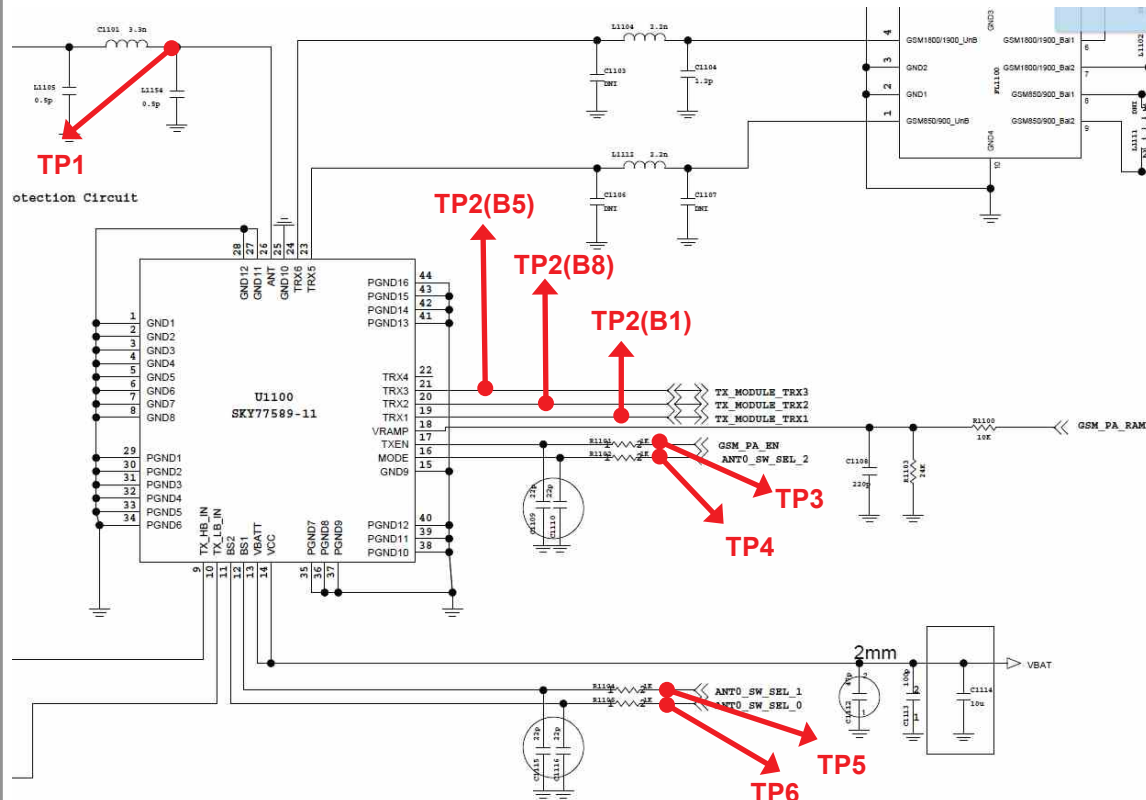
Main  
Top



## 3.1 WCDMA Rx Part

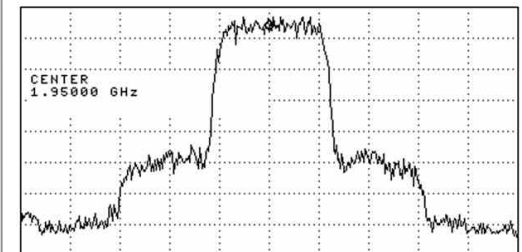
### Checking RF signal path (FEM)

#### Circuit Diagram



#### Image

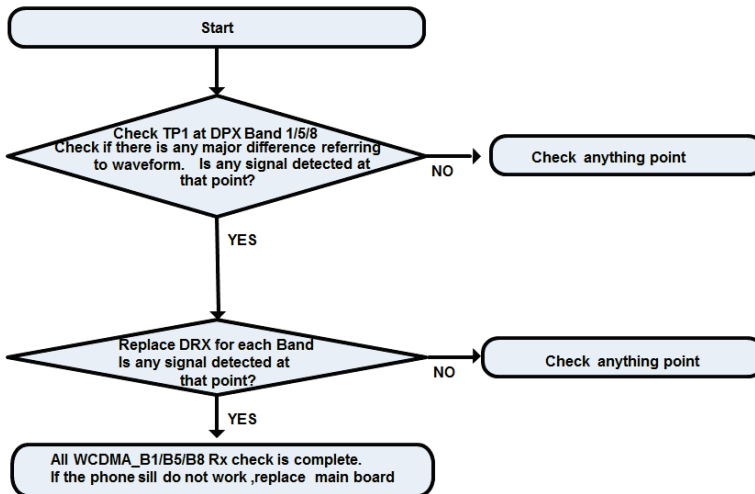
#### Waveform



## 3.1 WCDMA Rx Part

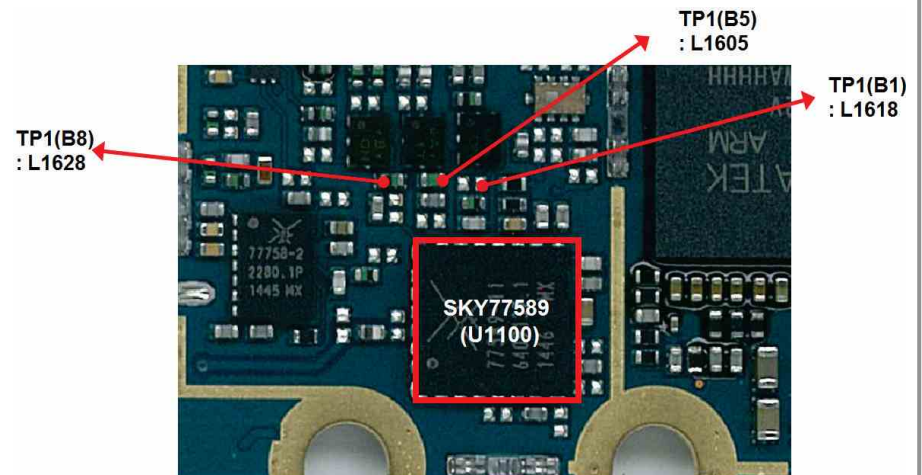
### Checking RF signal path (DPX)

#### Checking Flow



#### Image

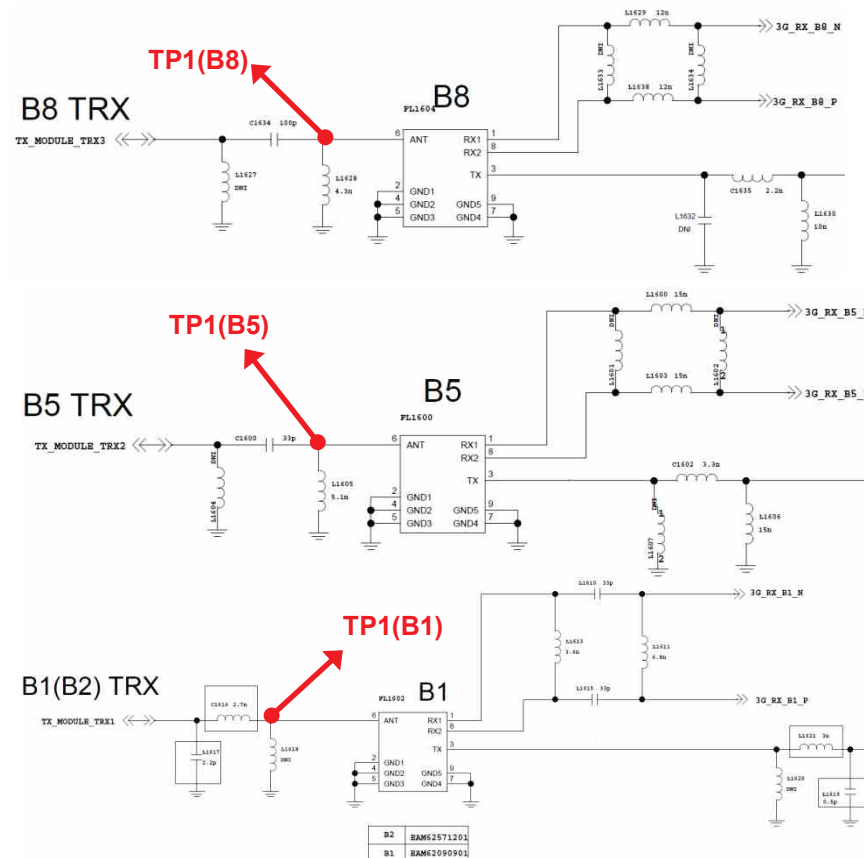
Main  
Top



## 3.1 WCDMA Rx Part

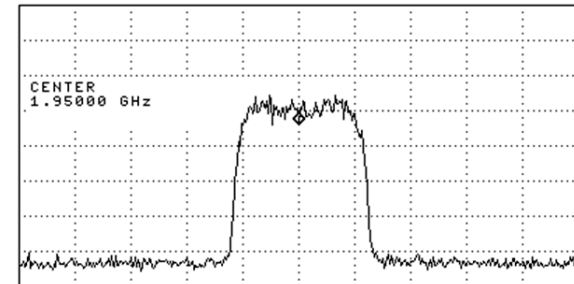
### Checking RF signal path (DPX)

#### Circuit Diagram



#### Image

#### Waveform

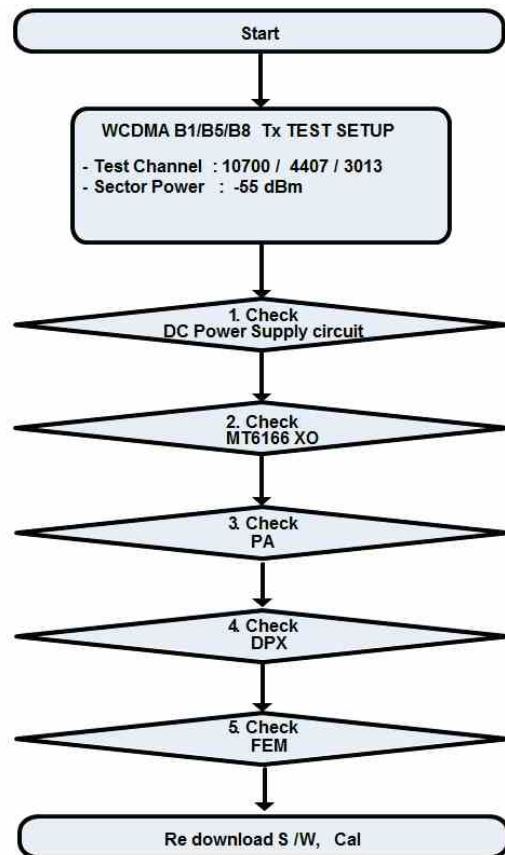


TP1\_RF IN

## 3.2 WCDMA Tx Part

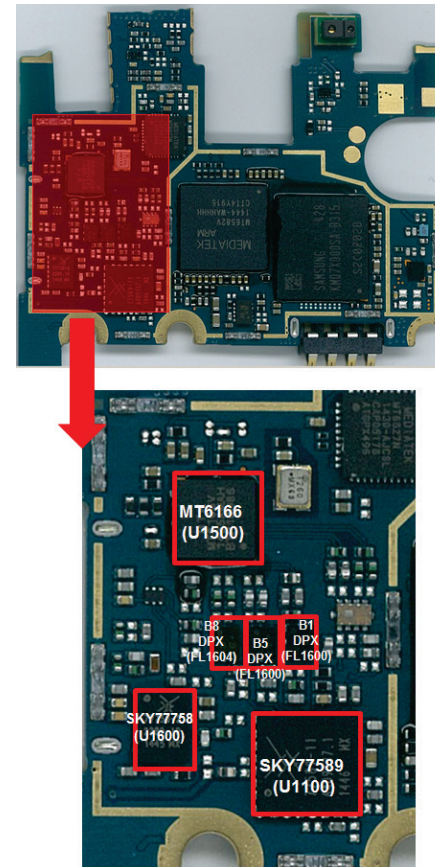
### WCDMA B1/B5/B8 Tx

#### Checking Flow



#### Image

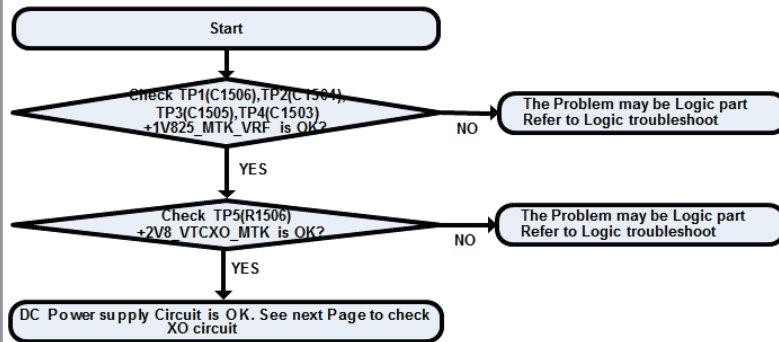
Main  
Top



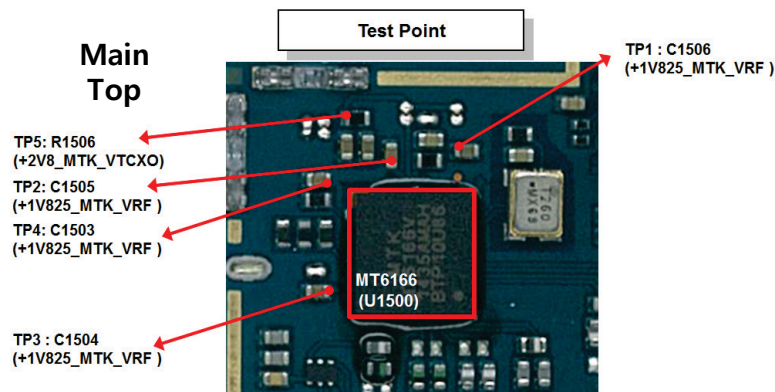
## 3.2 WCDMA Tx Part

### Checking DC Power supply circuit

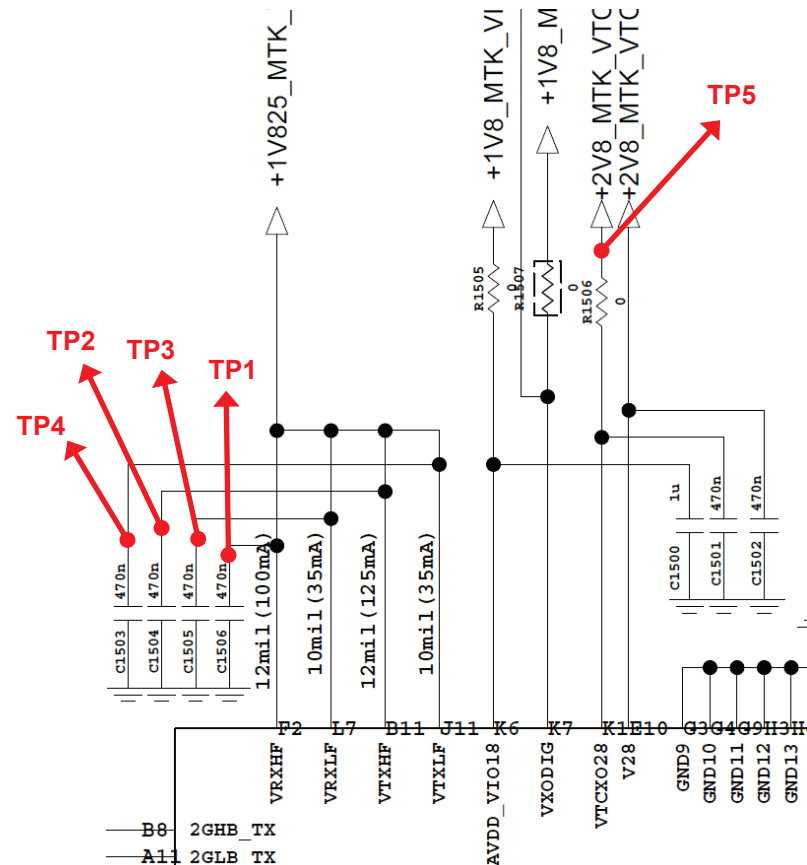
#### Checking Flow



#### Image



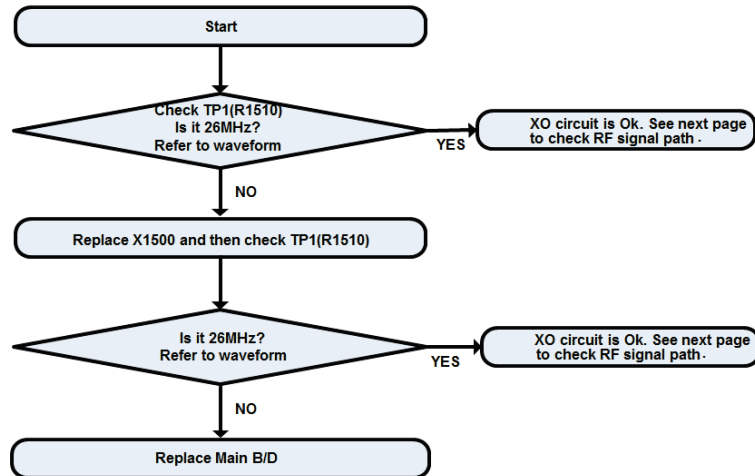
#### Circuit Diagram



## 3.2 WCDMA Tx Part

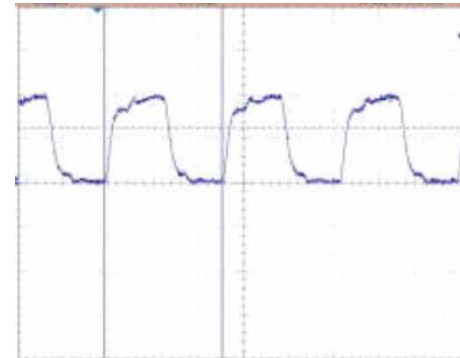
### Checking MT6166 XO circuit

#### Checking Flow



#### Image

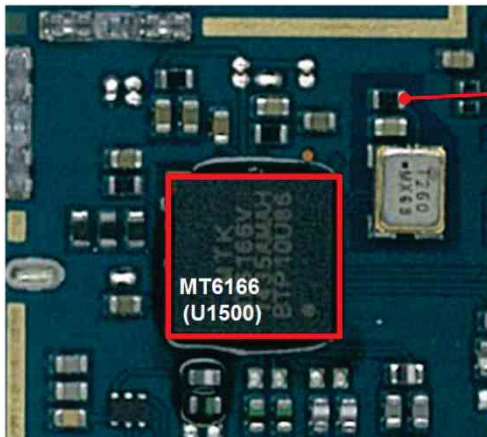
#### Waveform



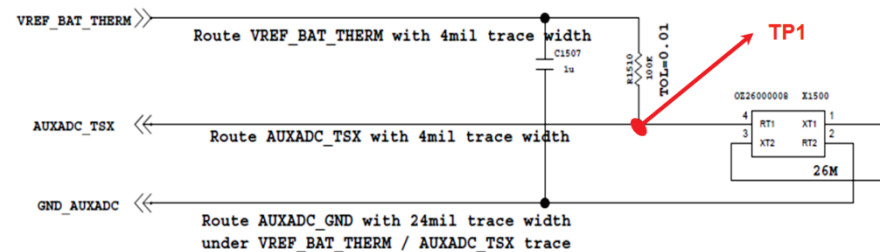
## 3.2 WCDMA Tx Part

### Checking MT6166 XO circuit

Image



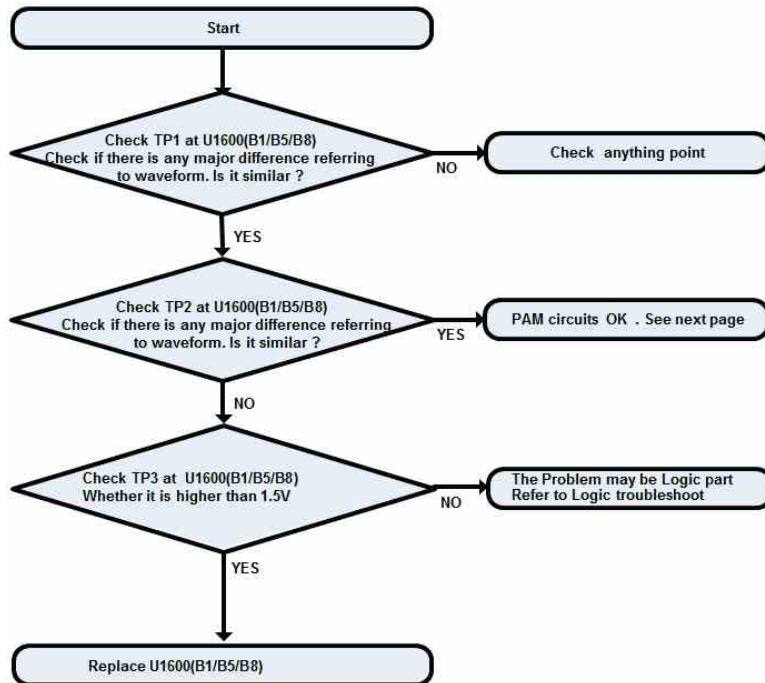
Circuit Diagram



## 3.2 WCDMA Tx Part

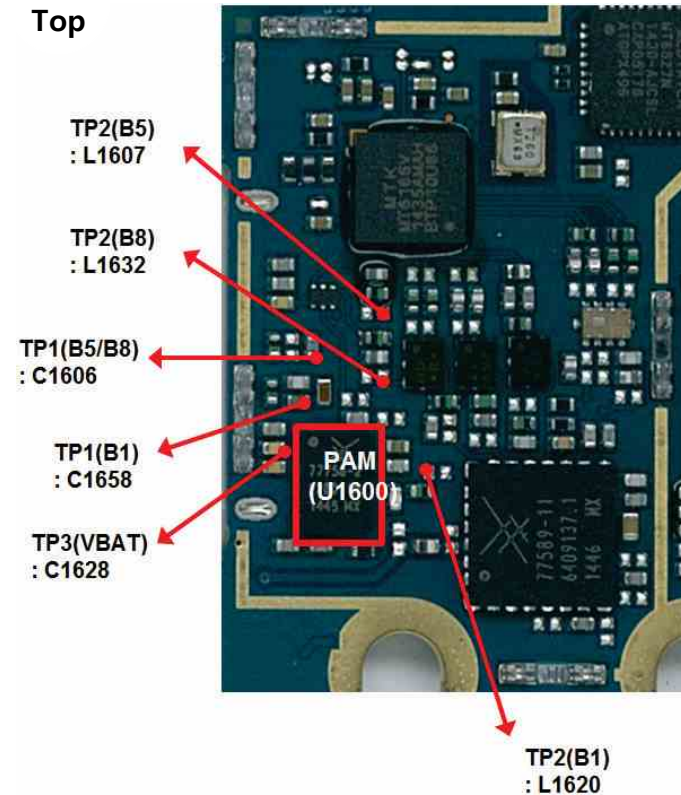
### Checking B1/B5/B8 TX PA

#### Checking Flow



#### Image

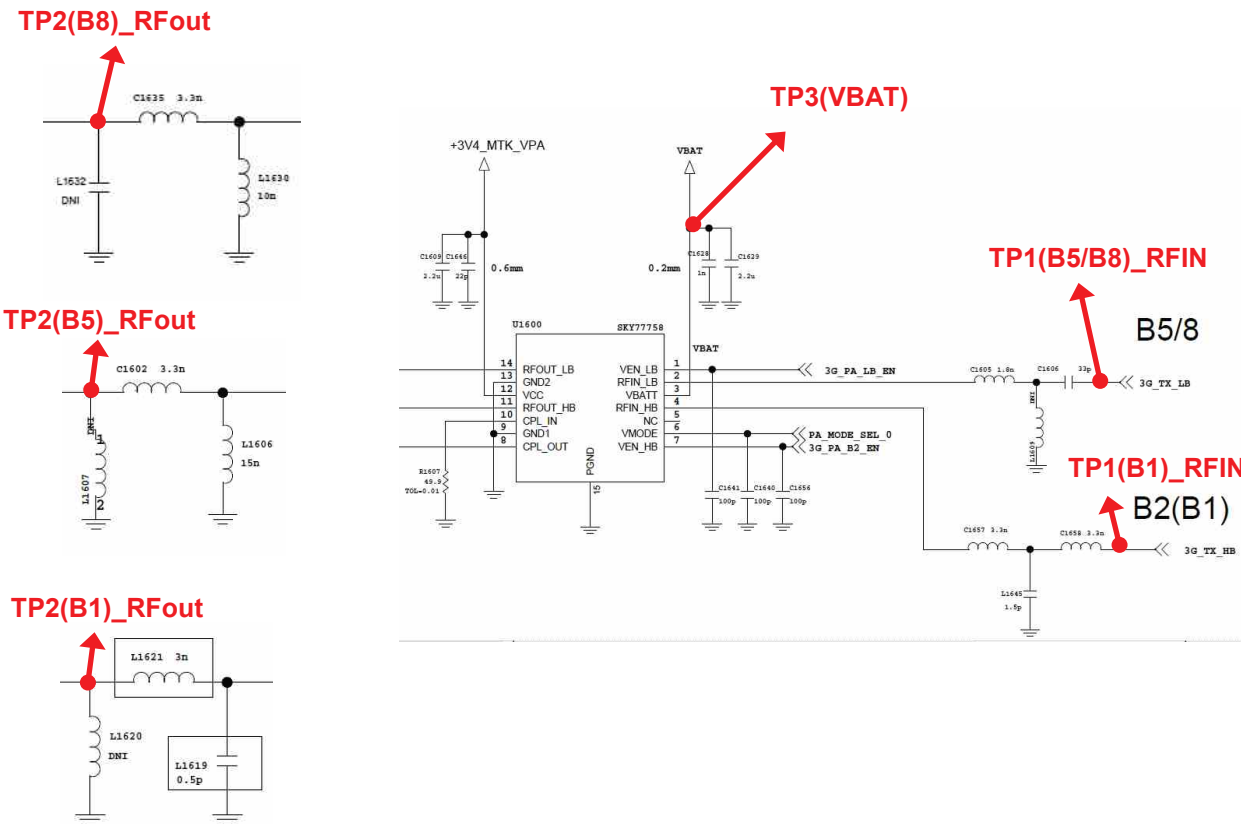
Main  
Top



## 3.2 WCDMA Tx Part

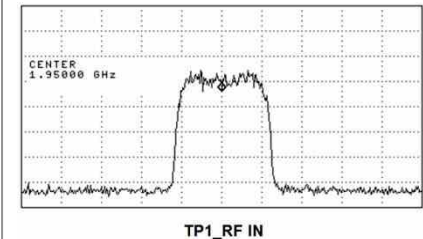
### Checking B1/B5/B8 TX PA

#### Circuit Diagram



#### Image

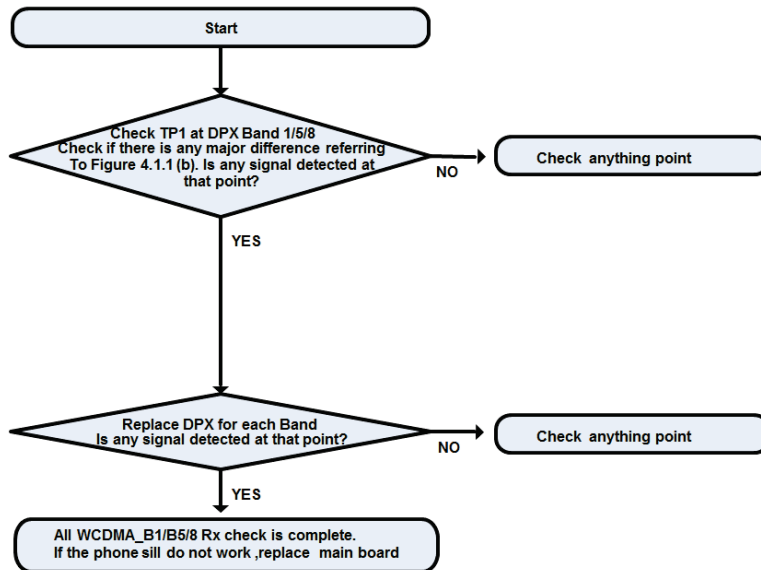
#### Waveform



## 3.2 WCDMA Tx Part

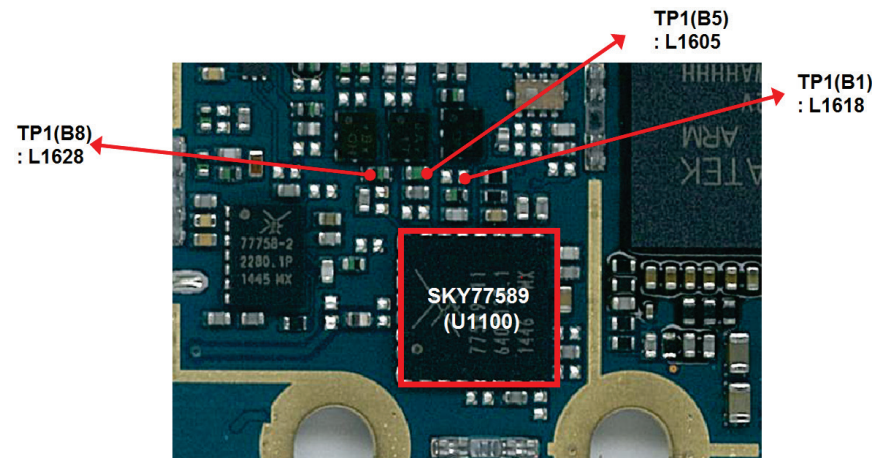
### Checking RF signal path (DPX)

#### Checking Flow



#### Image

#### Main Top

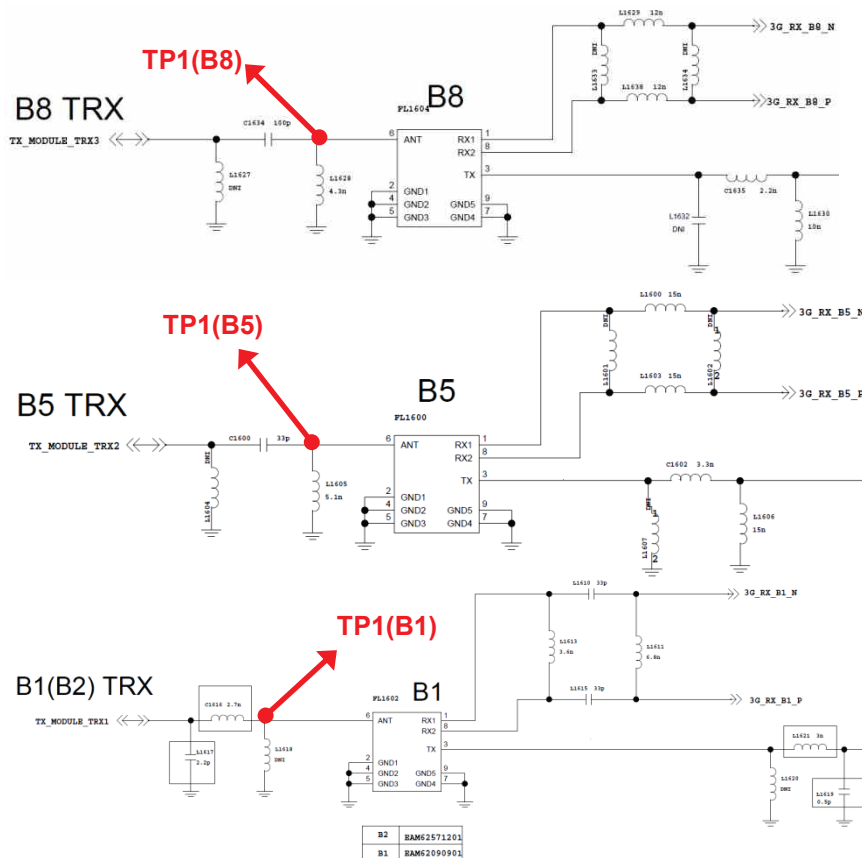


### 3. TROUBLE SHOOTING

### 3.2 WCDMA Tx Part

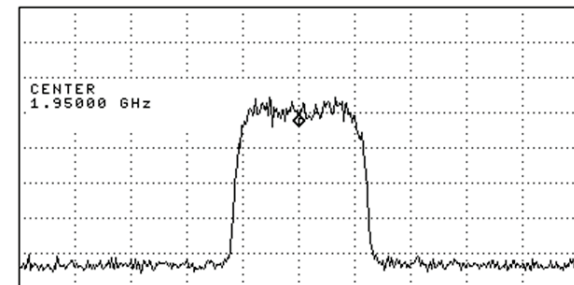
## Checking RF signal path (DPX)

### Circuit Diagram



## Image

## Waveform

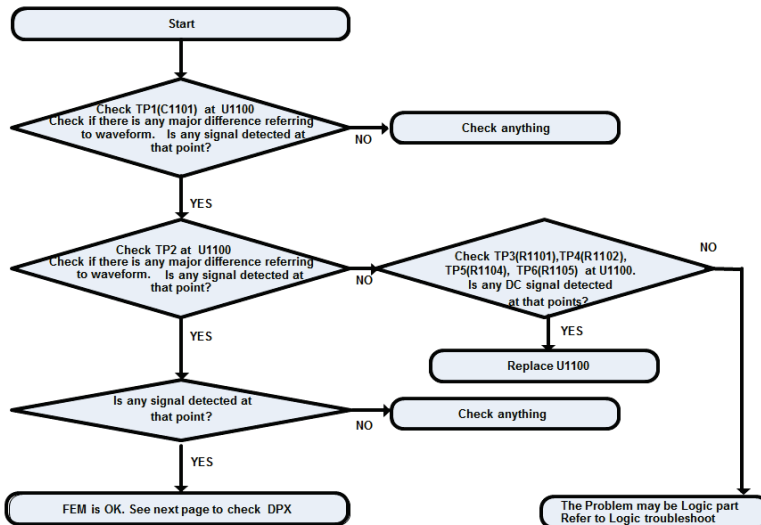


TP1\_RF IN

## 3.2 WCDMA Tx Part

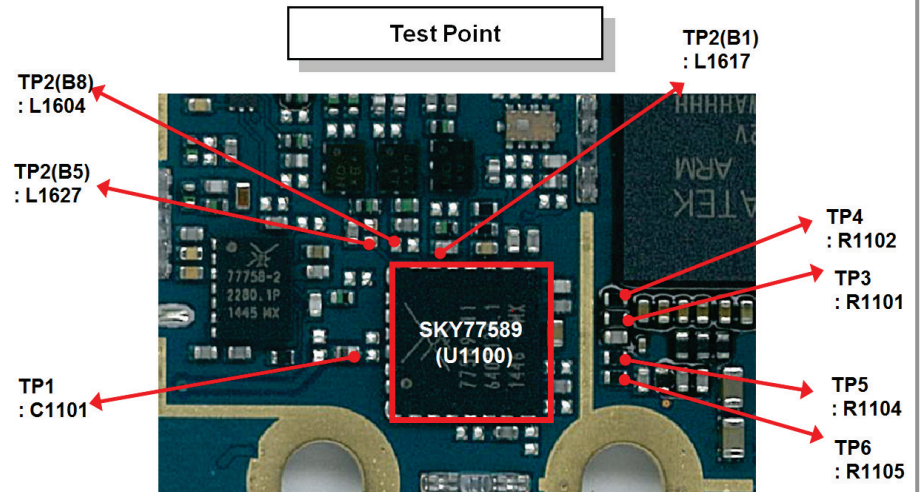
### Checking RF signal path (FEM)

#### Checking Flow



#### Image

#### Main Top

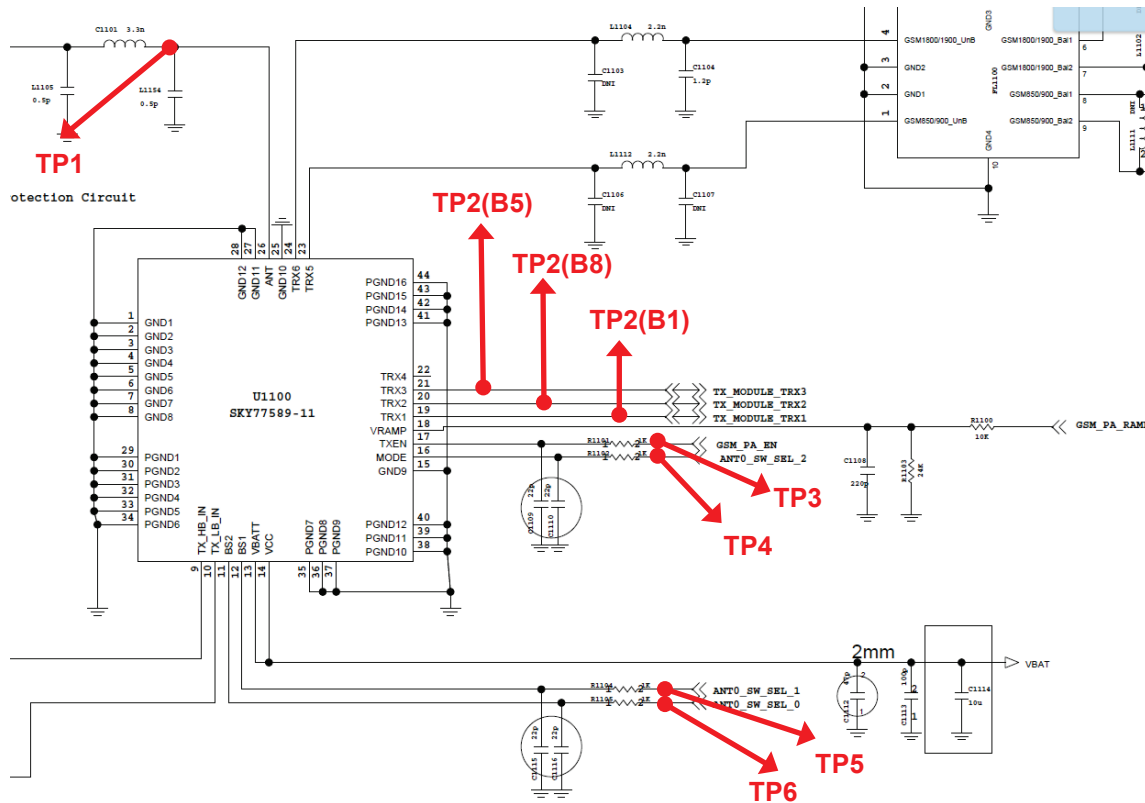


### 3. TROUBLE SHOOTING

### 3.2 WCDMA Tx Part

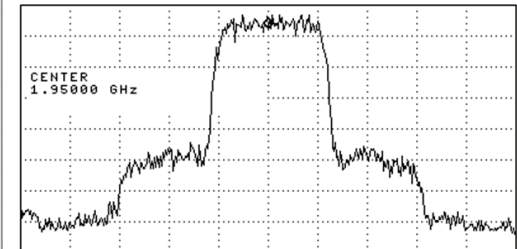
## Checking RF signal path (FEM)

### Circuit Diagram



## Image

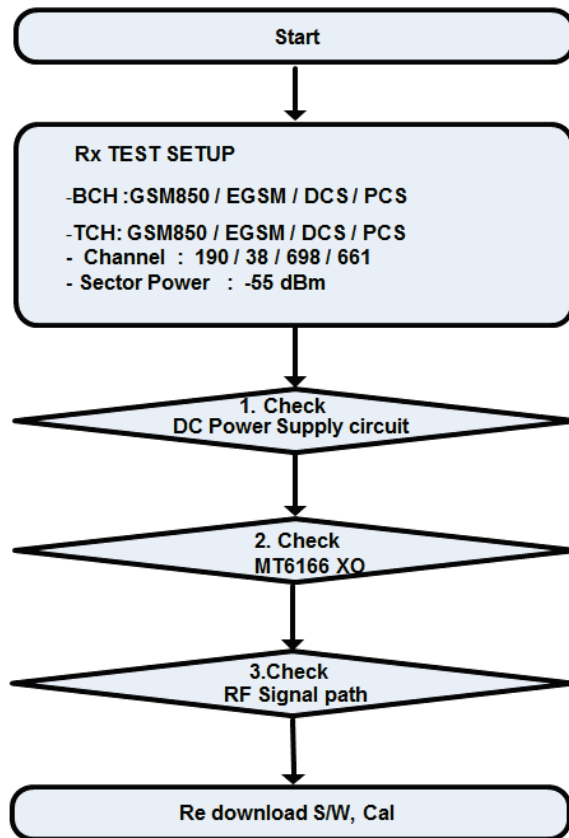
## Waveform



## 3.3 GSM Rx Part

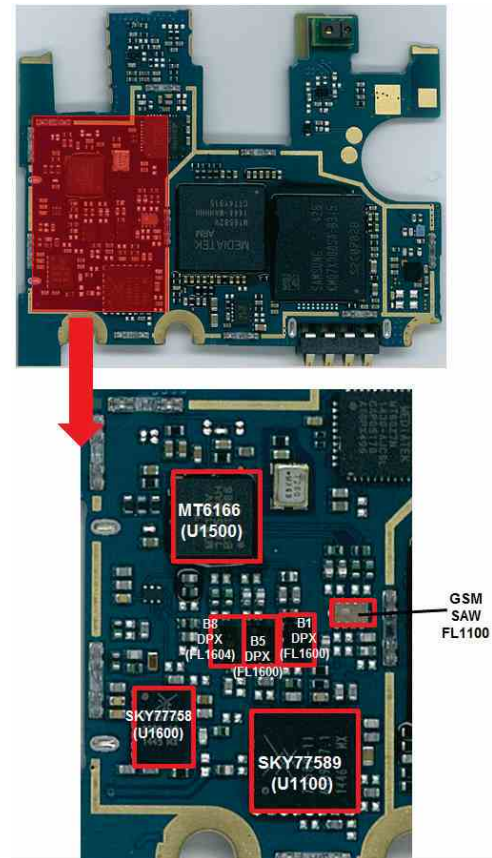
### GSM850/900/DCS/PCS Rx

#### Checking Flow



#### Image

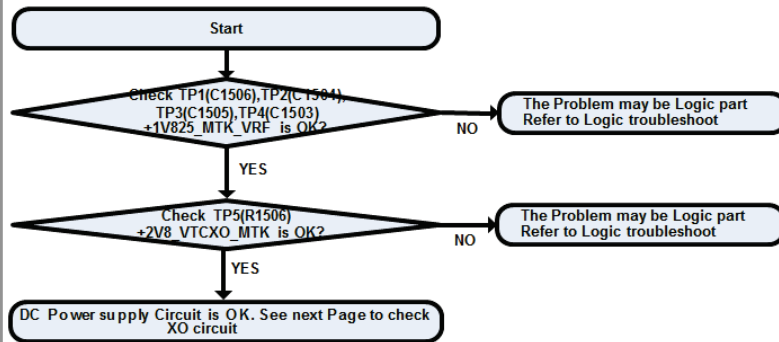
Main  
Top



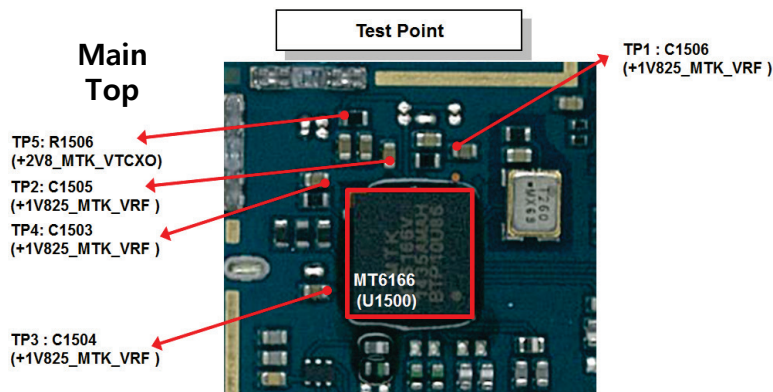
### 3.3 GSM Rx Part

## Checking DC Power supply circuit

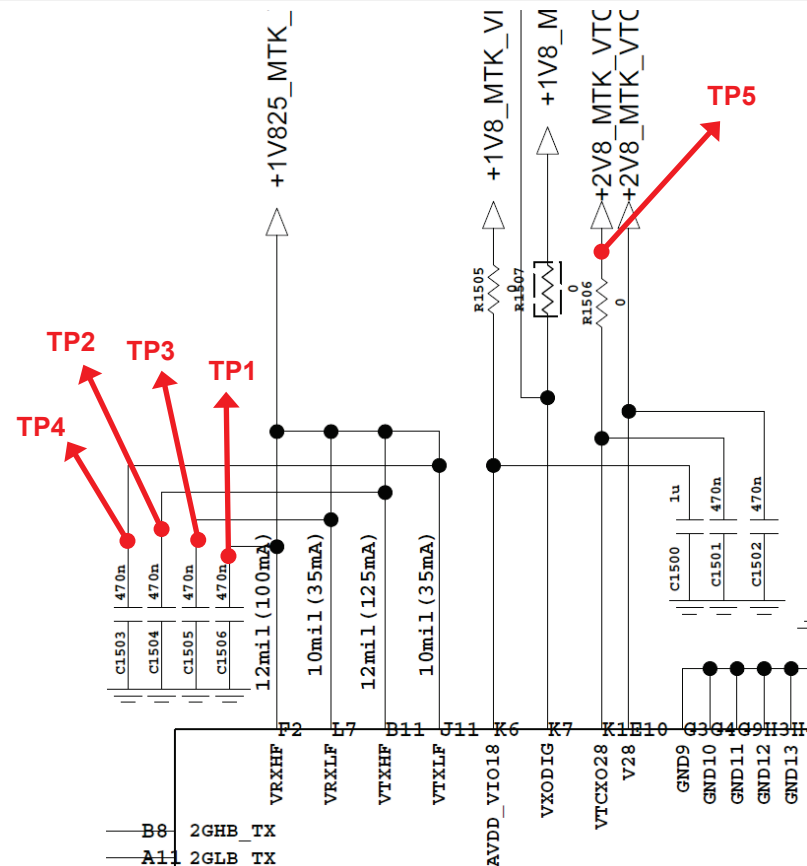
## Checking Flow



## Image



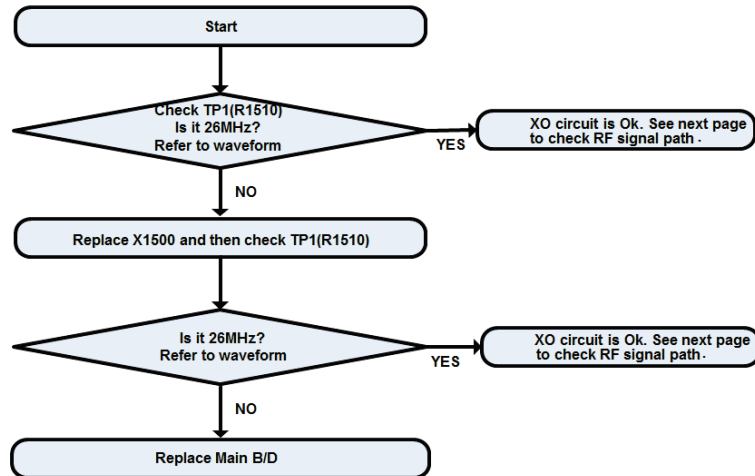
## Circuit Diagram



## 3.3 GSM Rx Part

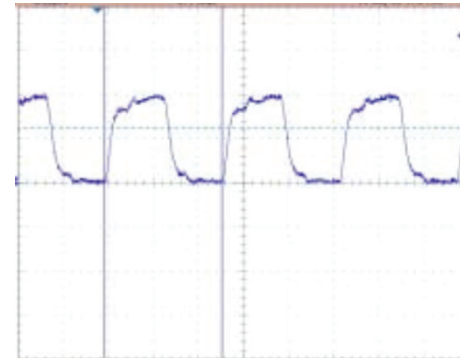
### Checking MT6166 XO circuit

#### Checking Flow



#### Image

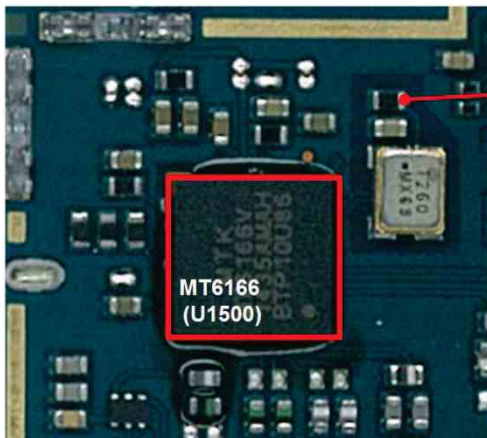
#### Waveform



## 3.3 GSM Rx Part

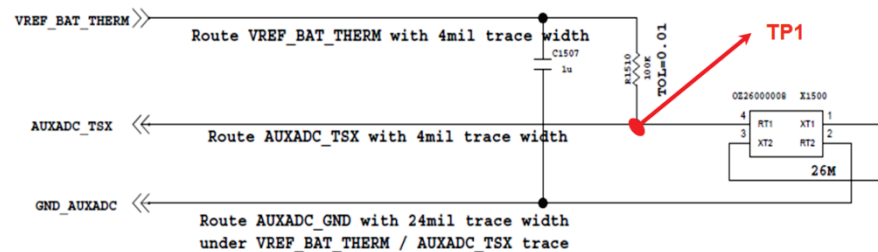
### Checking MT6166 XO circuit

Image



TP1: R1510  
(MAIN\_CLK\_26MHz out)

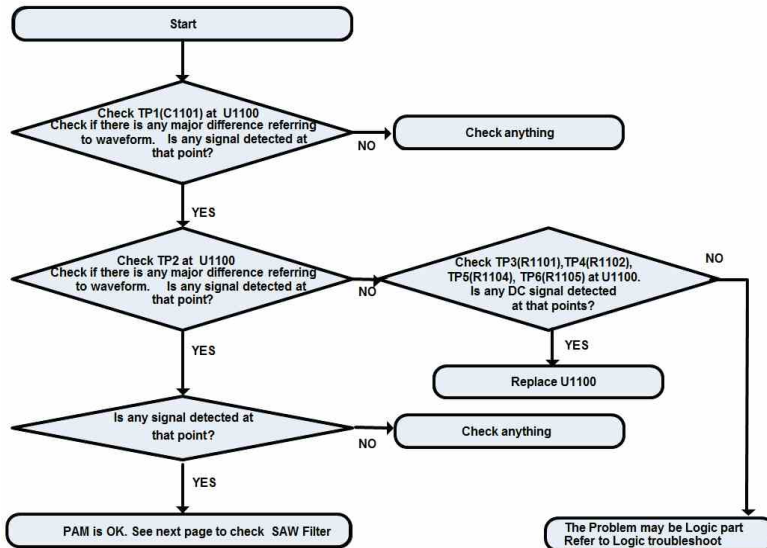
Circuit Diagram



## 3.3 GSM Rx Part

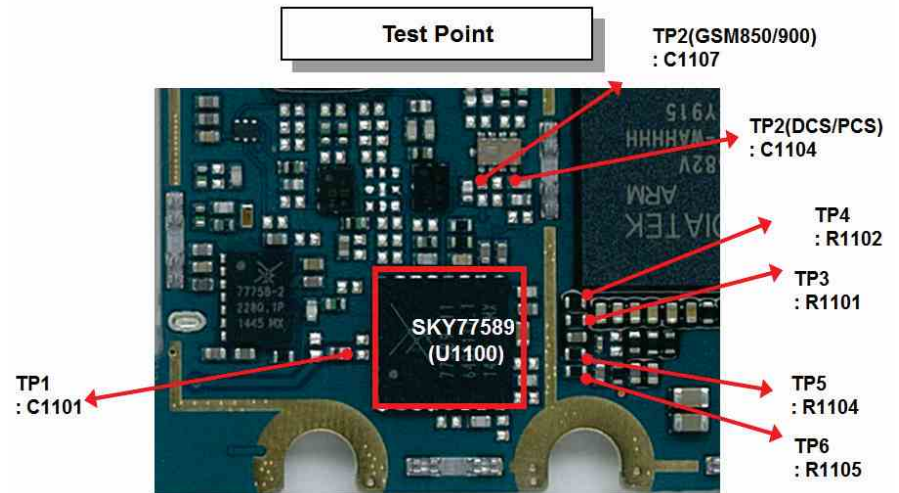
### Checking RF signal path (FEM)

#### Checking Flow



#### Image

#### Main Top

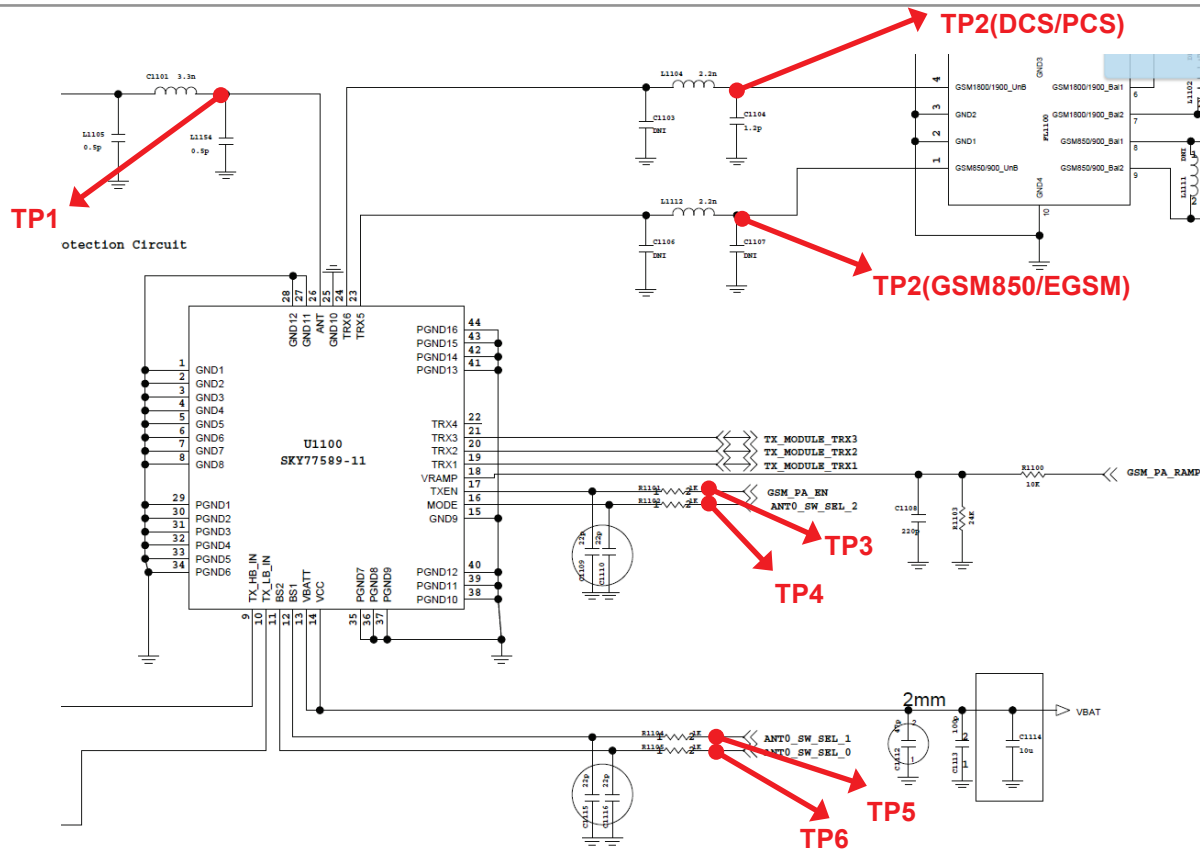


### 3.3 GSM Rx Part

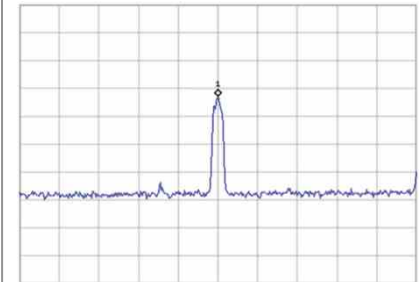
## Checking RF signal path (FEM)

### Circuit Diagram

## Image



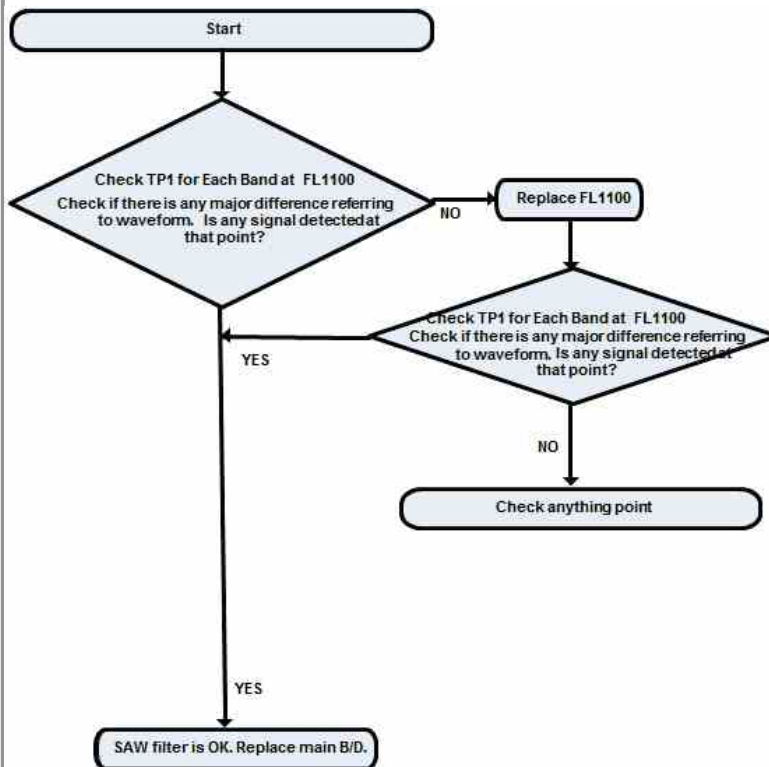
## Waveform



## 3.3 GSM Rx Part

### Checking RF Flow(SAW filter)

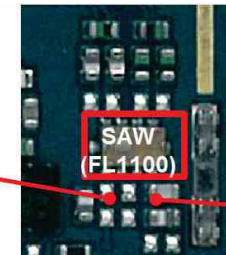
#### Checking Flow



#### Image

Main  
Top

TP1: L1112 (GSM850/EGSM)

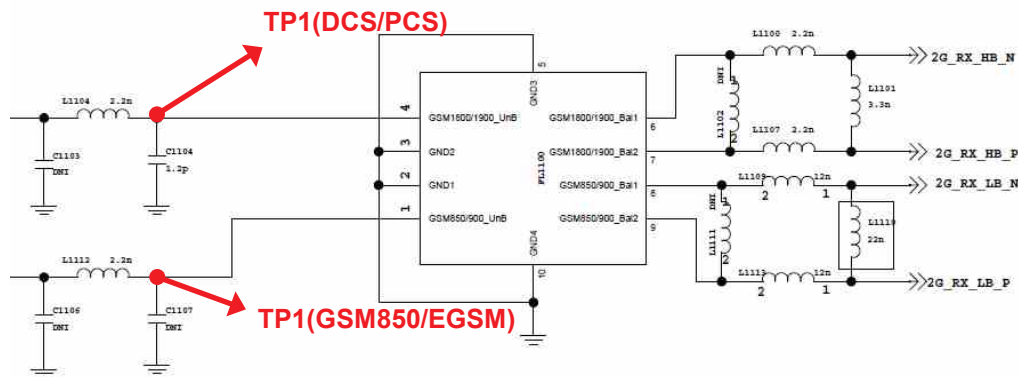


TP1: L1104 (DCS/PCS)

## 3.3 GSM Rx Part

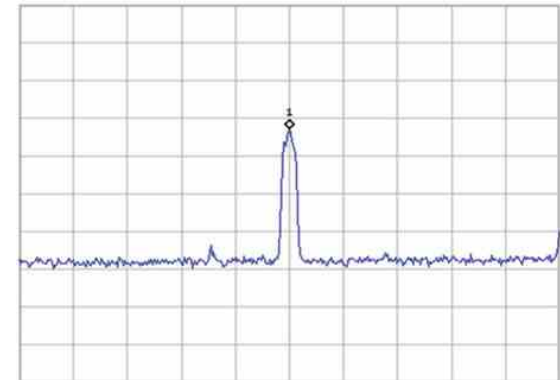
### Checking RF signal path (FEM)

#### Circuit Diagram



#### Image

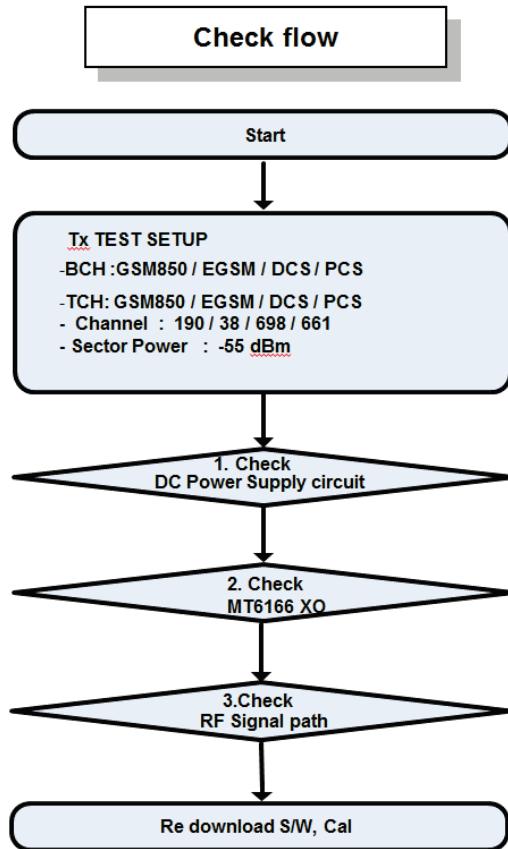
#### Waveform



## 3.4 GSM Tx Part

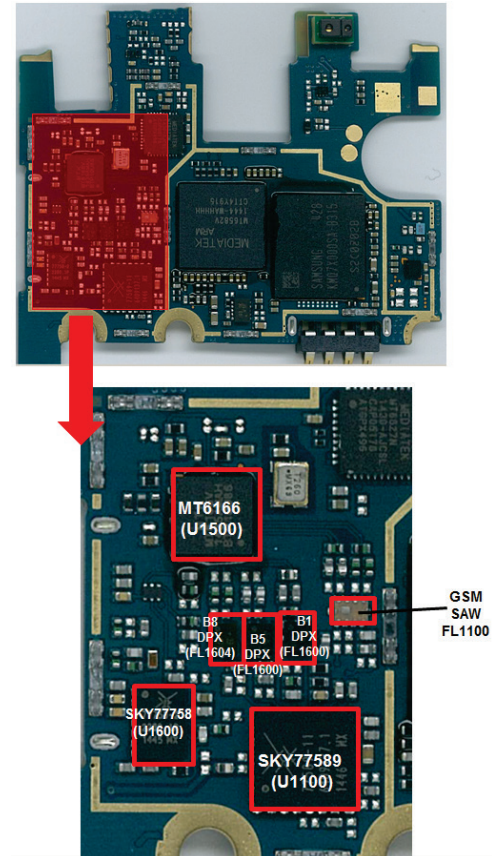
### GSM850/900/DCS/PCS Tx

#### Checking Flow



#### Image

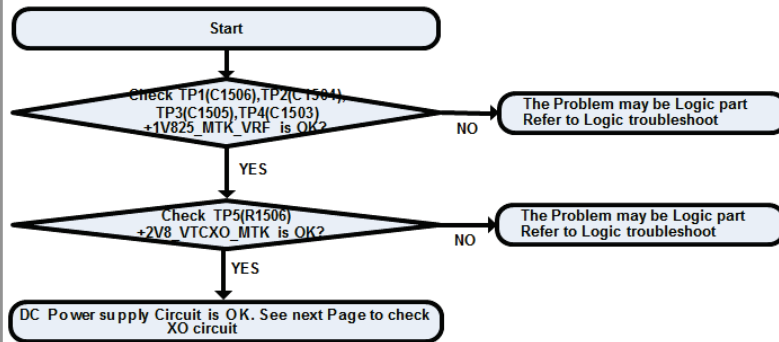
Main  
Top



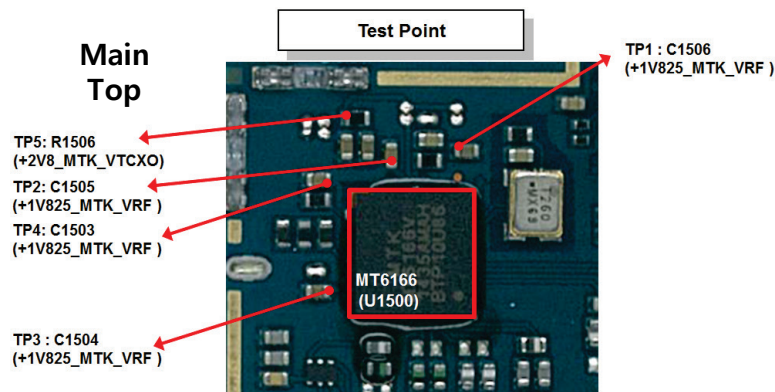
## 3.4 GSM Tx Part

### Checking DC Power supply circuit

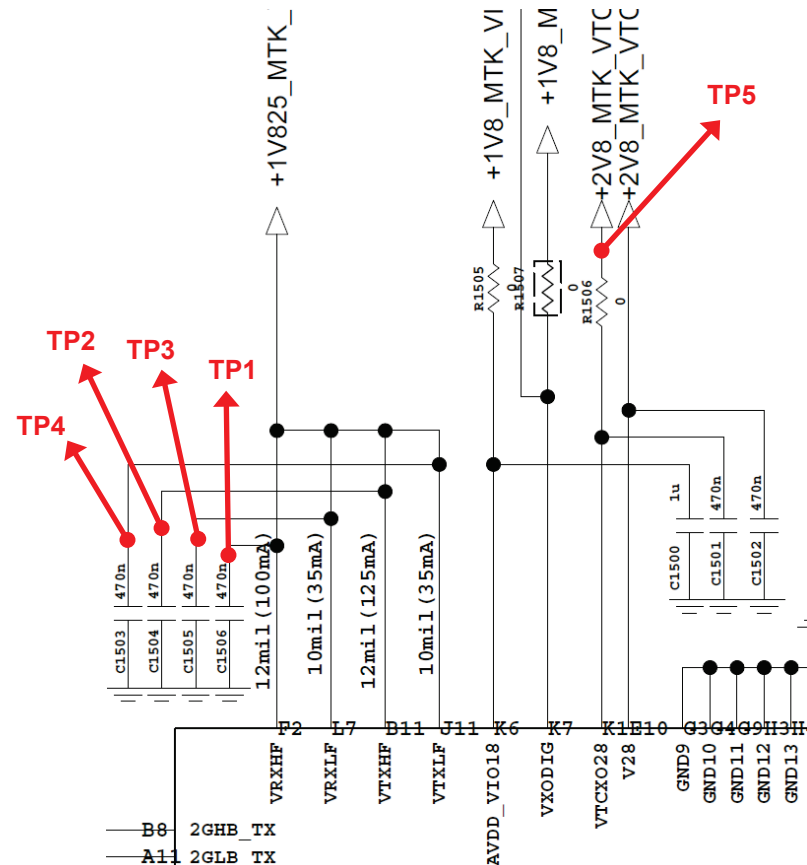
#### Checking Flow



#### Image



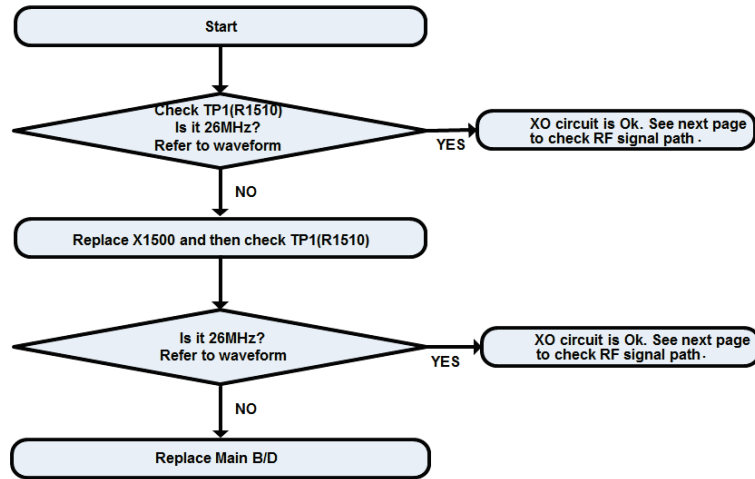
#### Circuit Diagram



## 3.4 GSM Tx Part

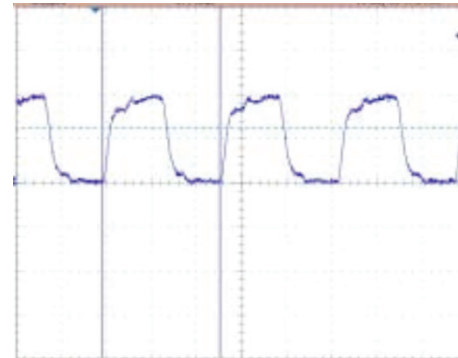
### Checking MT6166 XO circuit

#### Checking Flow



#### Image

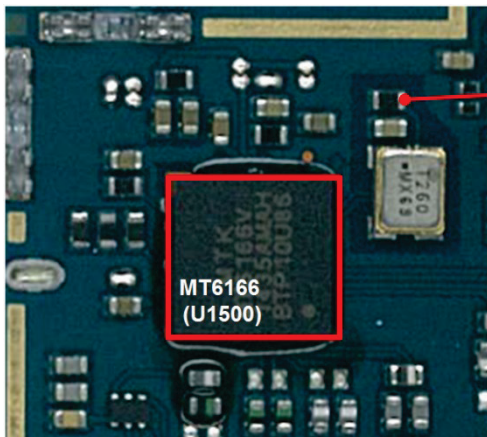
#### Waveform



## 3.4 GSM Tx Part

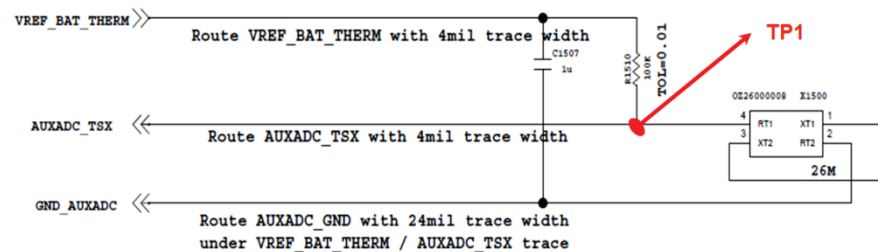
### Checking MT6166 XO circuit

Image



TP1: R1510  
(MAIN\_CLK\_26MHz out)

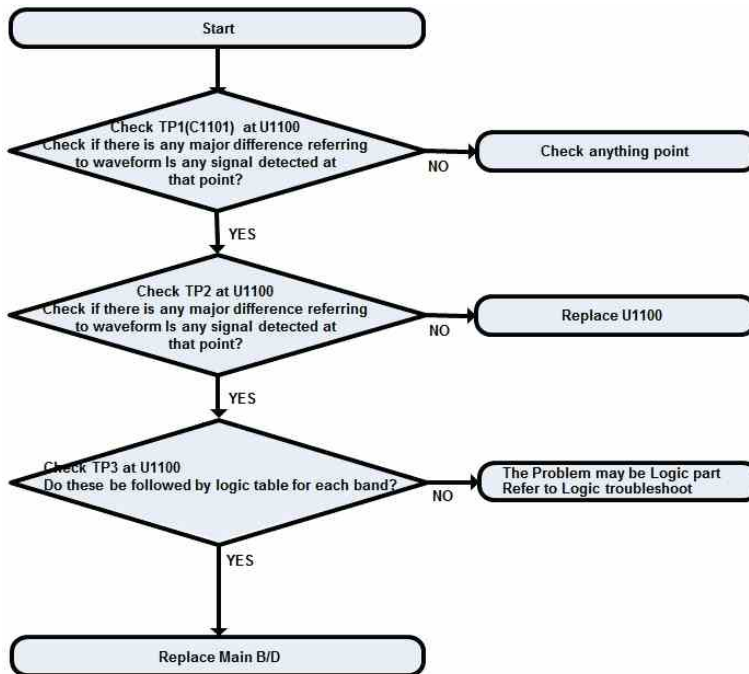
Circuit Diagram



## 3.4 GSM Tx Part

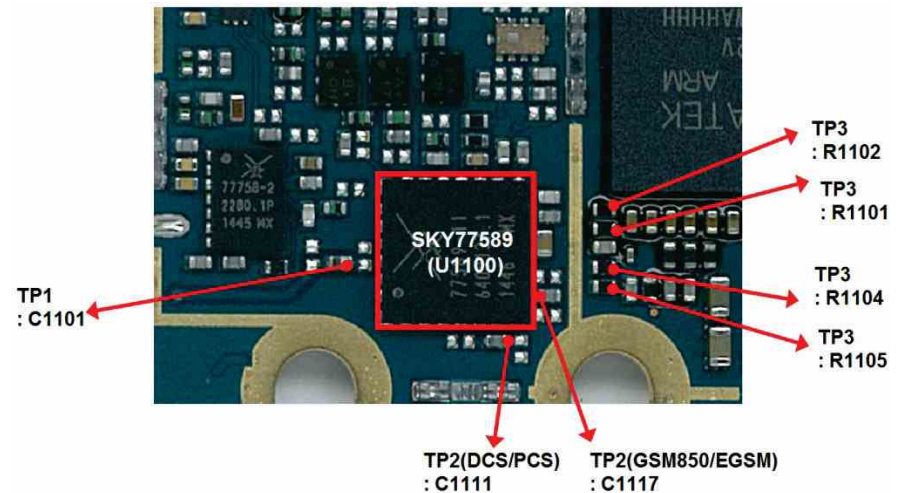
### Checking RF signal path (FEM)

#### Checking Flow



#### Image

#### Main Top

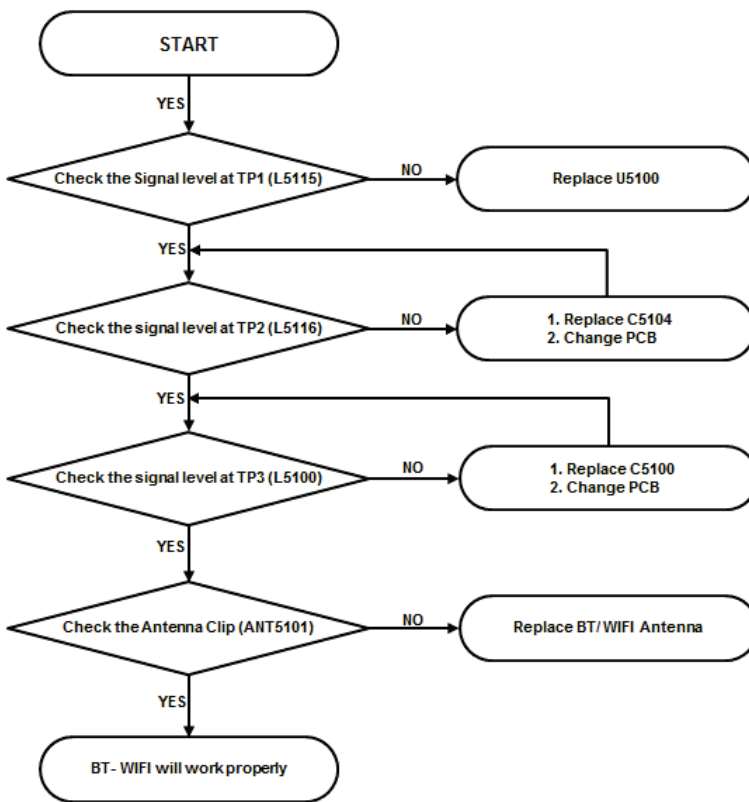


**LGE Internal Use Only**

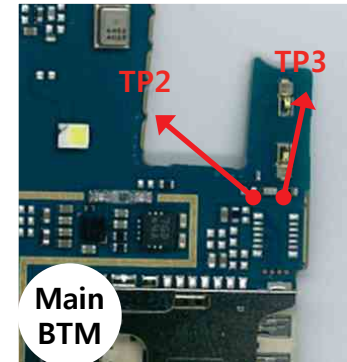
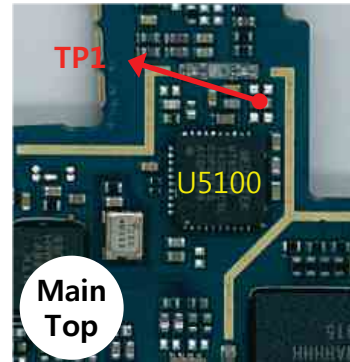
## 3.5 BT/WIFI Block

### Check BT/WIFI path

#### Checking Flow



#### Image



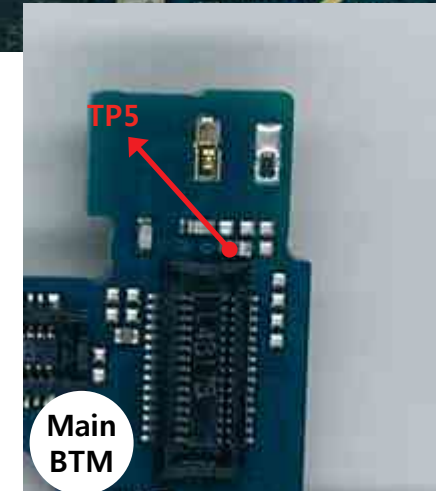
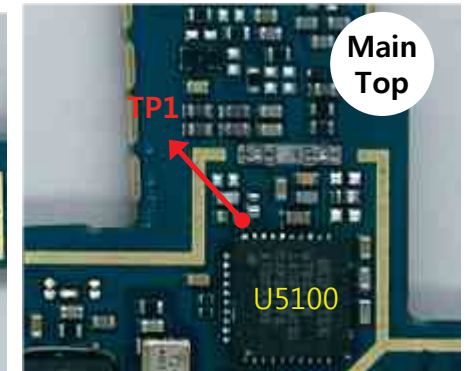
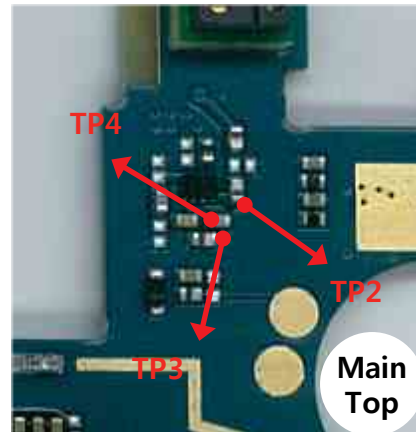
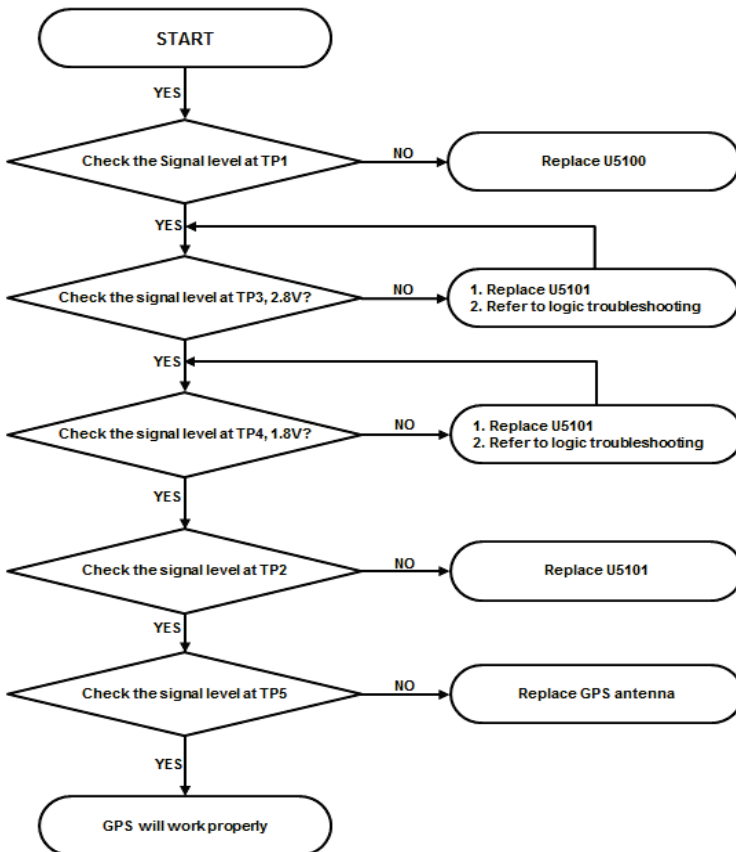
#### Circuit Diagram



## 3.6 GPS Block

### Check GPS path

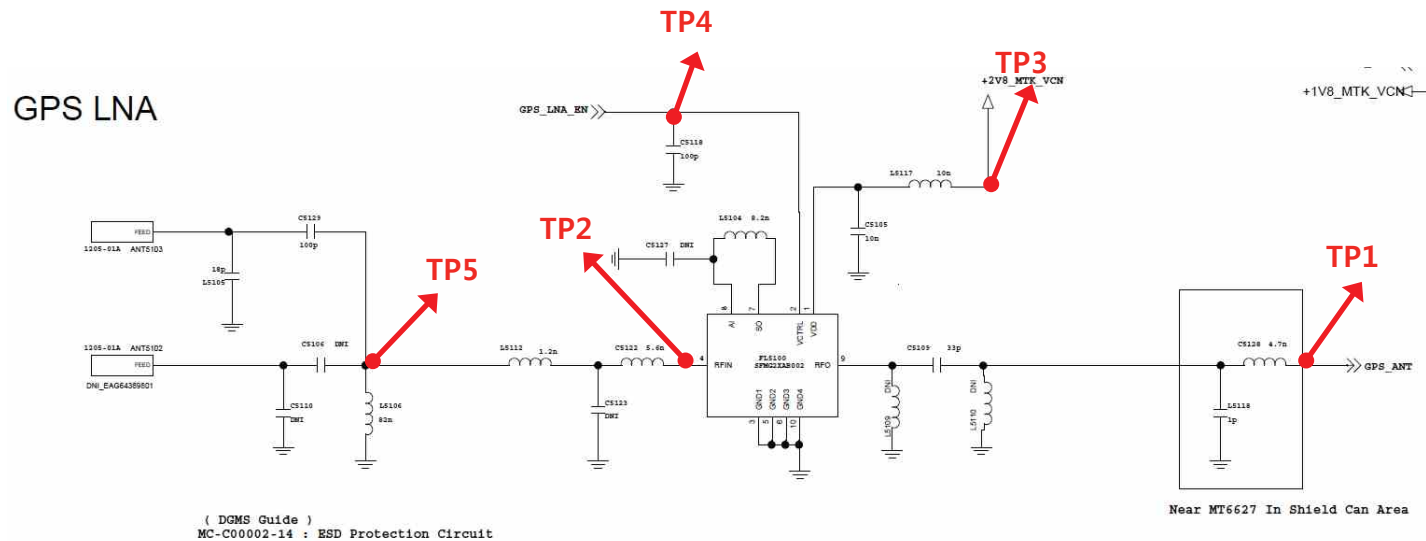
#### Checking Flow



### 3.6 GPS Block

## Check GPS path

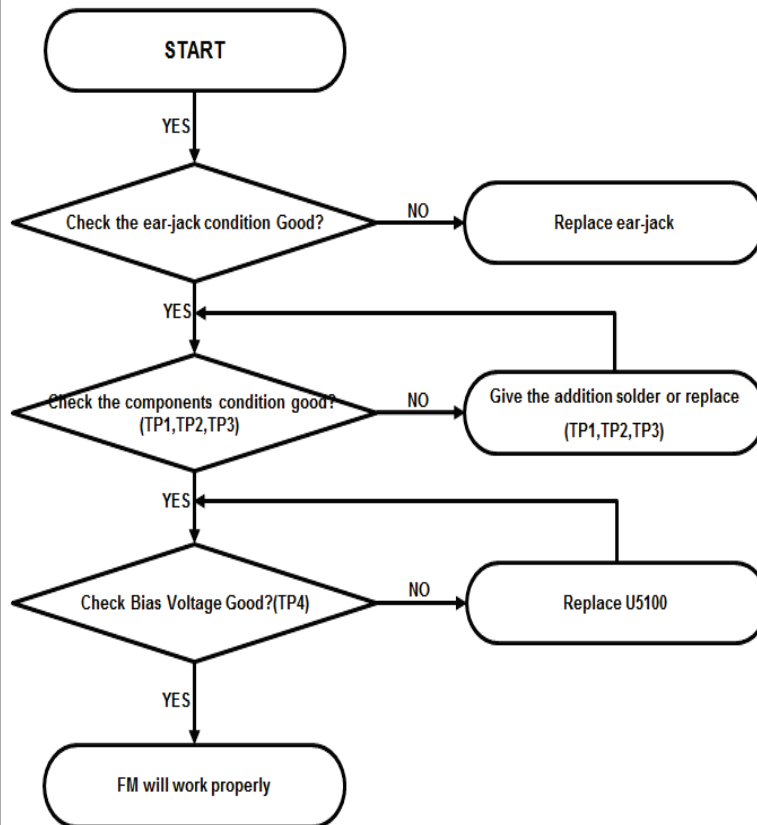
### Circuit Diagram



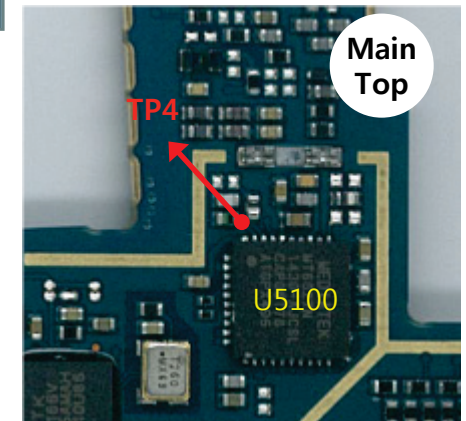
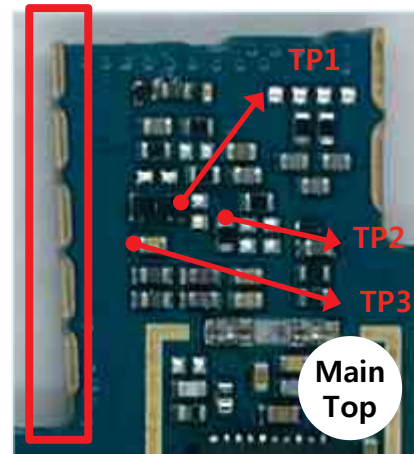
## 3.7 FM RADIO

### Check FM radio path

#### Checking Flow



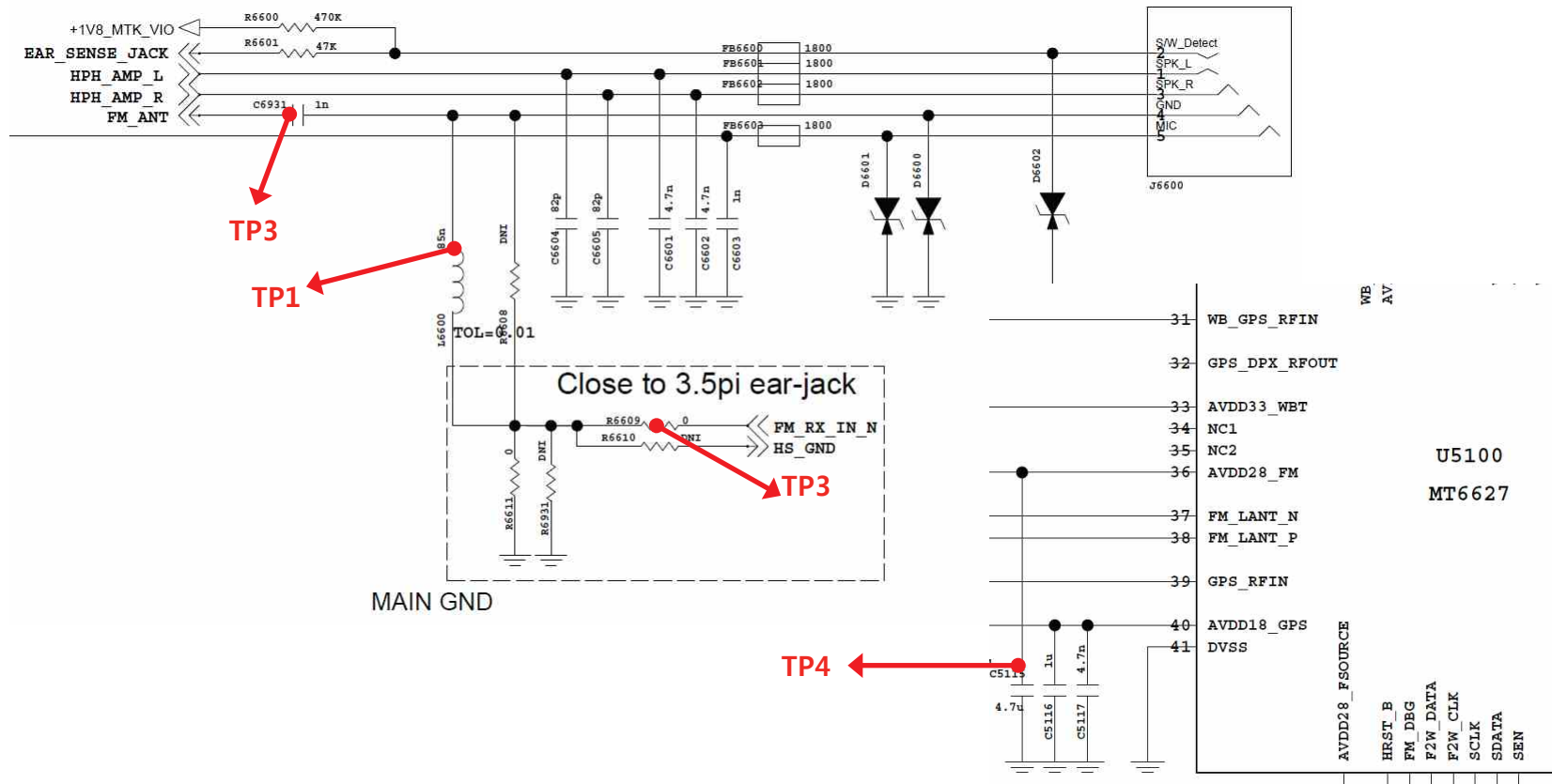
#### Image



### 3.7 FM RADIO

## Check FM radio path

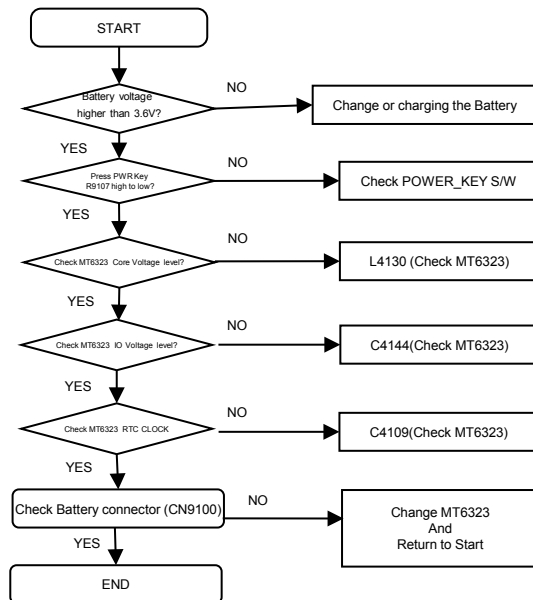
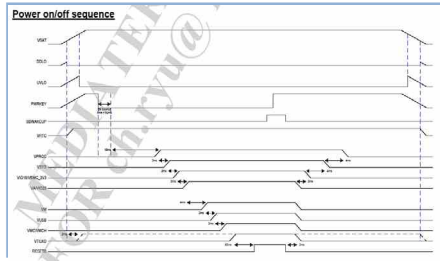
## Circuit Diagram



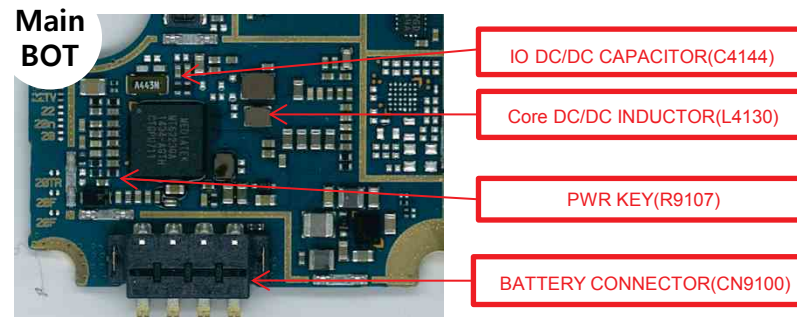
## 3.8 Power

The main power source of H422 is provided by 1 chip which is MT6323

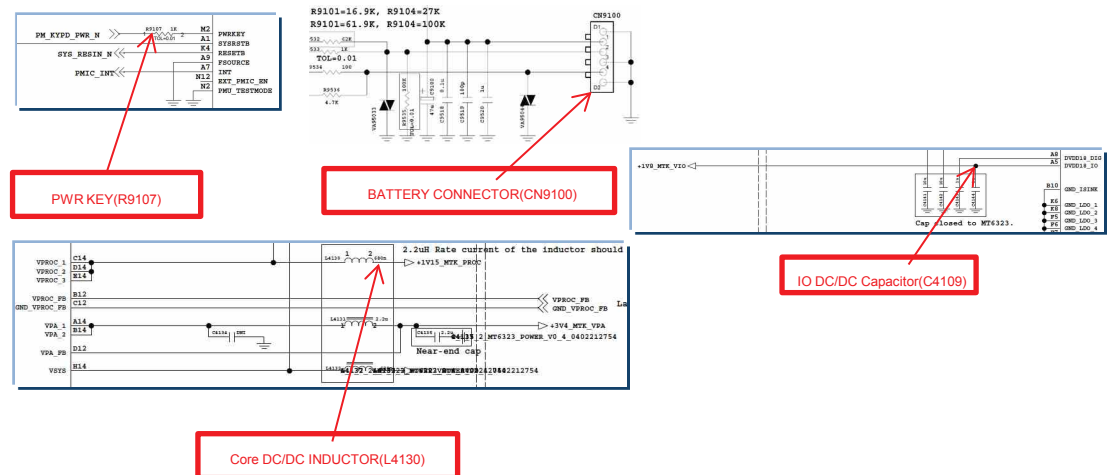
### Checking Flow



### Image



### Circuit Diagram



## 3.9 Charging

H422 uses Charger IC(liner Charger) for charging control. The Charger IC circuit includes the charging pass transistor detects the VDD and VBAT voltages, and its control driver. The Charger IC maximum charging current is set around 710mA.

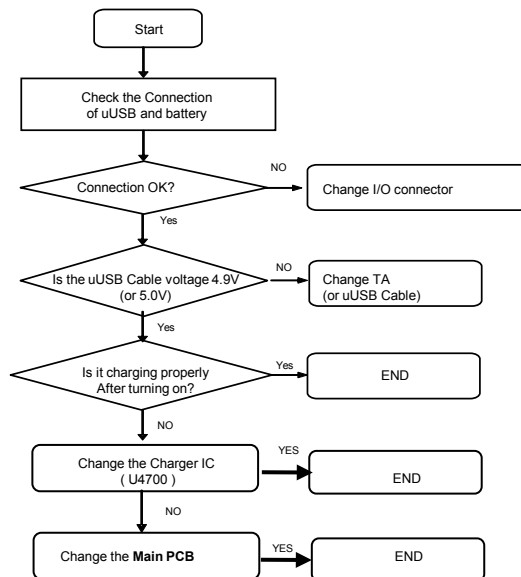
### Checking Flow

#### Charging Procedure

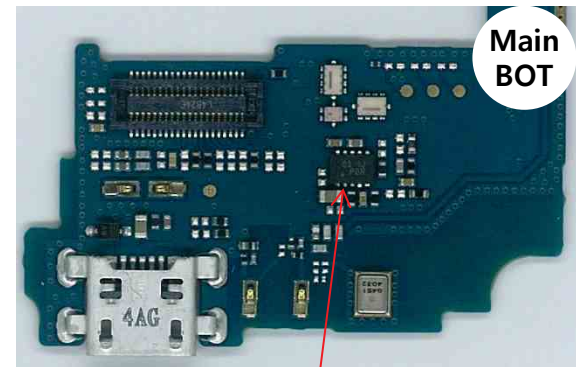
- Supply the VCHG use a TA or u-USB cable
- The Charger IC control the charging current
- Charging current flows into the main battery

#### Troubleshooting Check Point

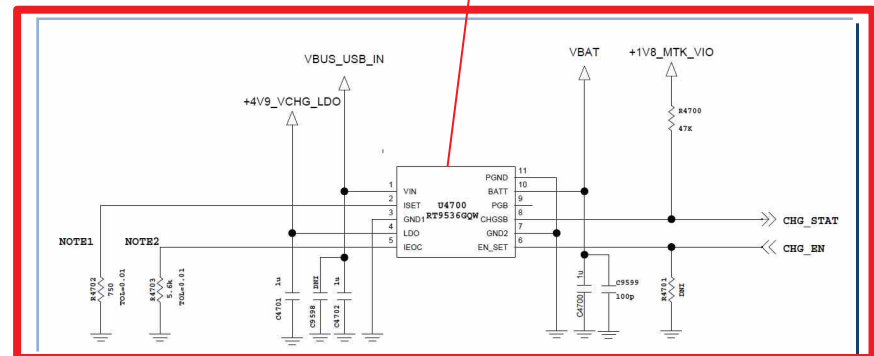
- Connection of the TA or u-USB cable
- Main battery



### Image



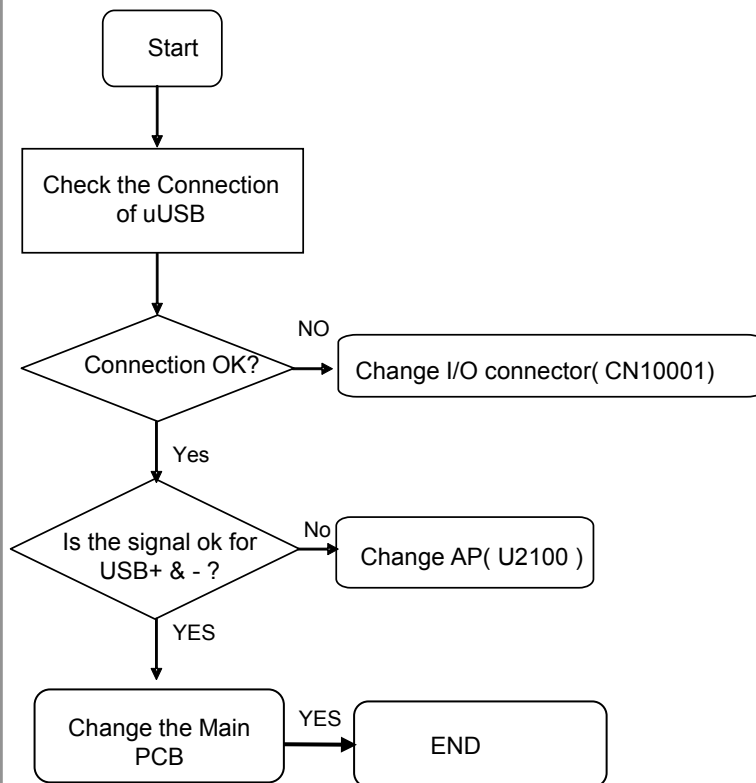
### Circuit Diagram



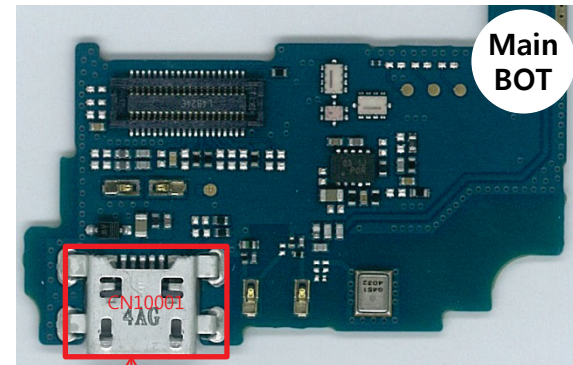
## 3.10 USB

### USB

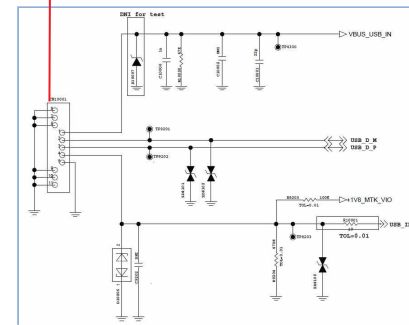
#### Checking Flow



#### Image



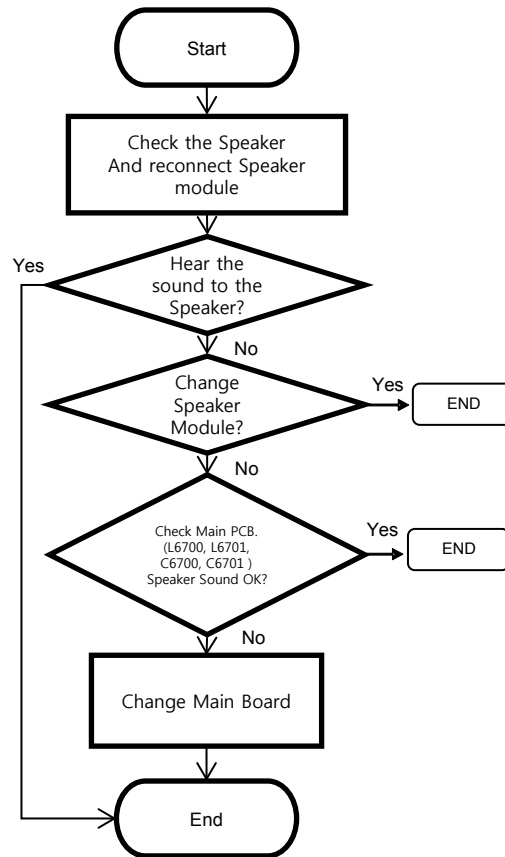
#### Circuit Diagram



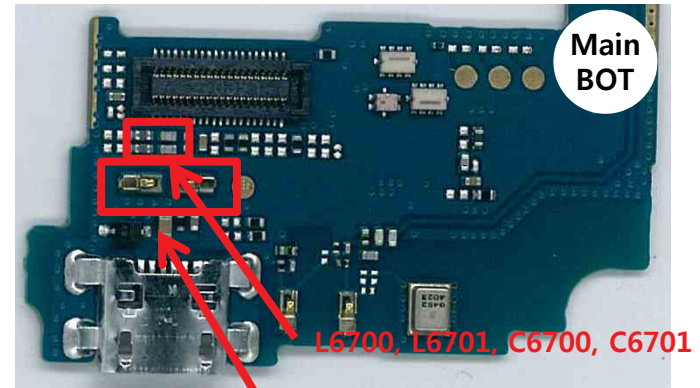
## 3.11.1 Speaker

It's trouble shooting guide for no sound case. Speaker control signals are generate by MT6323(U4100)  
If low level volume or breaking the sound, change speaker module

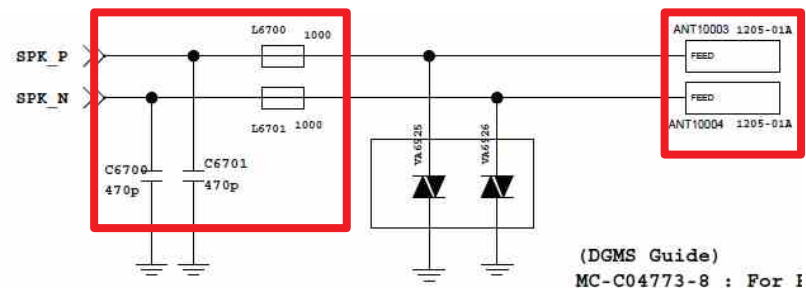
### Checking Flow



### Image



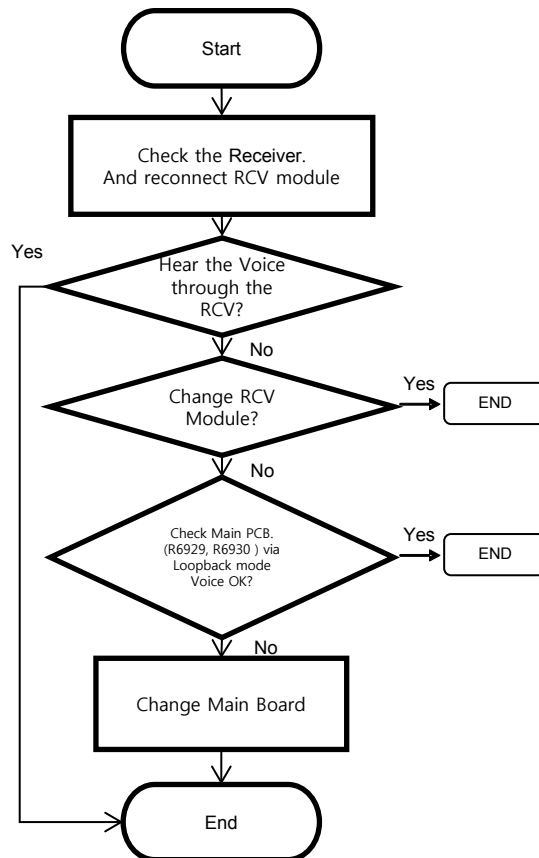
### Circuit Diagram



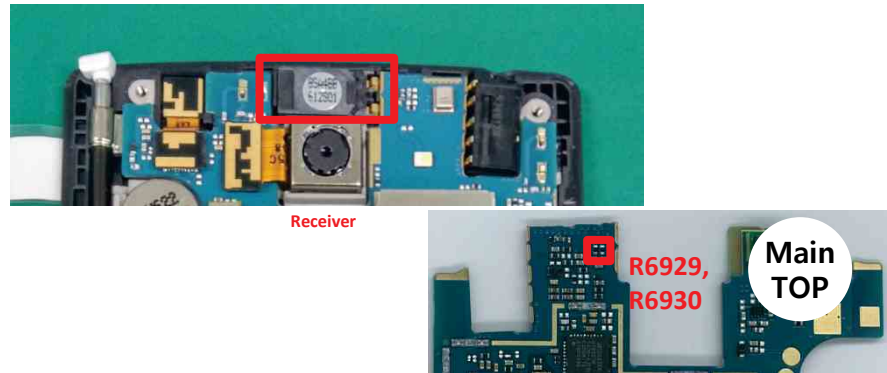
## 3.11.2 Receiver

It's trouble shooting guide for no sound case. Receiver control signals are generate by MT6323(U4100)  
If low level volume or breaking the voice, change RCV module

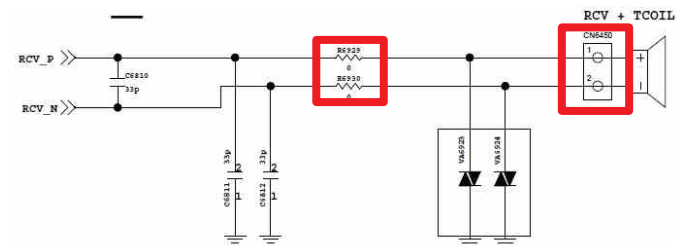
### Checking Flow



### Image



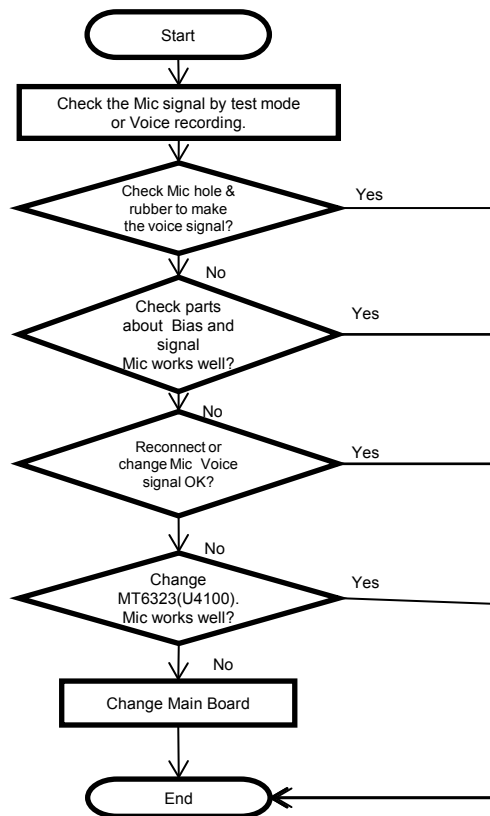
### Circuit Diagram



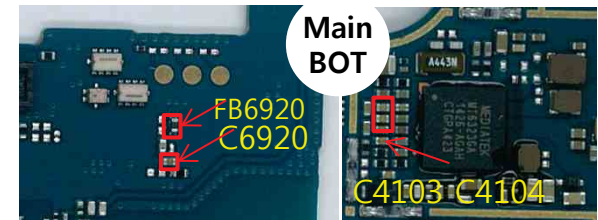
## 3.11.3 Main Mic

It's operating voice call(except speakerphone), voice recording, camcorder recording

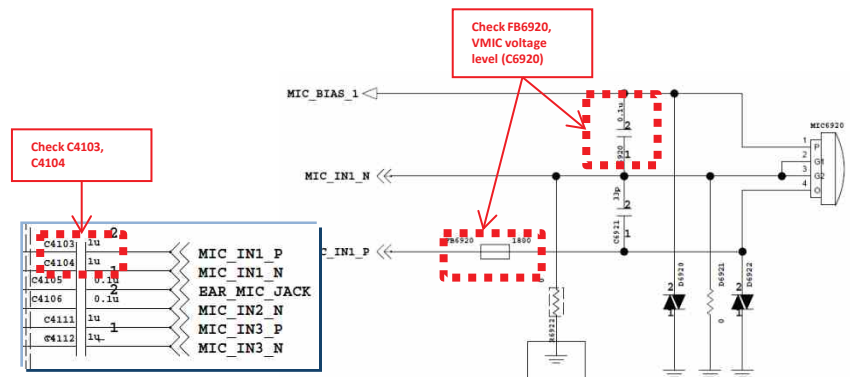
### Checking Flow



### Image



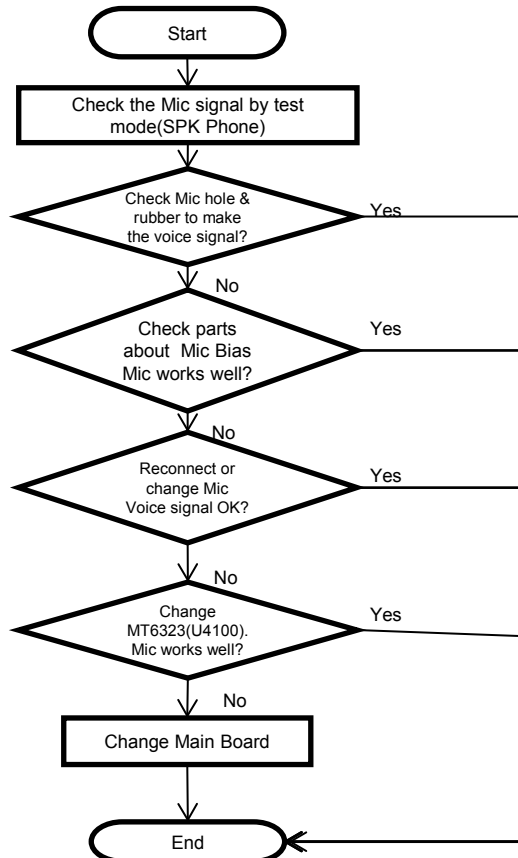
### Circuit Diagram



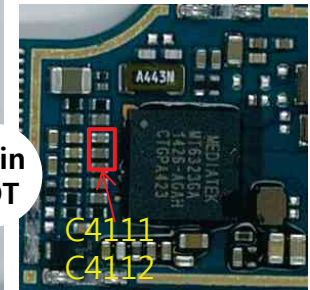
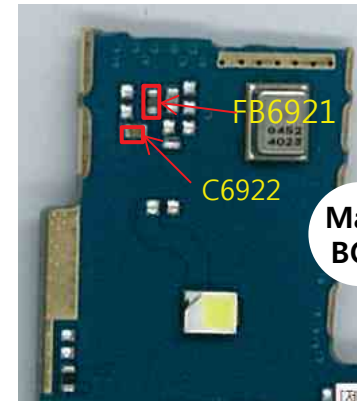
## 3.11.4 Sub Mic

It's operating Speakerphone call

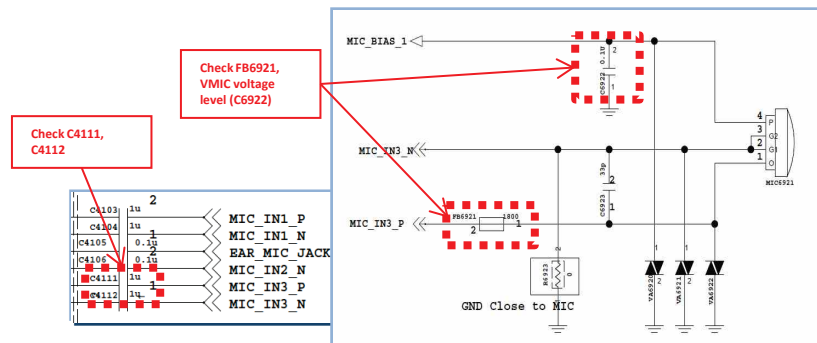
### Checking Flow



### Image



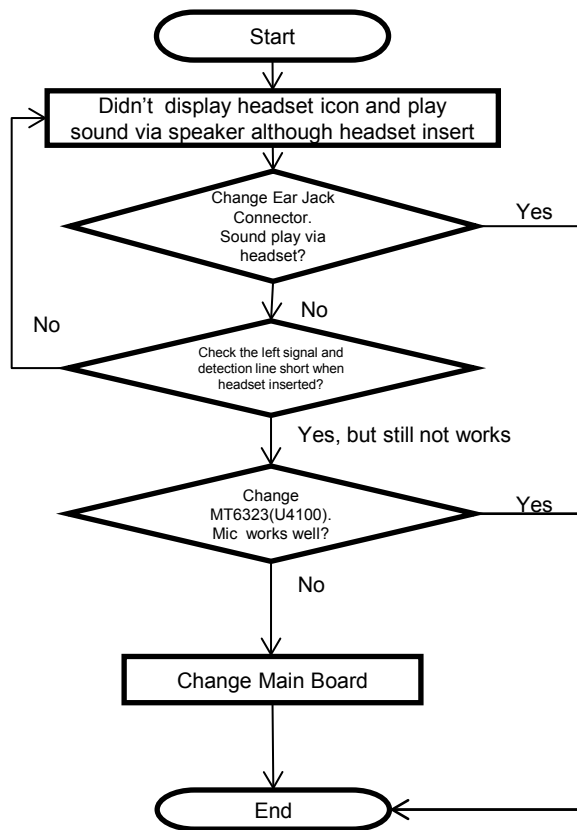
### Circuit Diagram



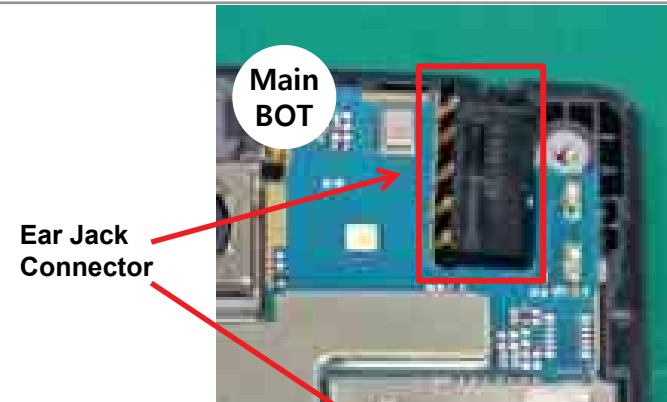
## 3.11.5 Ear-Mic -Case A

Ear MIC control signals are generate by MT6323(U4100)  
Case A. Disable detected headset insert.

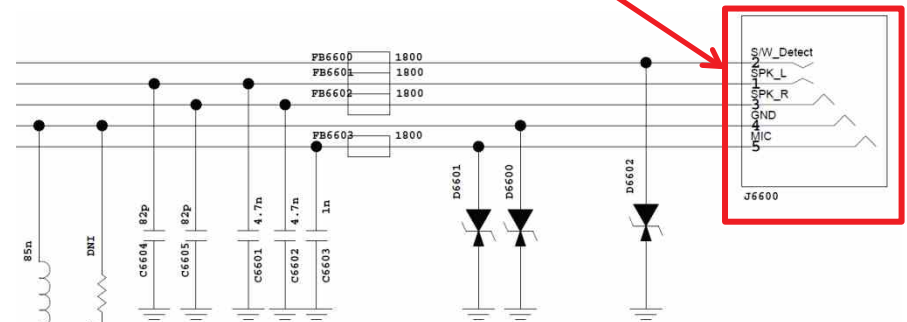
### Checking Flow



### Image



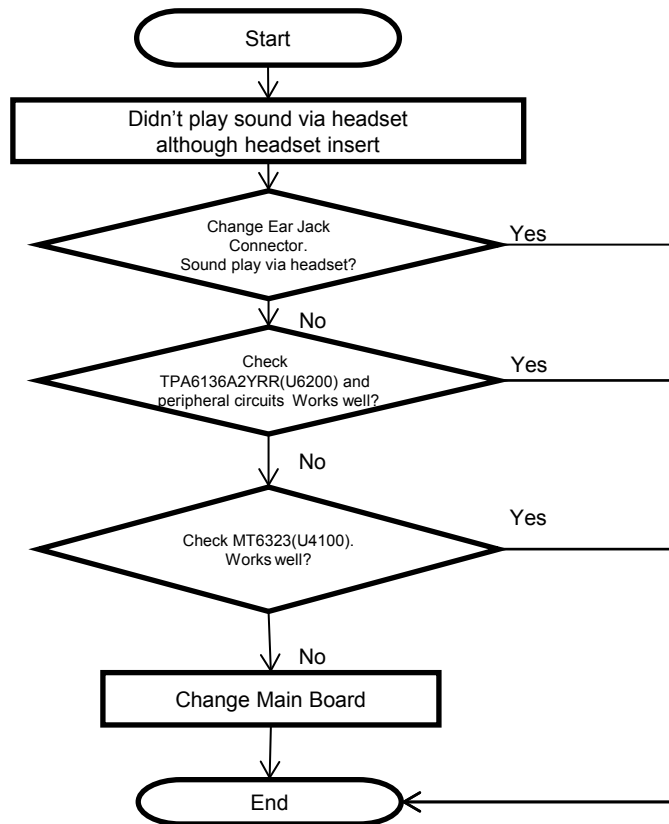
### Circuit Diagram



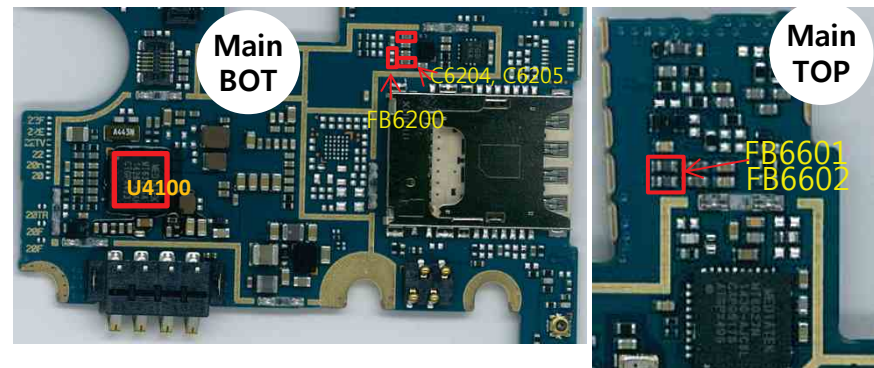
## 3.11.6 Ear-Mic- Case B

**Case B. No Sound from Headset. Check the input power VBAT (FB6200)**  
**In case of you can't heard anything, check C6204, C6205, FB6601, FB6602.**

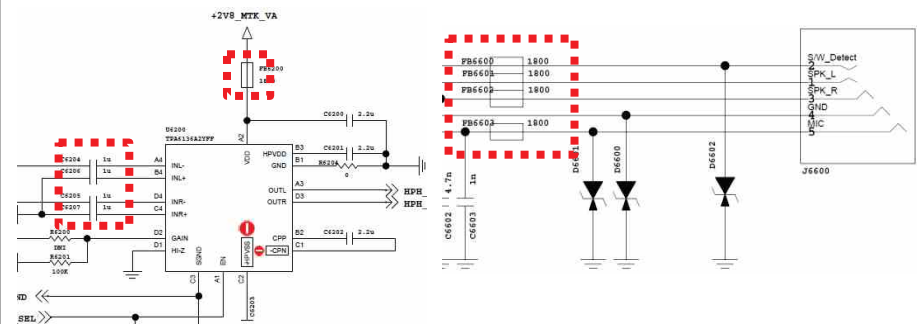
### Checking Flow



### Image



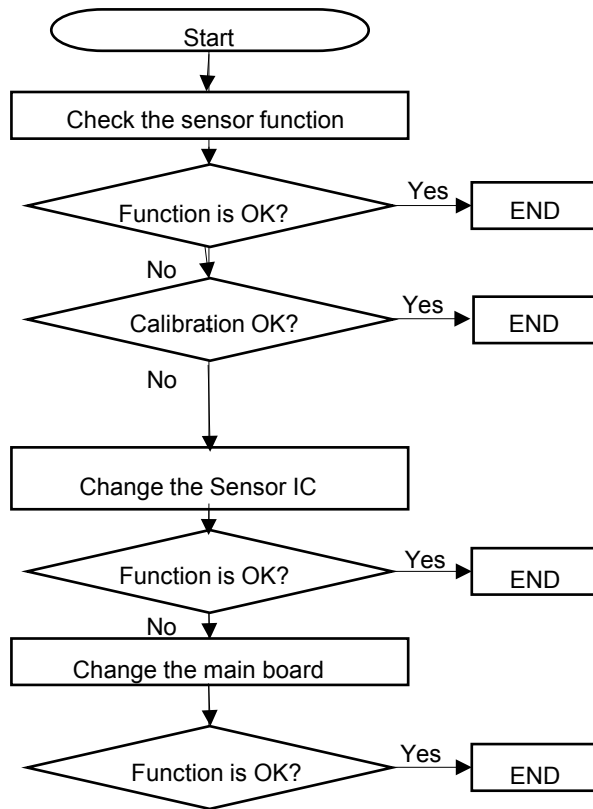
### Circuit Diagram



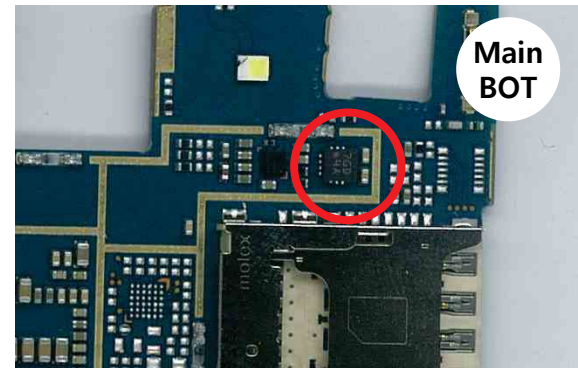
## 3.12.1 Accelerometer & Compass Sensor on/off

The Accl. & compass sensors are calibrated by using SW algorithm.

### Checking Flow



### Image

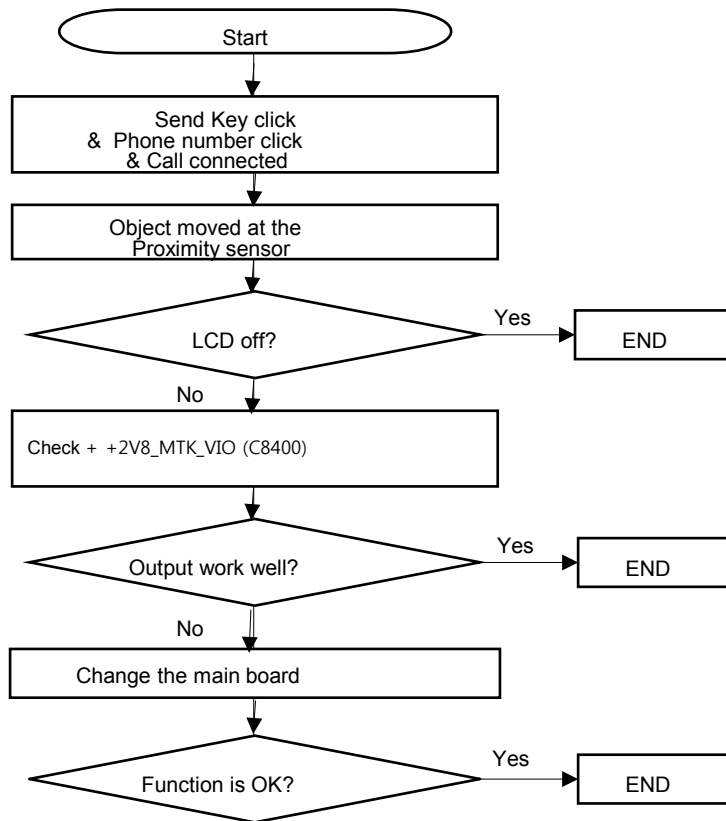


**Accelerometer  
&  
Compass sensor**

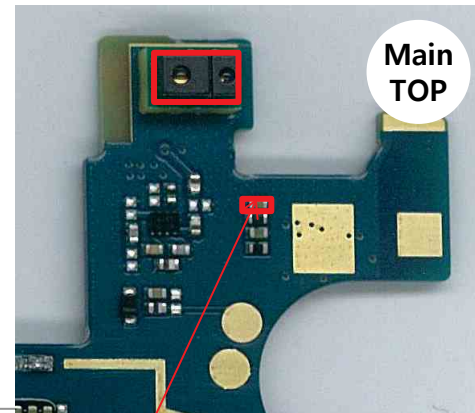
### 3.12.2 Proximity Sensor on/off

**Proximity Sensor is worked as below**  
**Control the screen's on/off operation automatically**

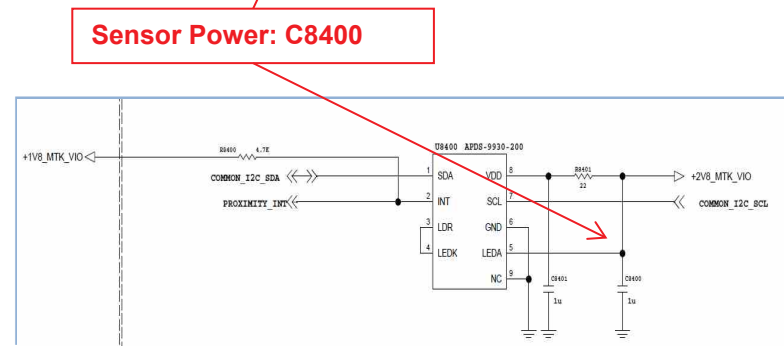
## Checking Flow



## Image



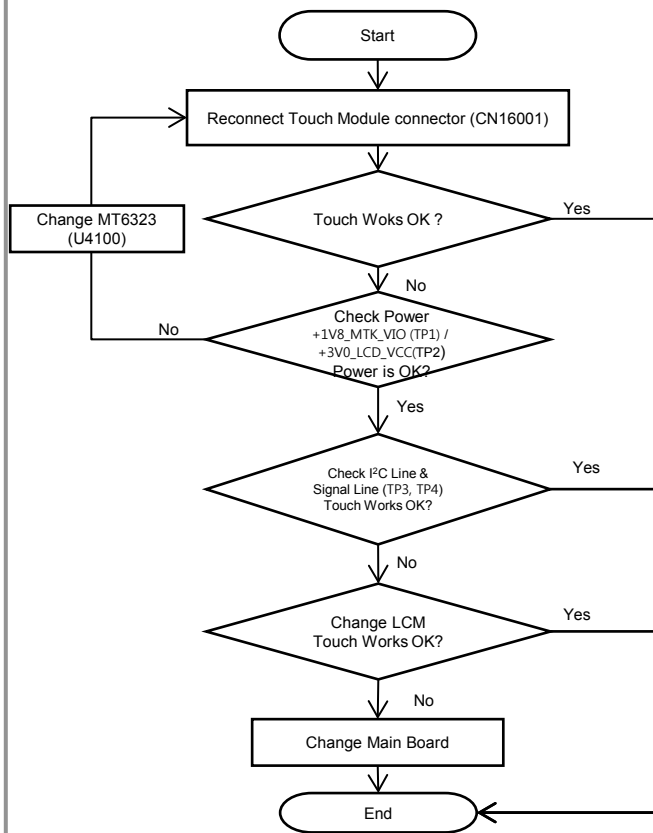
### Circuit Diagram



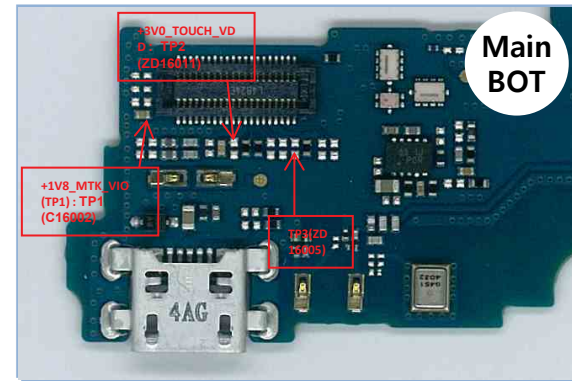
**LGE Internal Use Only**

## Touch Sensor

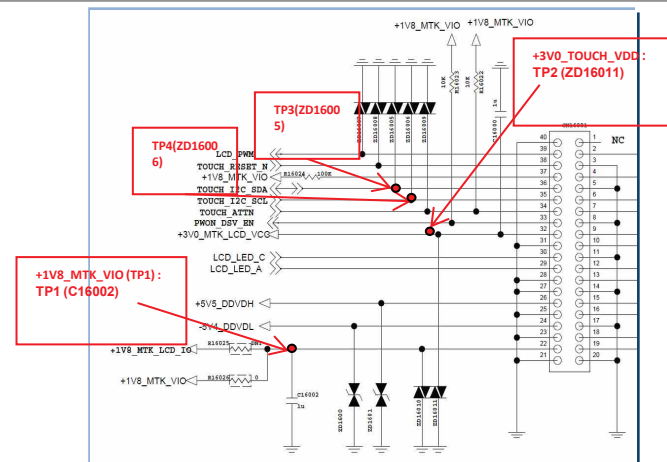
## Checking Flow



## Image



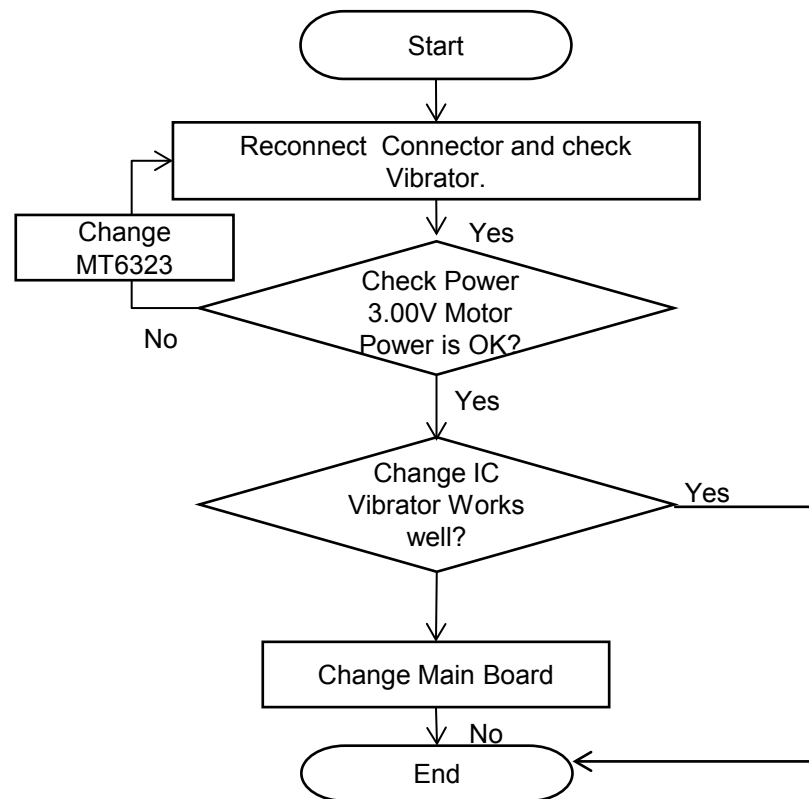
### Circuit Diagram



## 3.14 Vibrator

Motor Type is small cylinder.

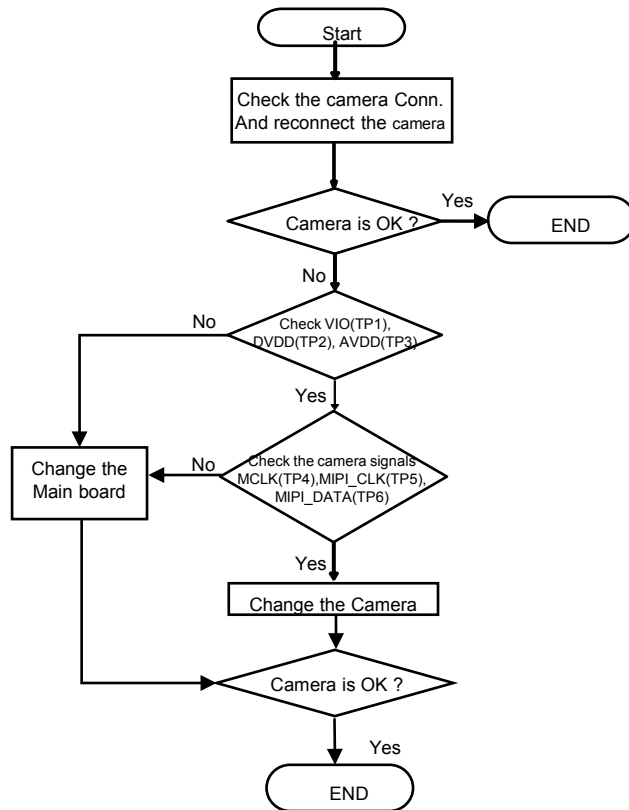
### Checking Flow



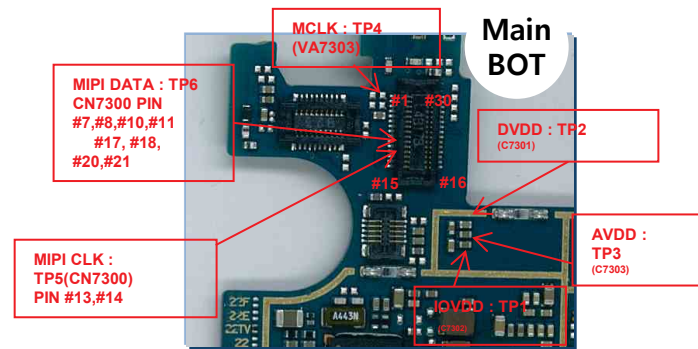
## 3.15.1 8M Main Camera

8M camera control signals are generated by MT6582 (U2100 : Main Chipset).  
And powered by voltage regulator (U4100)

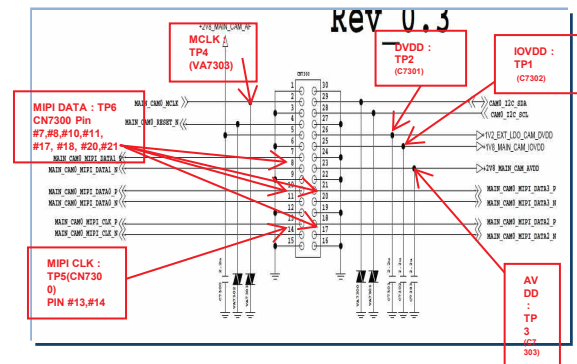
### Checking Flow



### Image



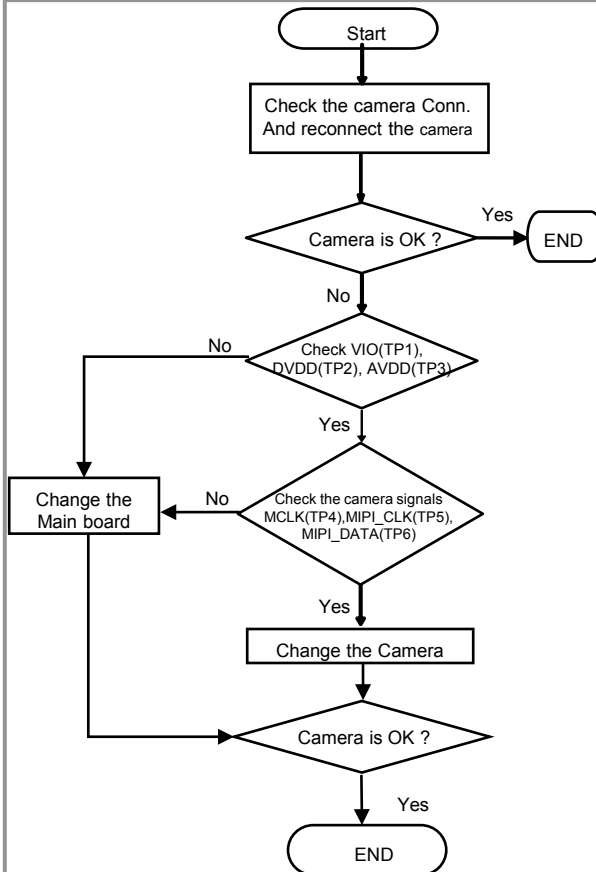
### Circuit Diagram



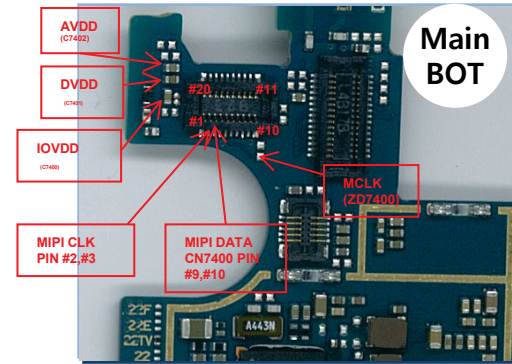
## 3.15.2 VT Camera

VGA camera control signals are generated by MT6582 (U2100 : Main Chipset).  
And powered by voltage regulator (U4100)

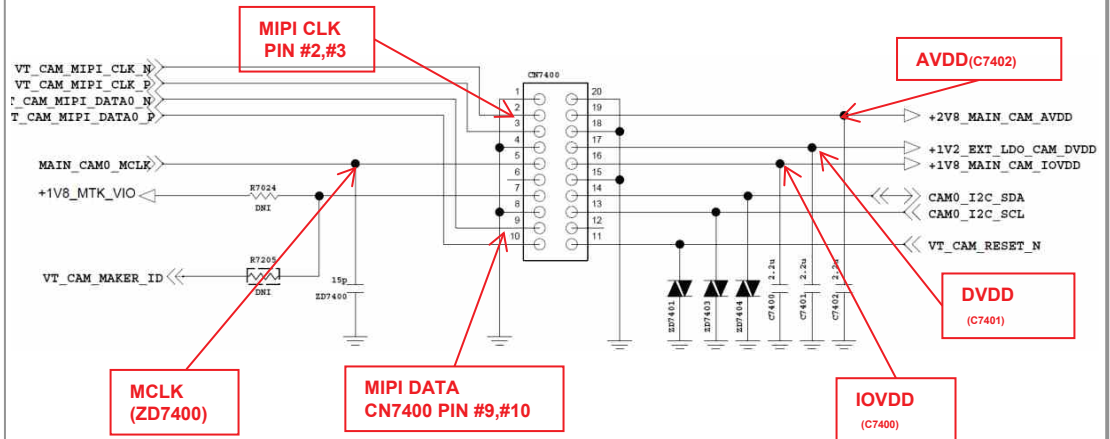
### Checking Flow



### Image



### Circuit Diagram

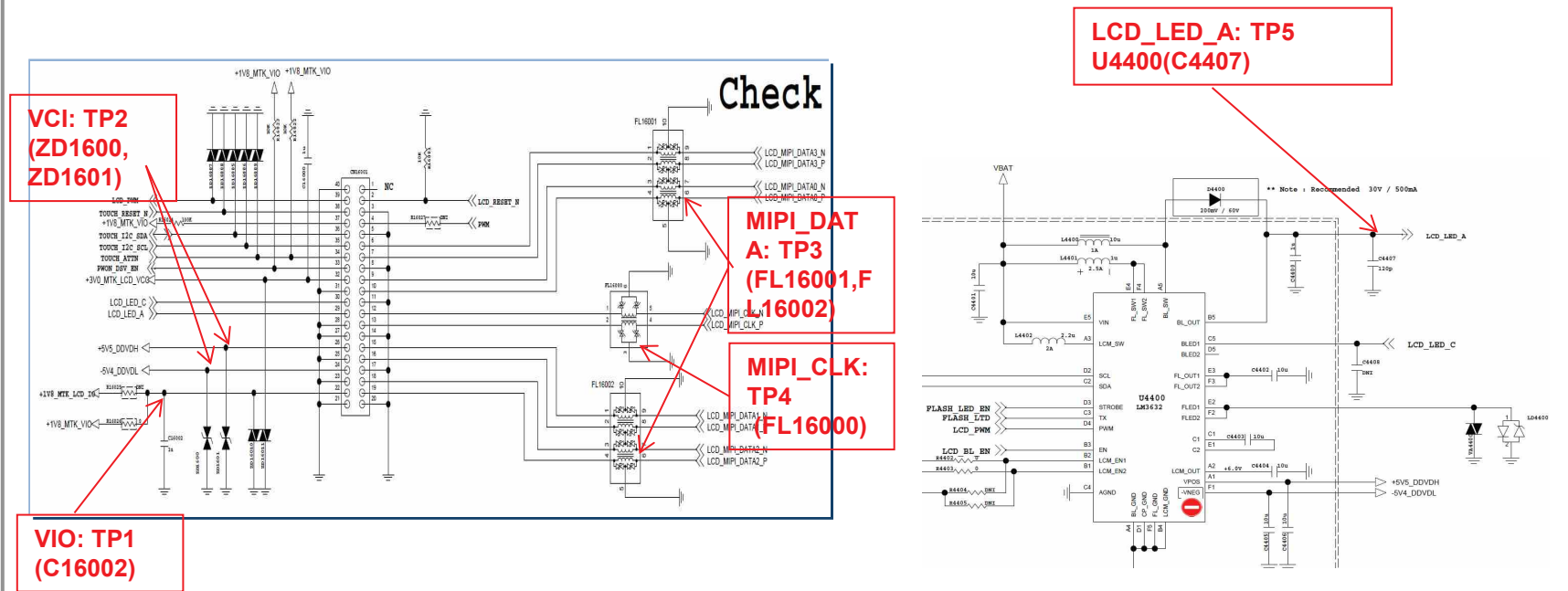


### 3. TROUBLE SHOOTING

### 3.16 Main LCD

The LCD control signals are generated by MT6582. It needs three external powers like VIO, VCI. And its interface is MIPI having three data lanes and one clock lane.

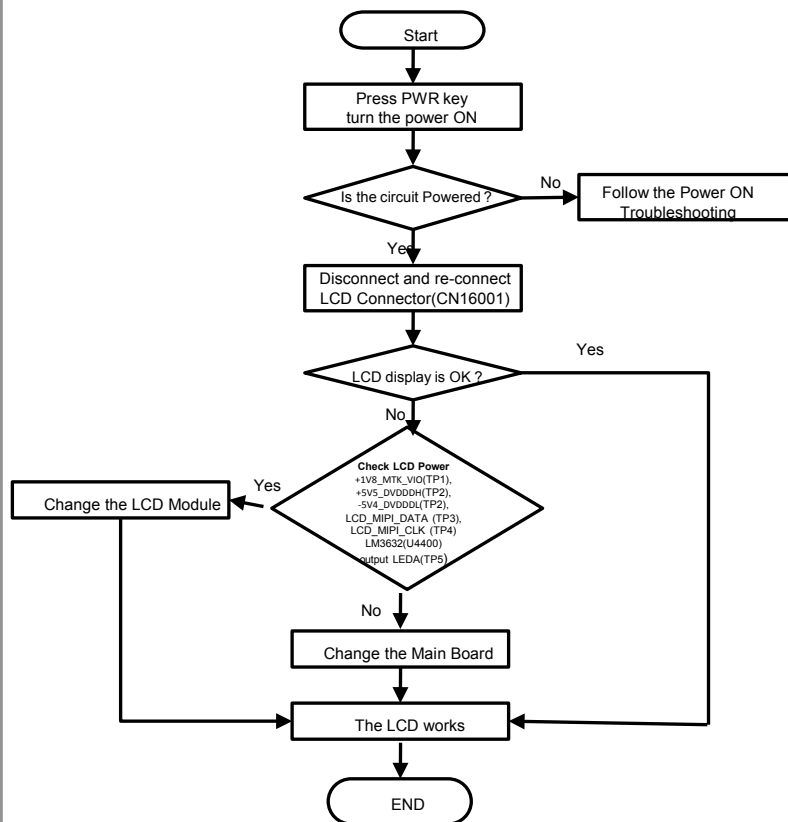
## Circuit Diagram



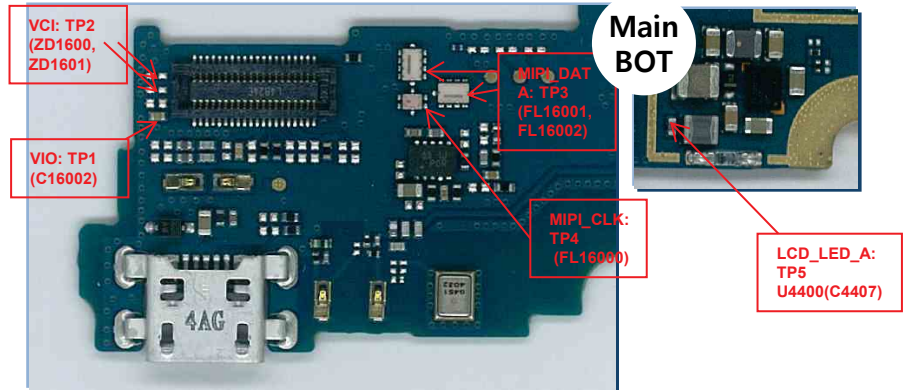
## 3.16 Main LCD

The LCD control signals are generated by MT6582. It needs three external powers like VIO, VCI. And its interface is MIPI having three data lanes and one clock lane.

### Checking Flow

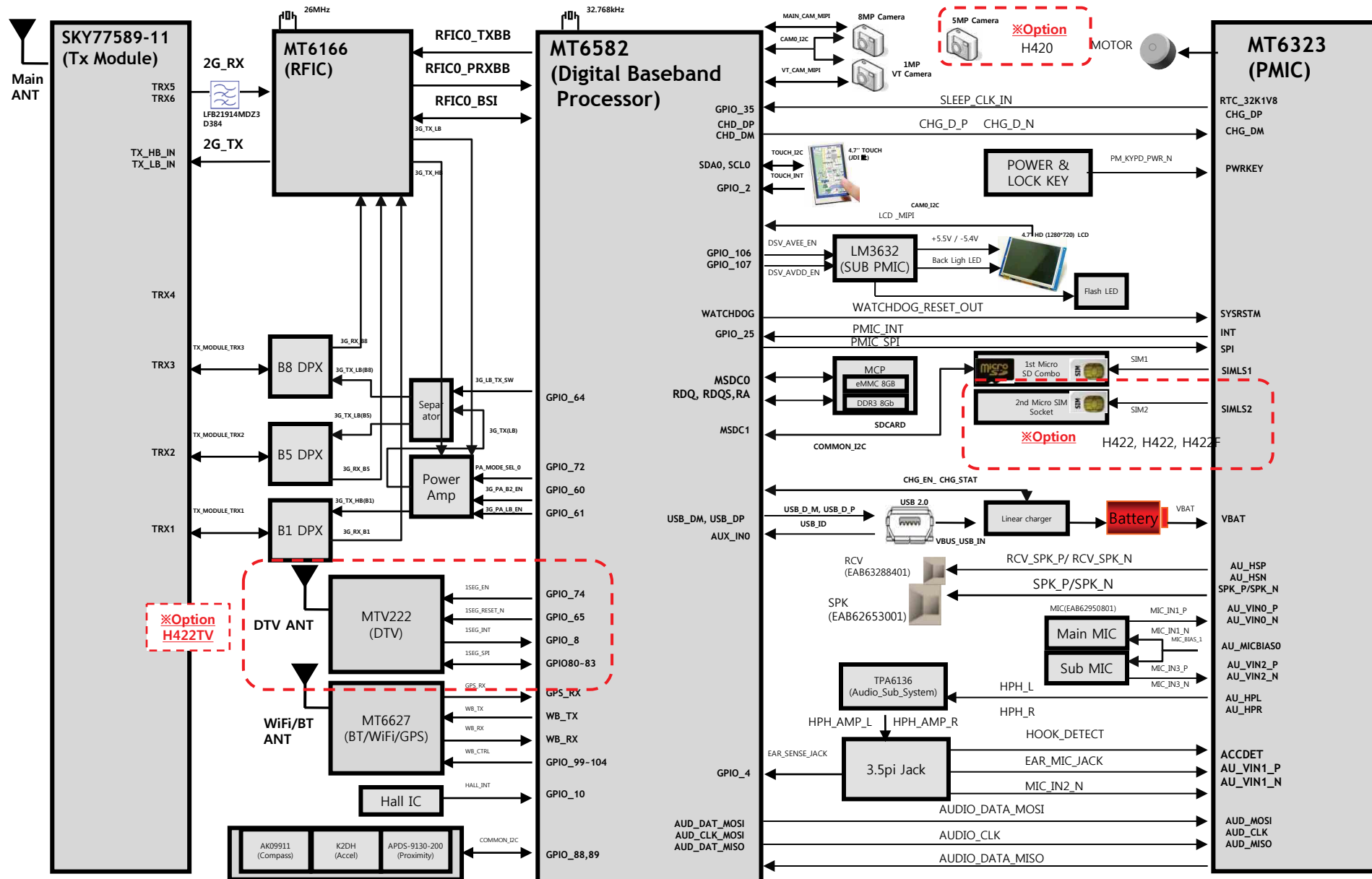


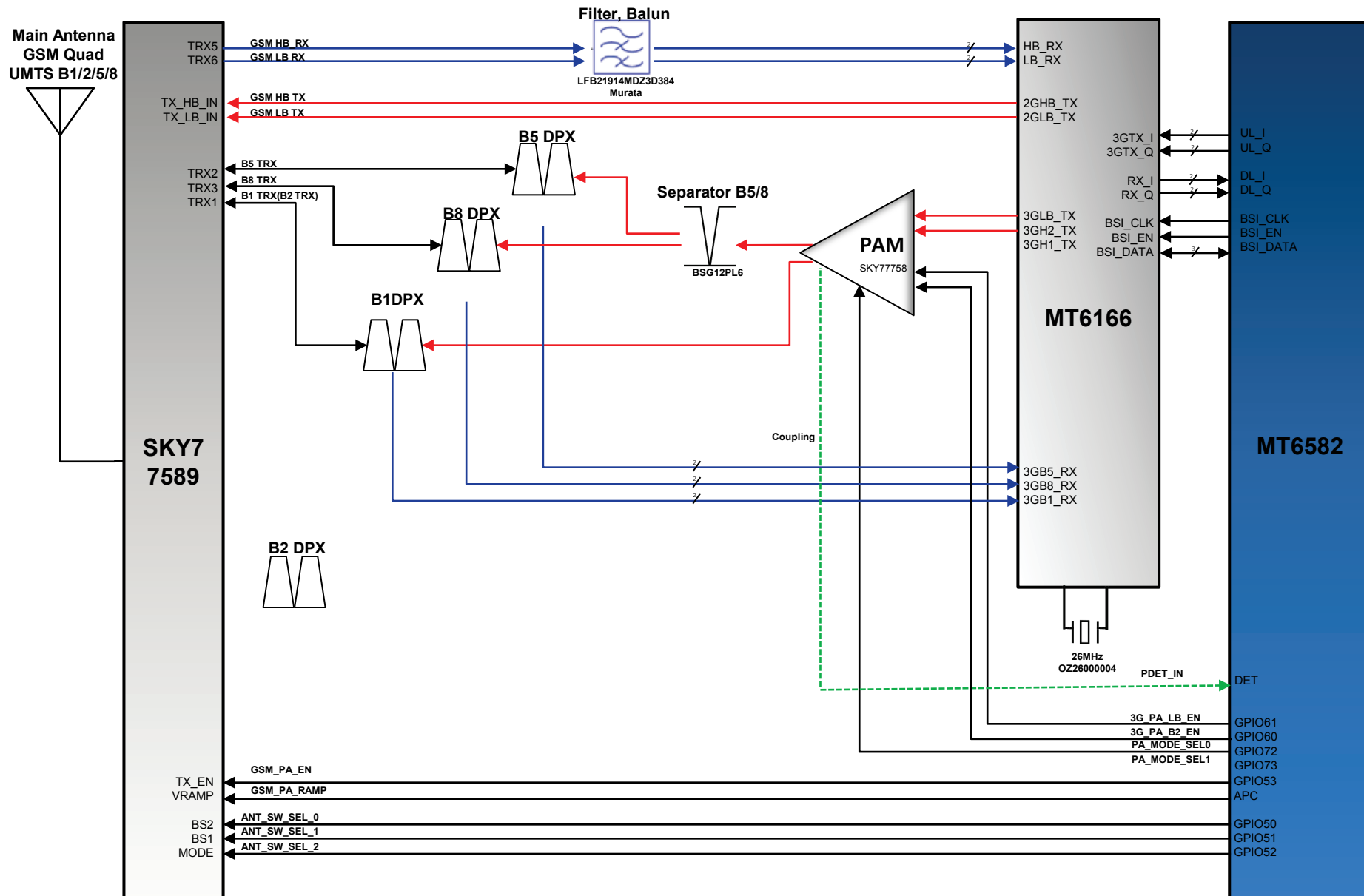
### Image

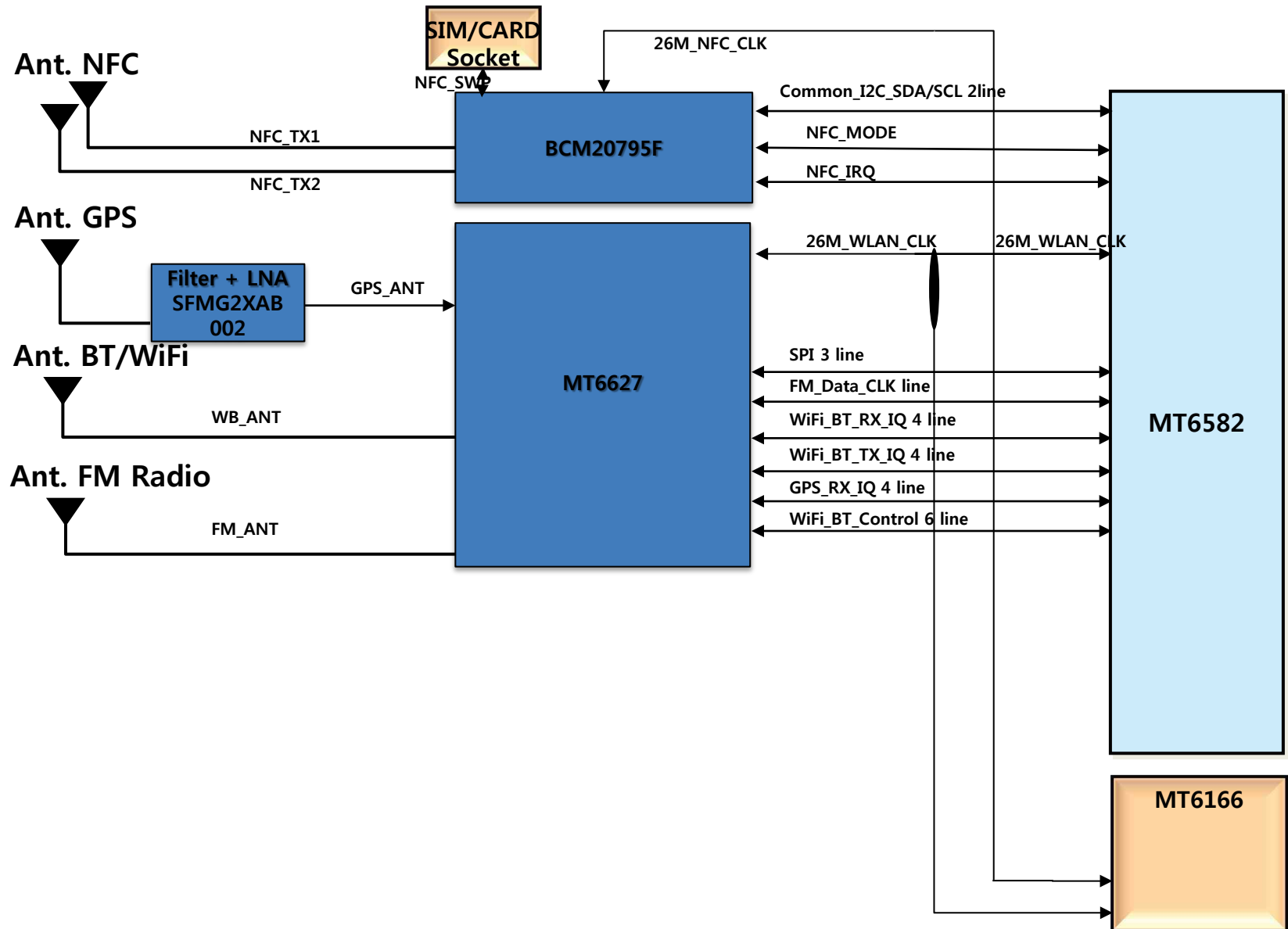


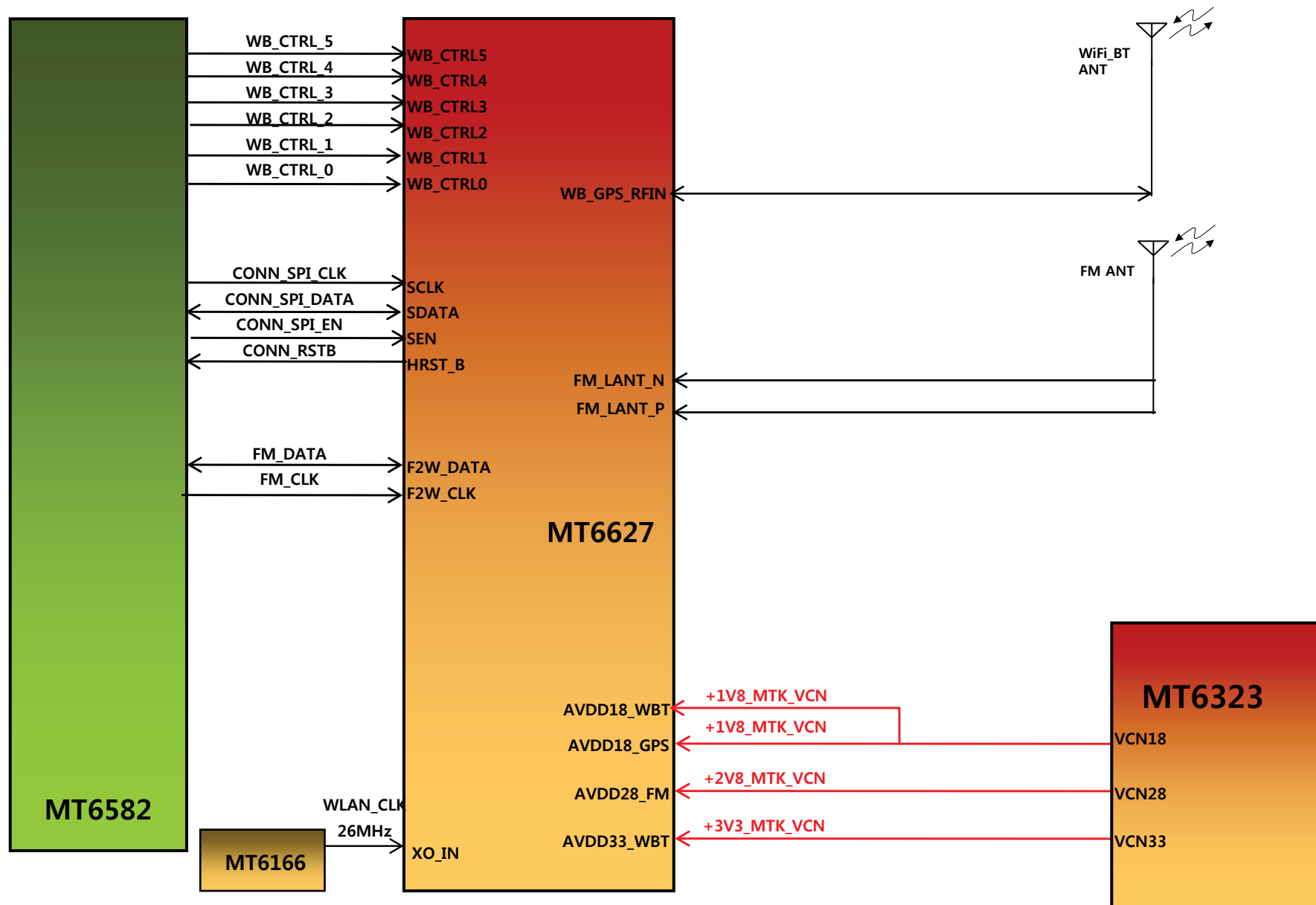
# 1. Total block diagram

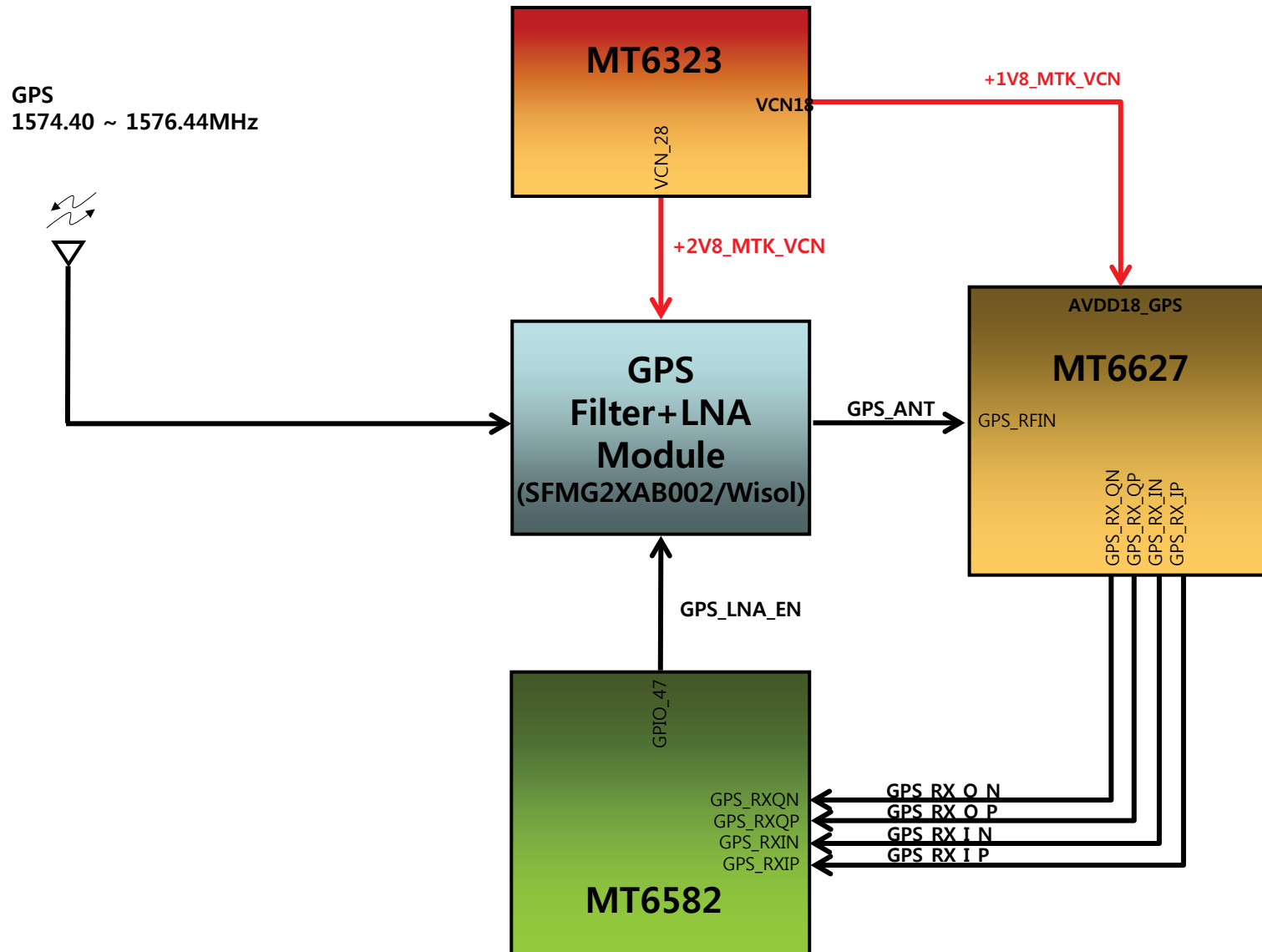
# 4. BLOCK DIAGRAM





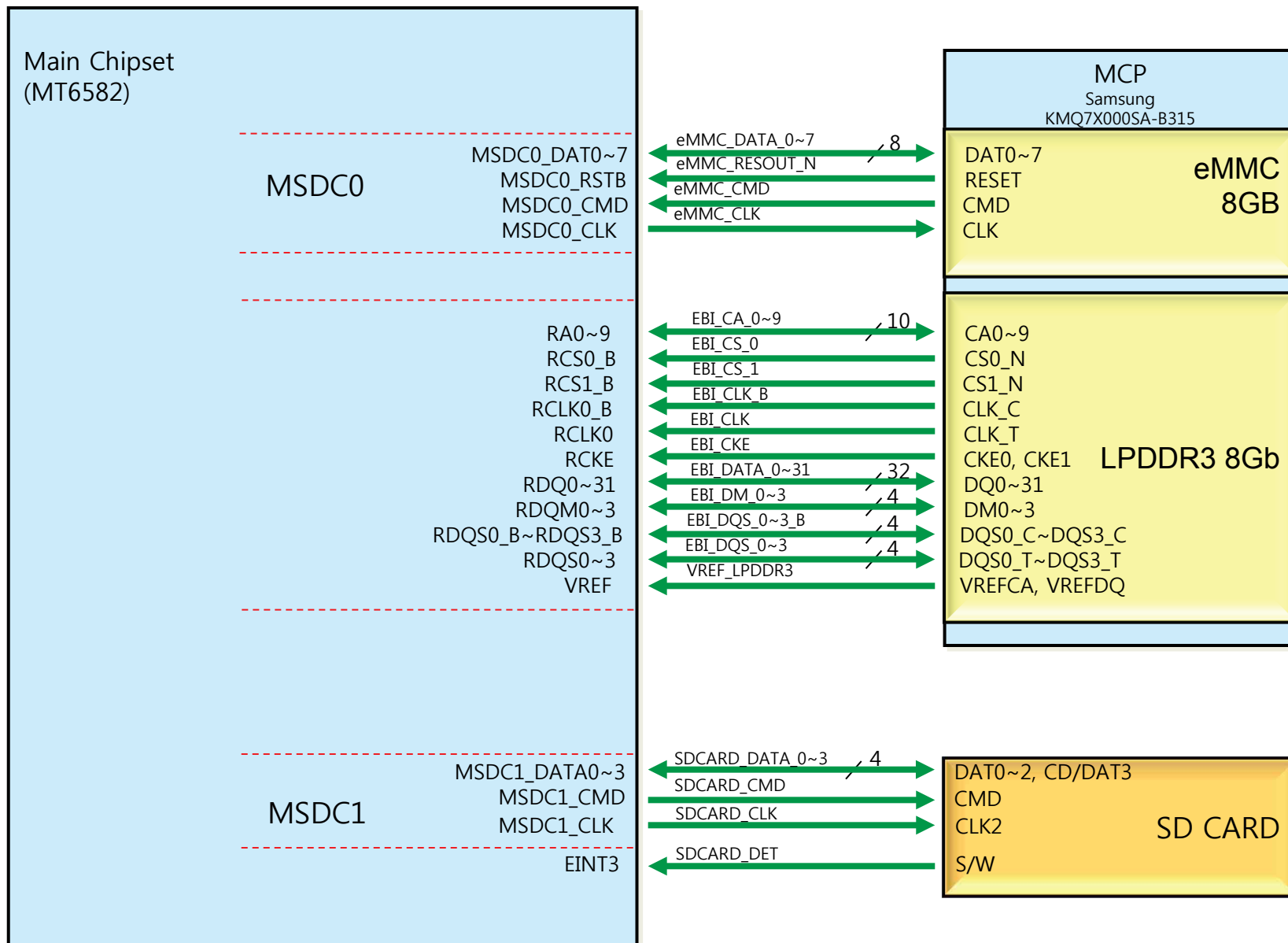


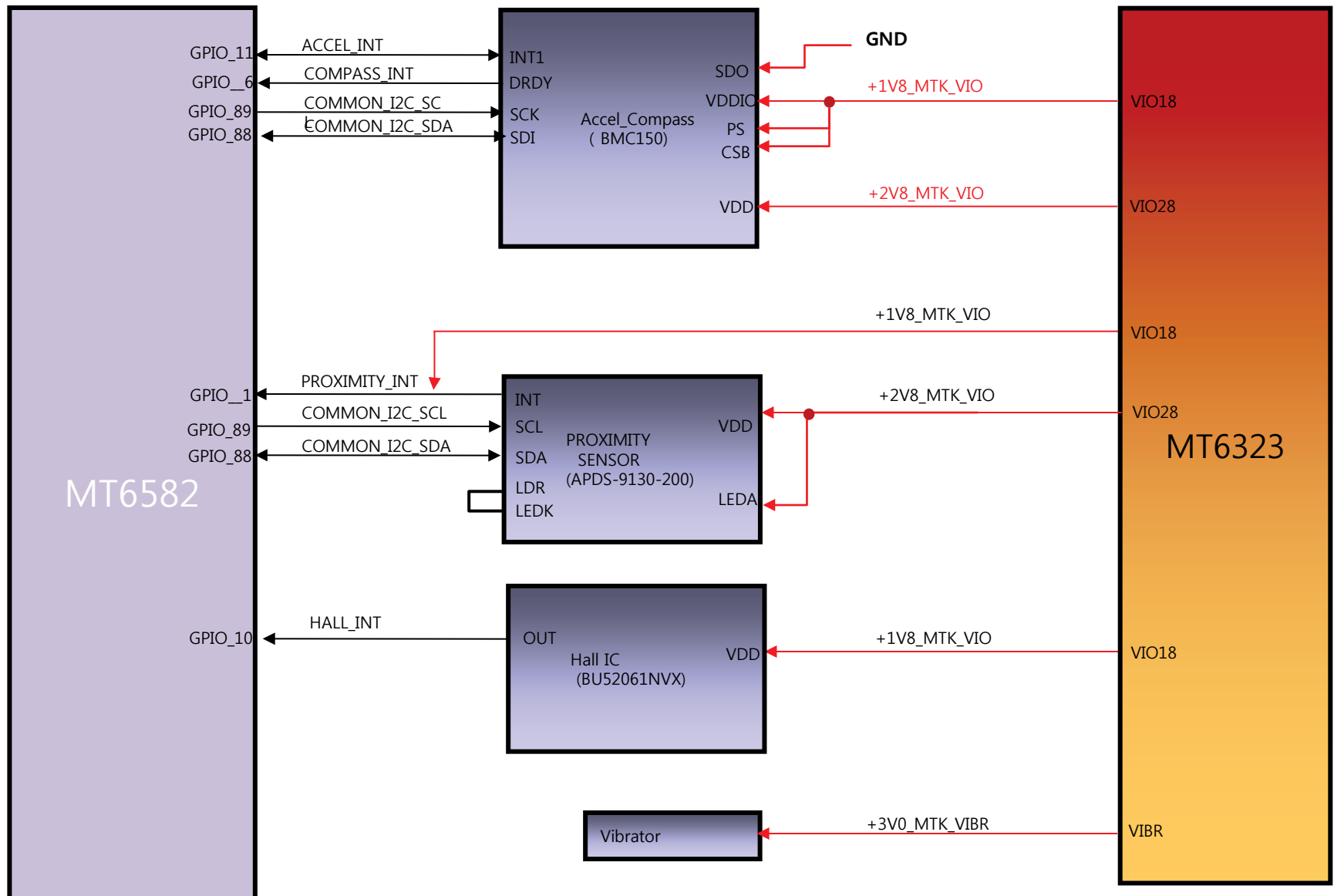


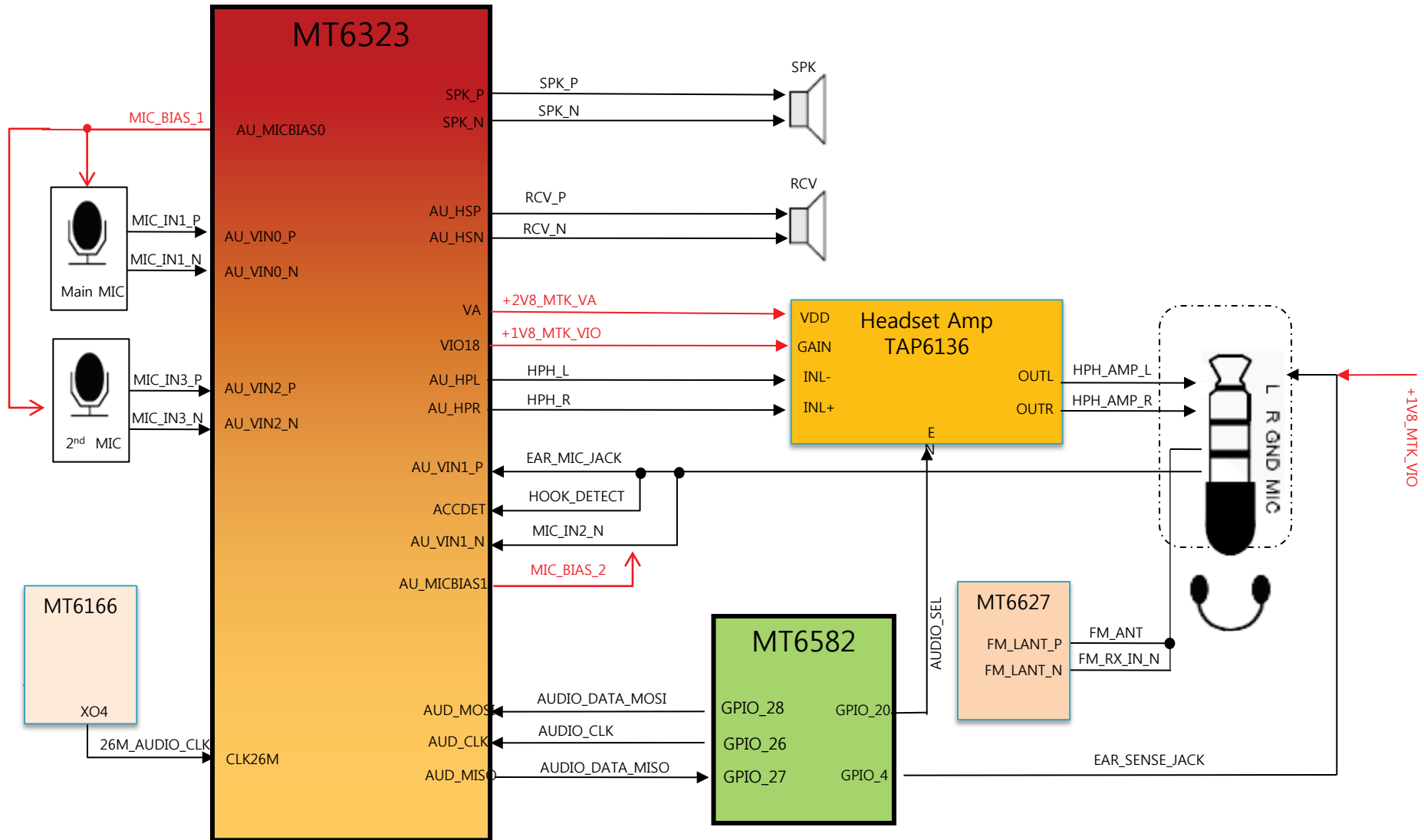


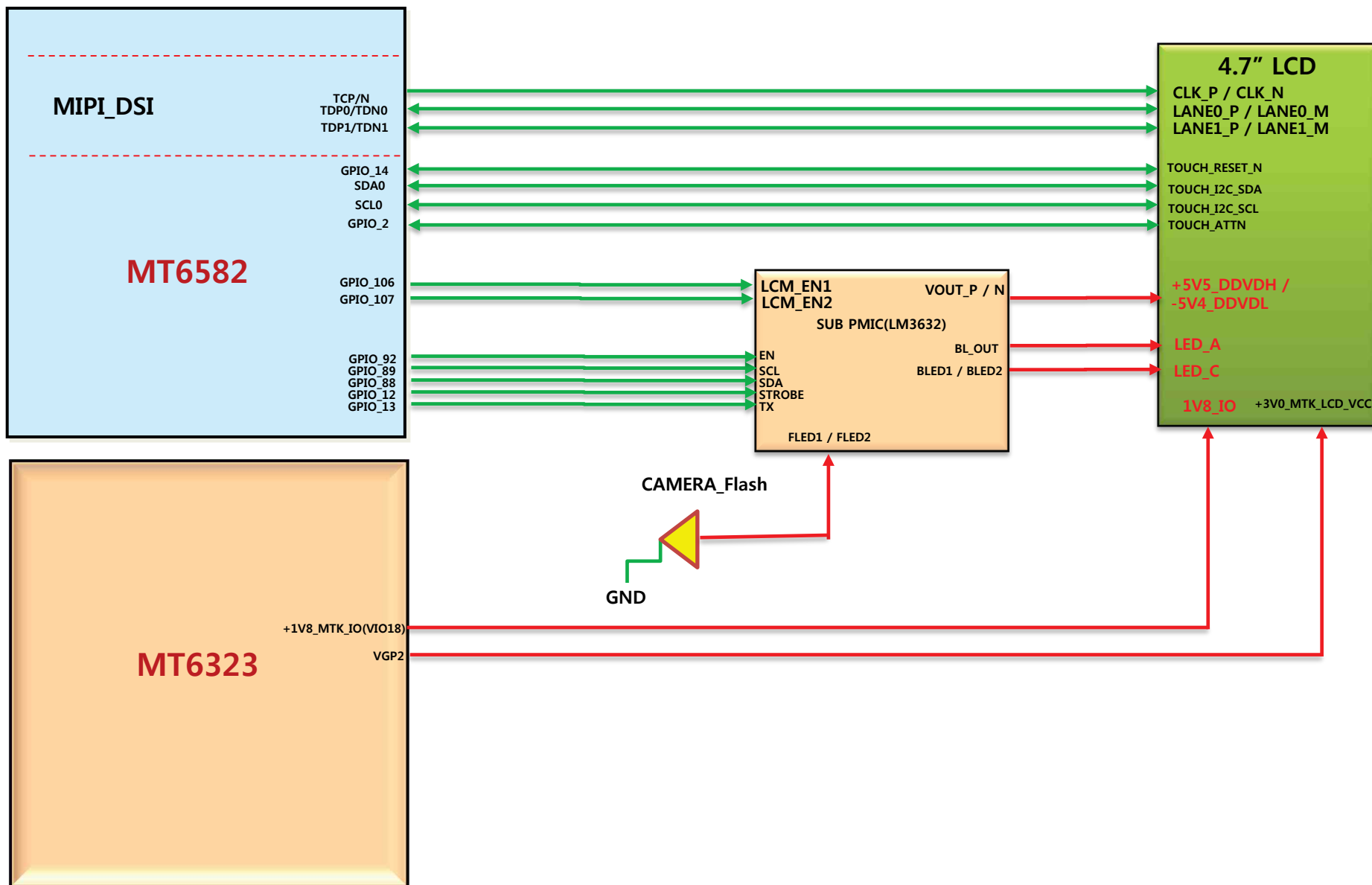
#### 4. BLOCK DIAGRAM

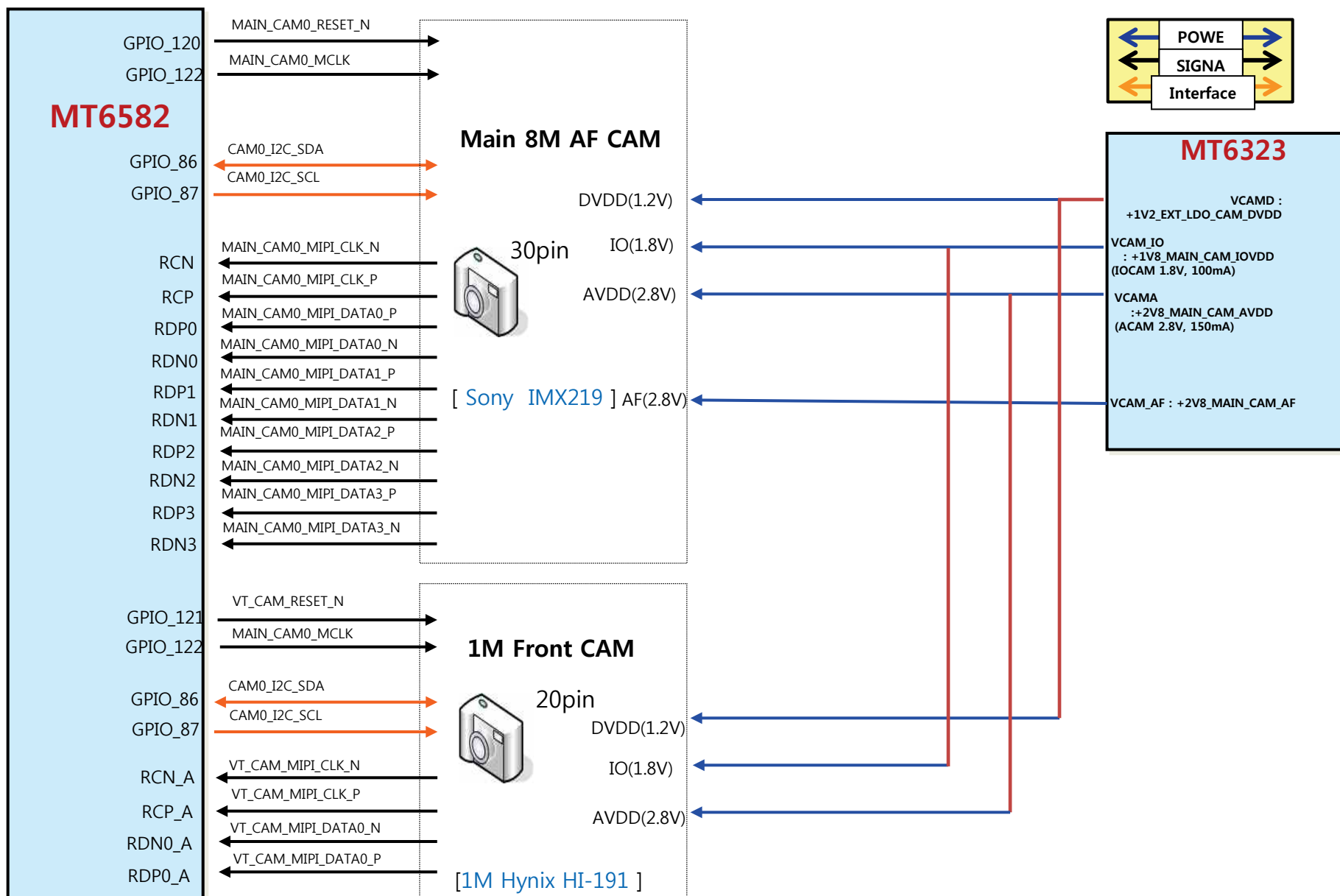










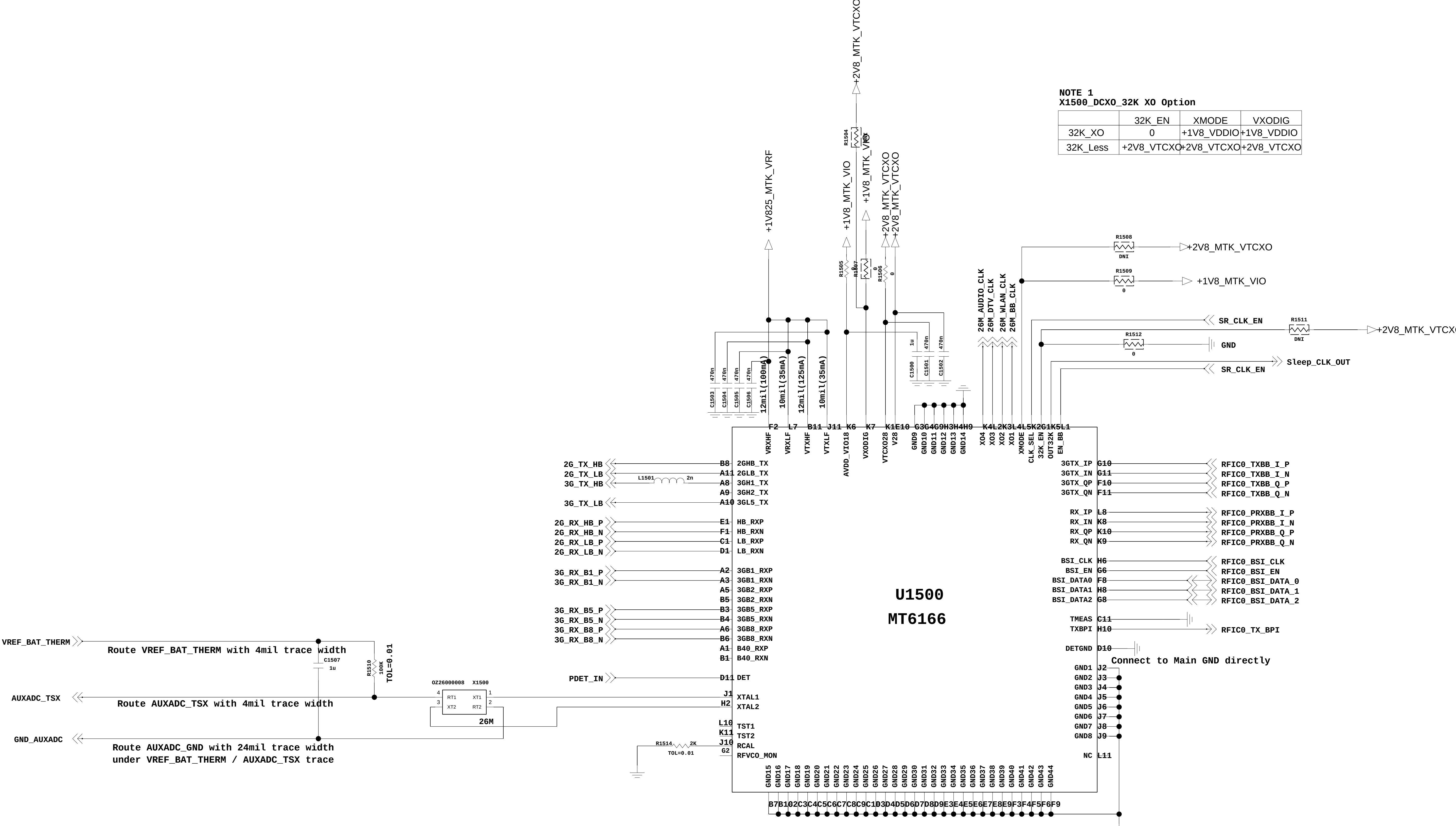


# CIRCULT DIAGRAM



**< 1-4-2-1\_RFIC\_MT6166**

**Rev\_0.3**



L

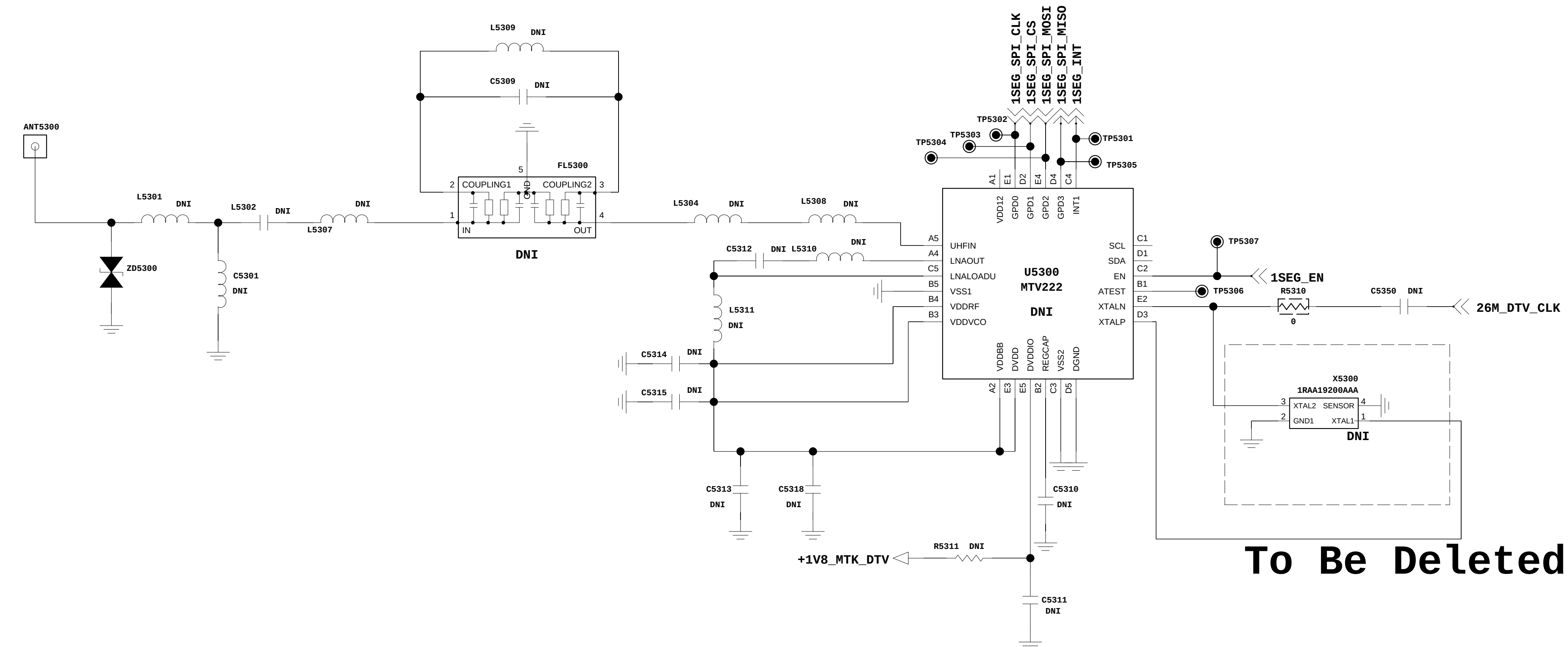
## H

**A**



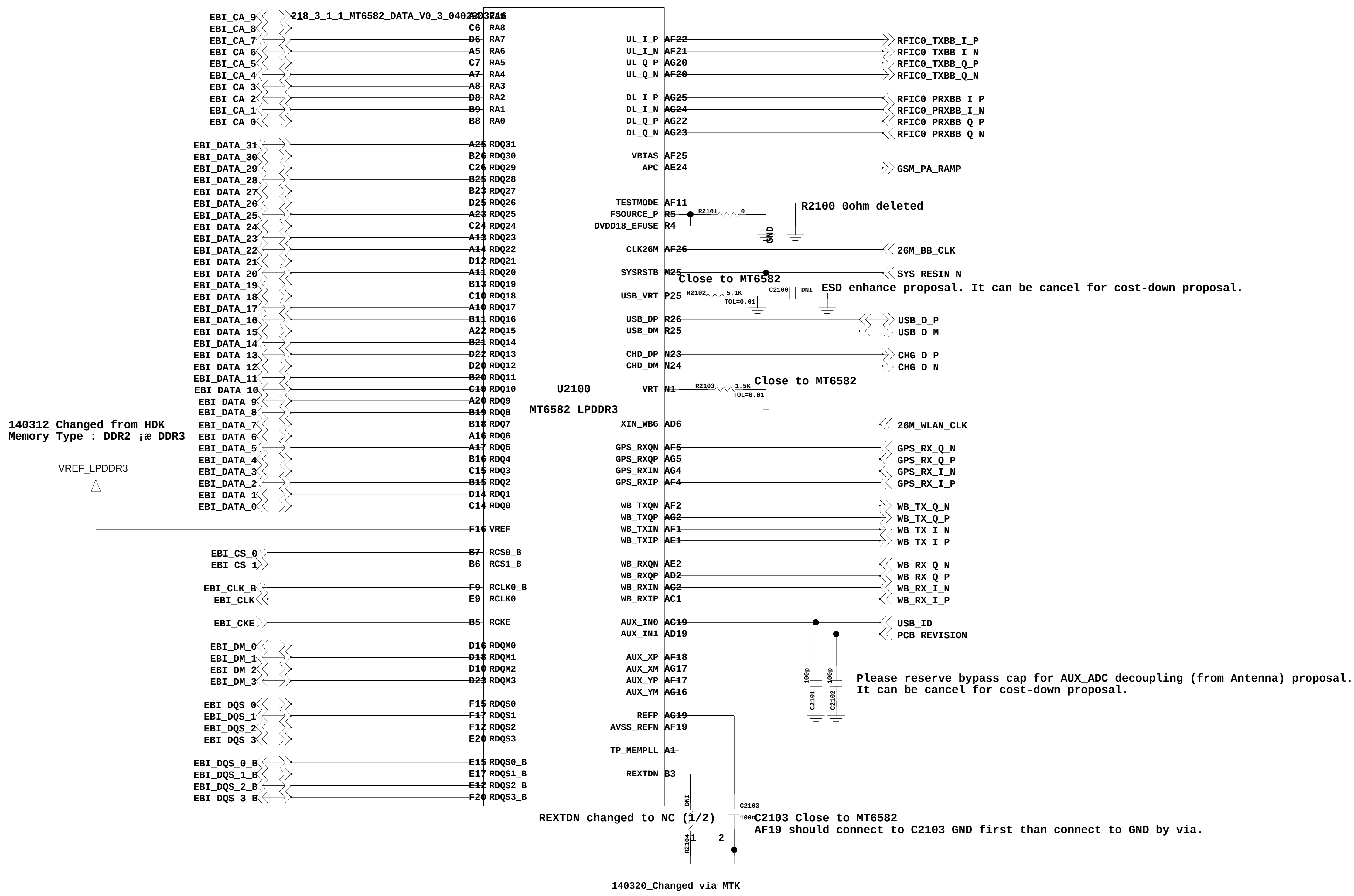


**DTV Only** 5-3-1-3\_Broadcasting\_ONE-SEG\_MTV222>



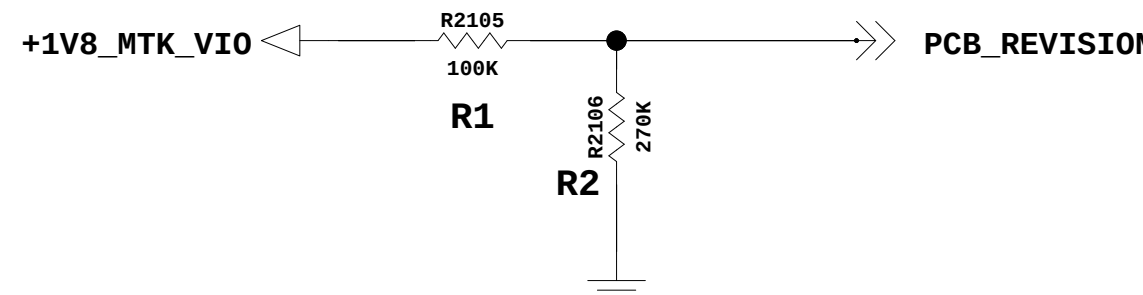
**LGE Internal Use Only**

# <2-3-3-1-1\_MT6582\_DATA> Rev.0.3



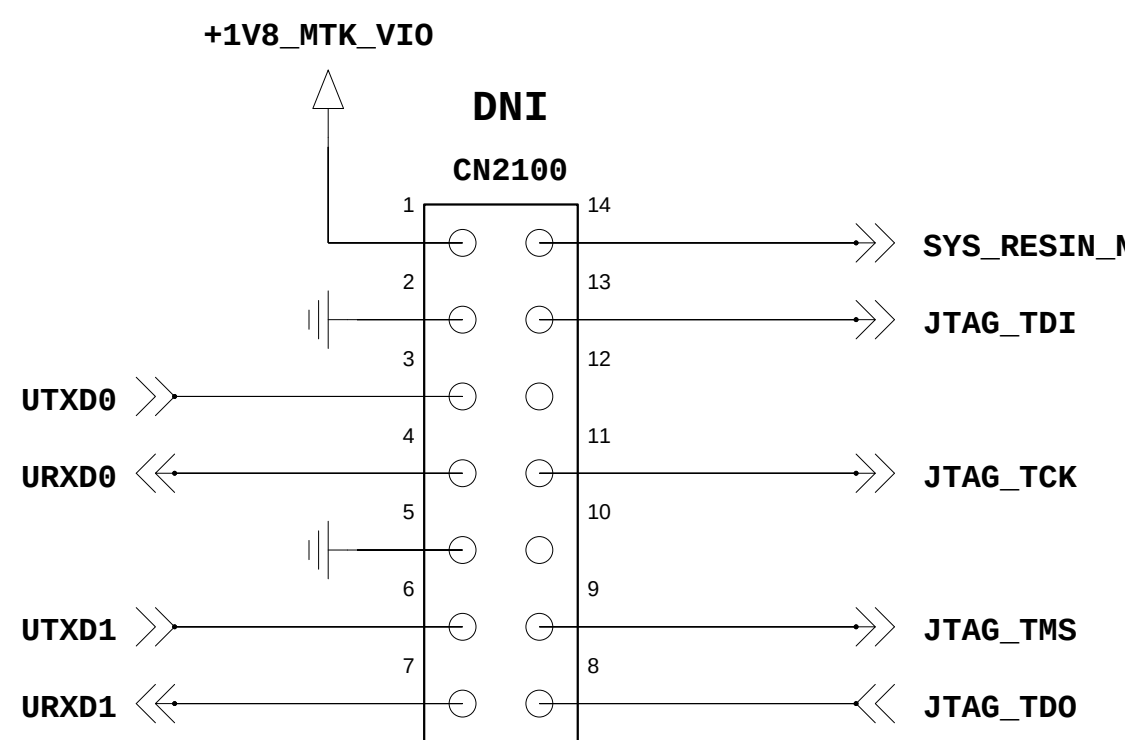
## Rev\_0.6

### ADC

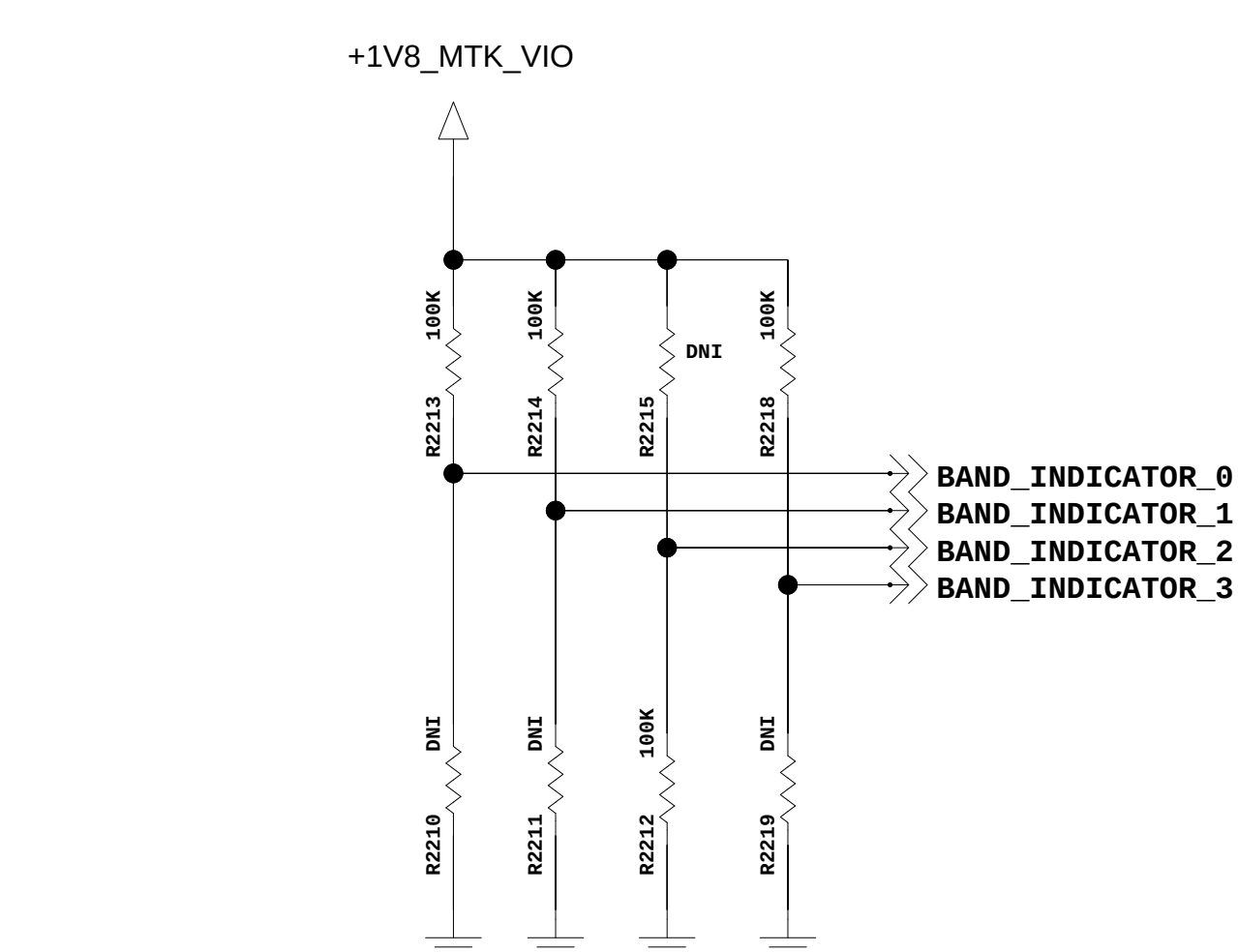


| Revision | R1   | R2   | ADC Range(HEX.) |
|----------|------|------|-----------------|
| 0.0      | 100K | 9.1K | 0.150           |
| A        | 100K | 30K  | 0.415           |
| B        | 100K | 51K  | 0.608           |
| C        | 100K | 82K  | 0.811           |
| 1.0      | 100K | 270K | 1.313           |
| 1.1      | 100K | 130K | 1.017           |
|          |      |      |                 |
|          |      |      |                 |

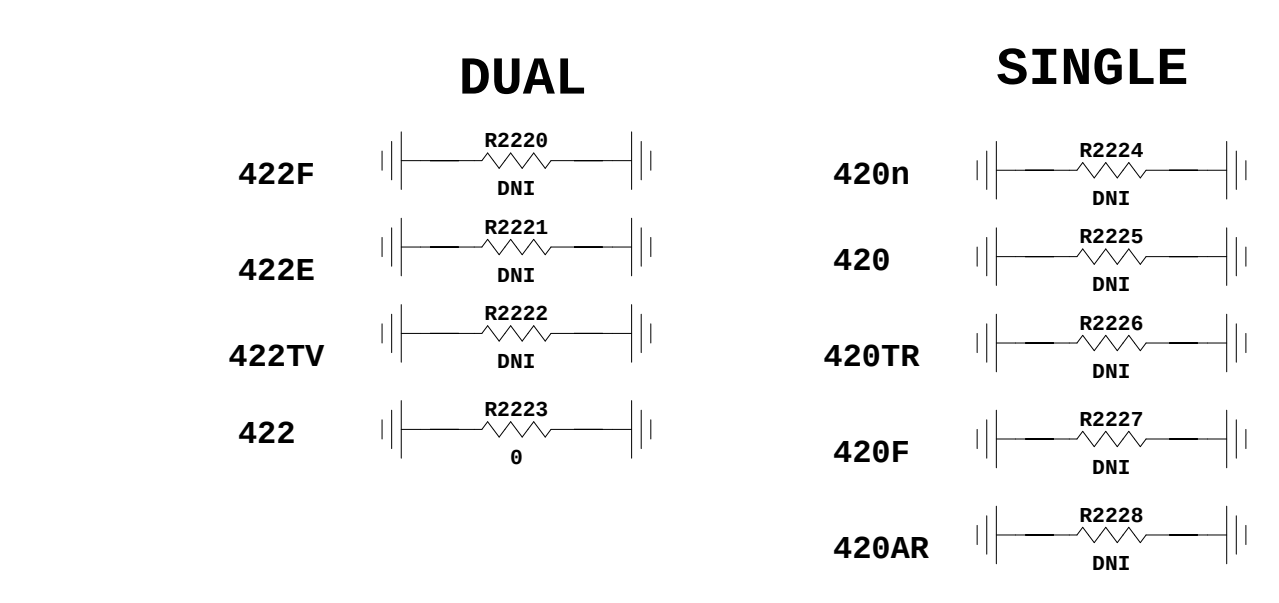
## < 9-3-1\_JTAG > Rev\_0.4



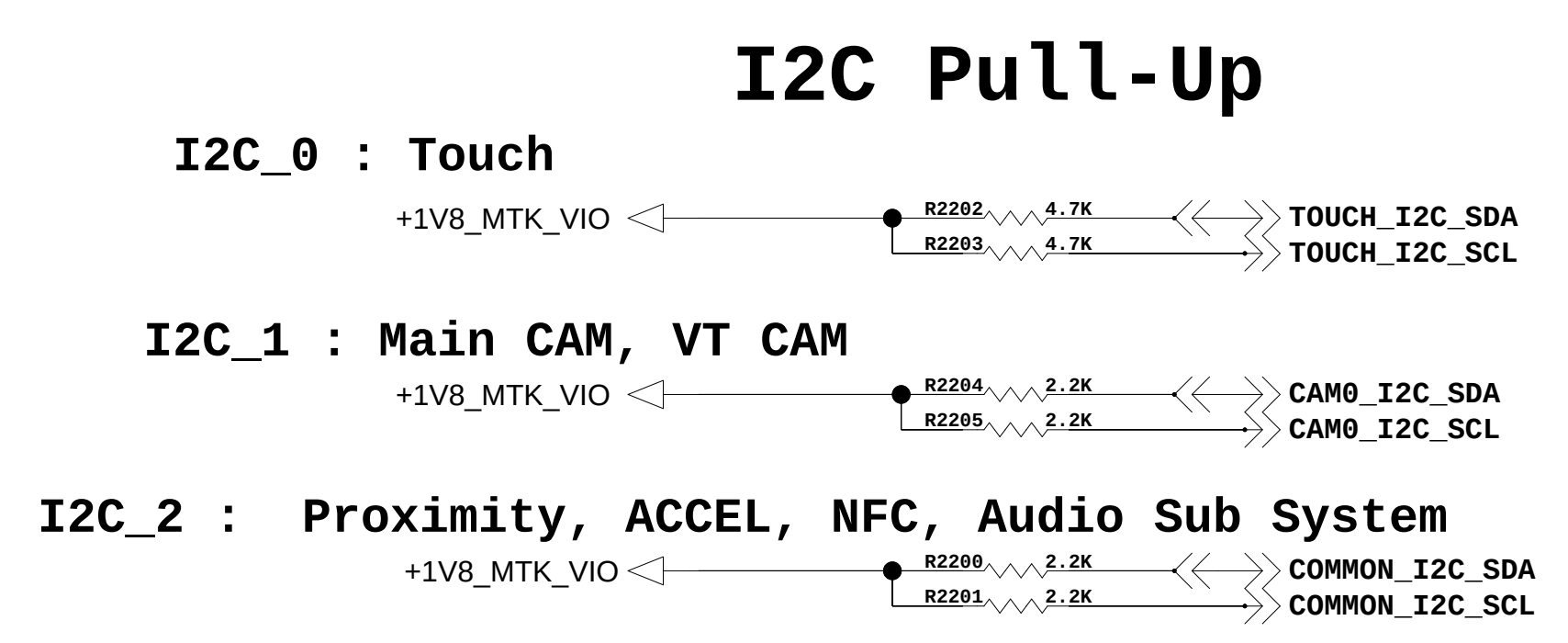
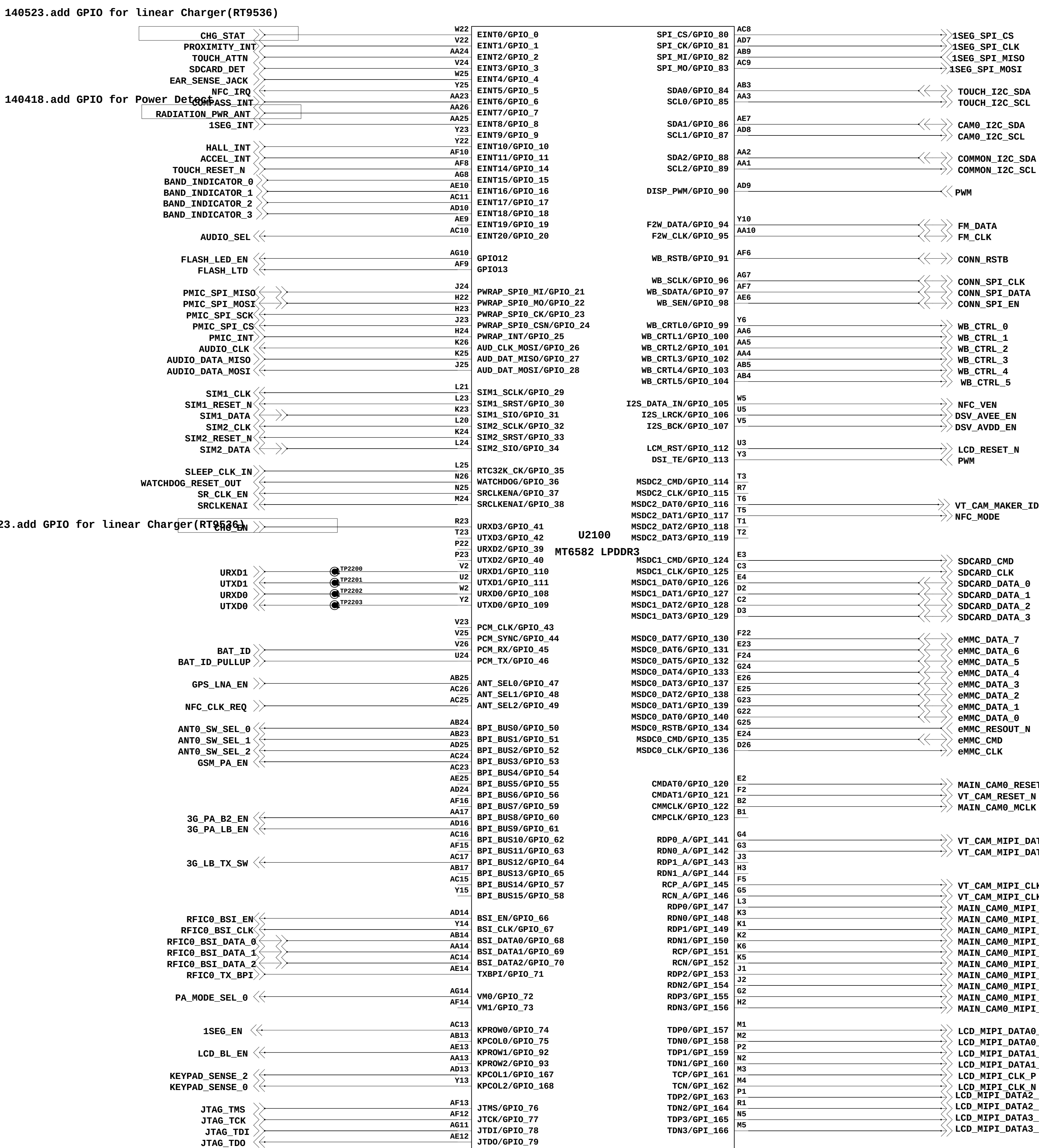
Band\_Indicator



|        |            | BI_0    | BI_1    | BI_2    | BI_3    |
|--------|------------|---------|---------|---------|---------|
|        |            | GPIO 15 | GPIO 16 | GPIO 17 | GPIO 18 |
| H420n  | SINGLE SIM | 0       | 0       | 0       | 0       |
| H420   | SINGLE SIM | 0       | 0       | 0       | 0       |
| H420TR | SINGLE SIM | 1       | 1       | 0       | 0       |
| H420F  | SINGLE SIM | 0       | 1       | 1       | 0       |
| H420AR | SINGLE SIM | 1       | 1       | 1       | 0       |
|        |            |         |         |         |         |
| H422F  | DUAL SIM   | 0       | 1       | 1       | 0       |
| H422TV | DUAL SIM   | 1       | 1       | 1       | 1       |
| H422   | DUAL SIM   | 1       | 1       | 0       | 1       |



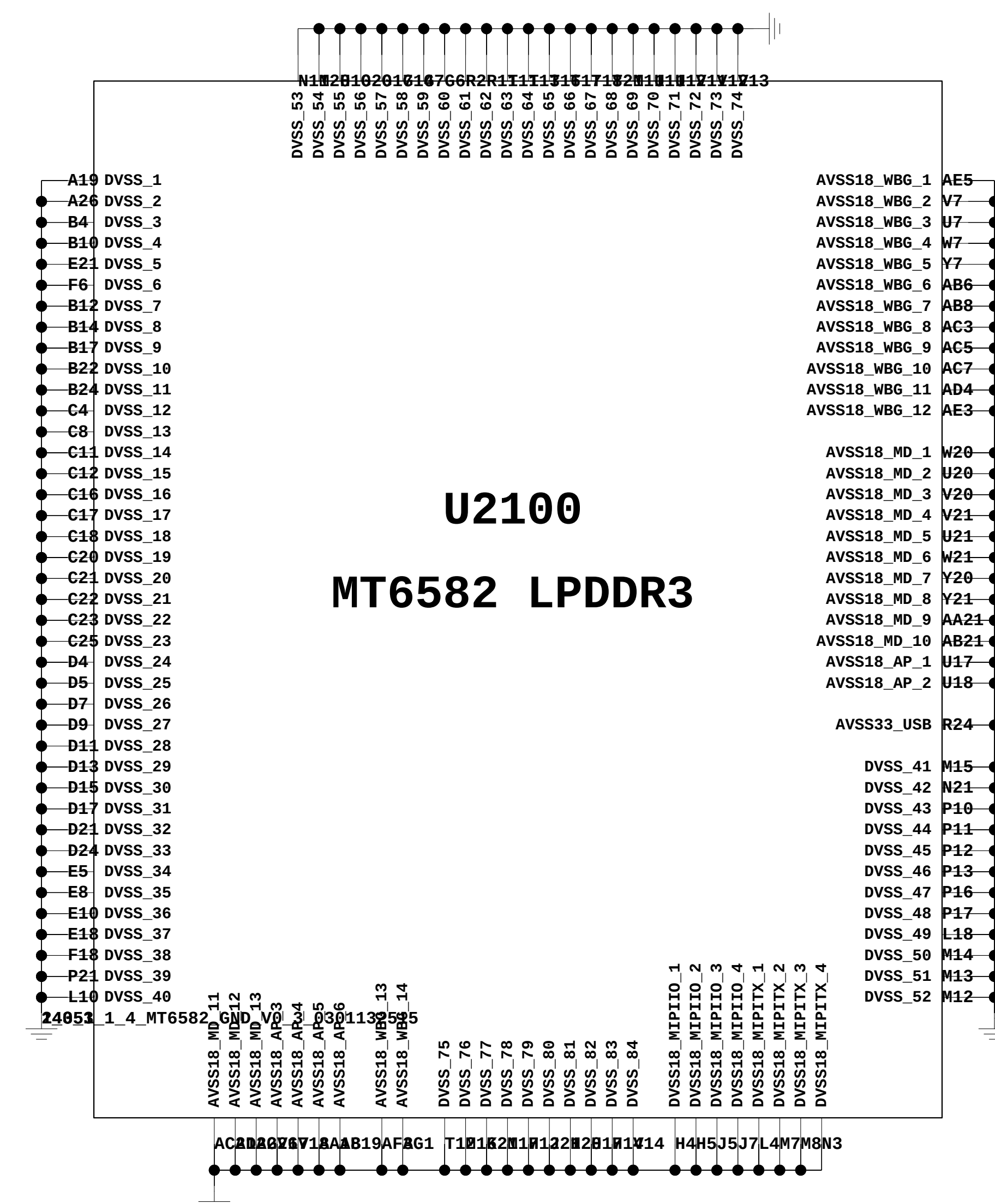
LGE Internal Use Only



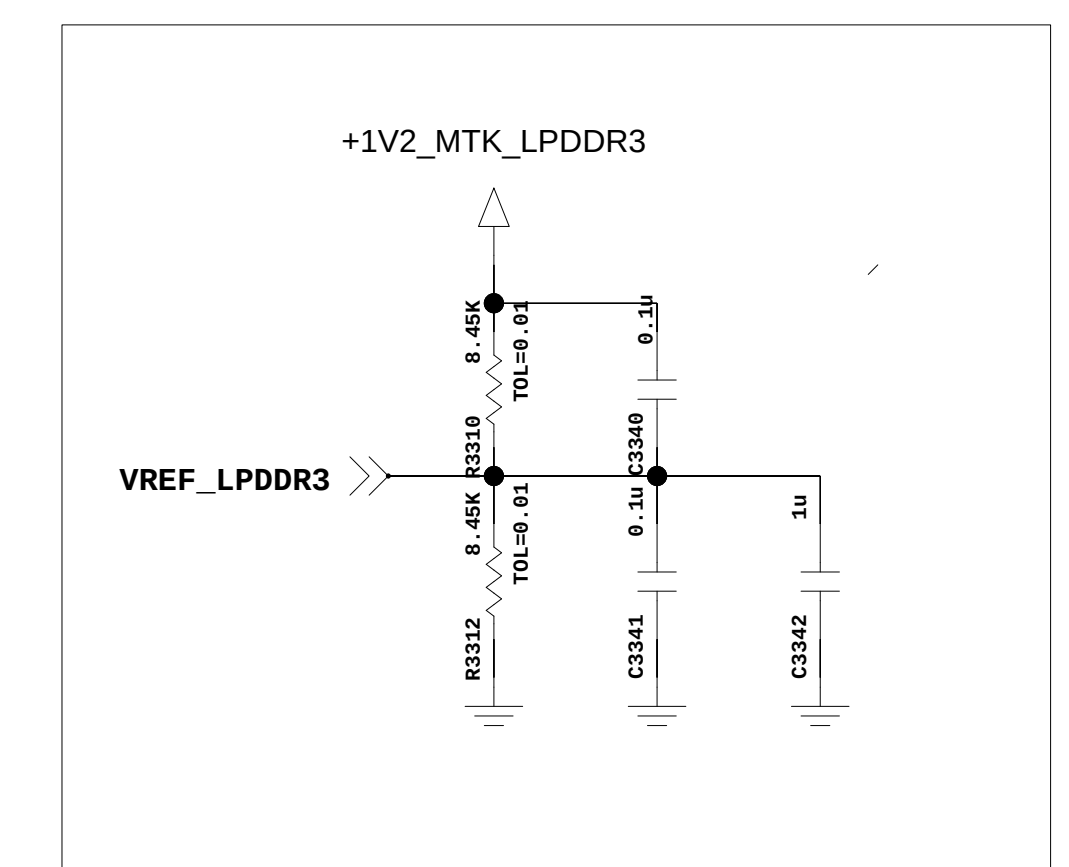
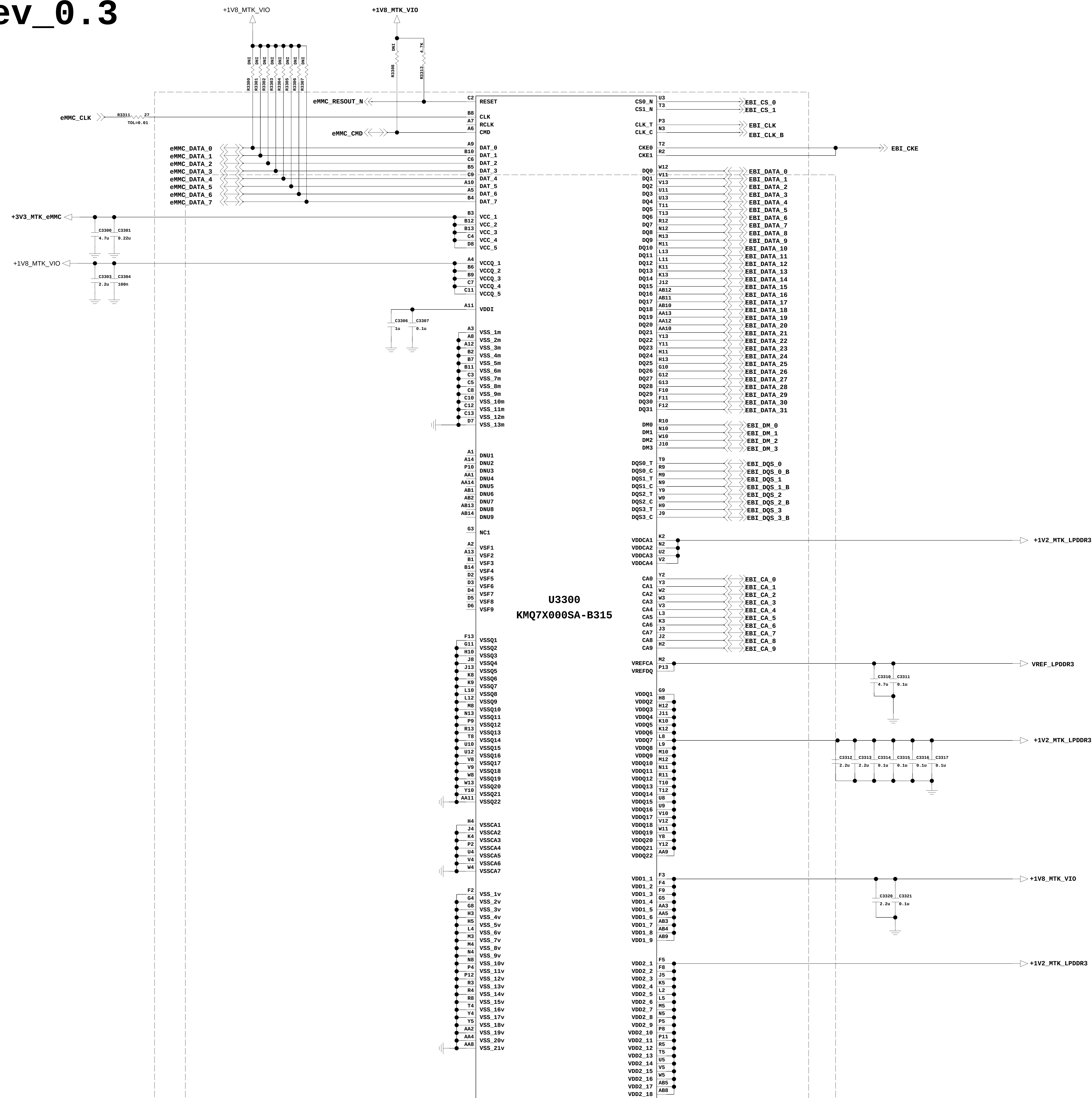


**<2-3-3-1-4\_MT6582\_GND>**

**Rev . 0 . 3**



**< 3-3-3-2-1\_MCP\_eMMC\_5\_0\_8G\_DDR3\_8G\_Samsung >**  
**Rev\_0.3**

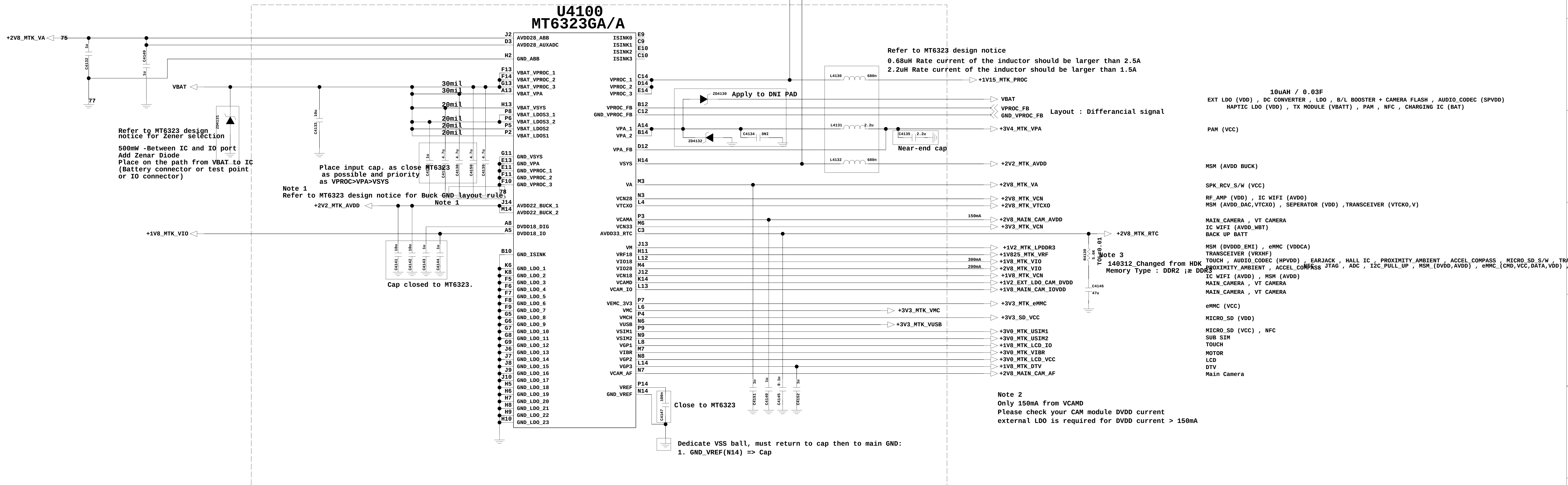


# A



87

# <4-1-7-2\_MT6323\_POWER> Rev\_0.4



Buck converter brief specifications

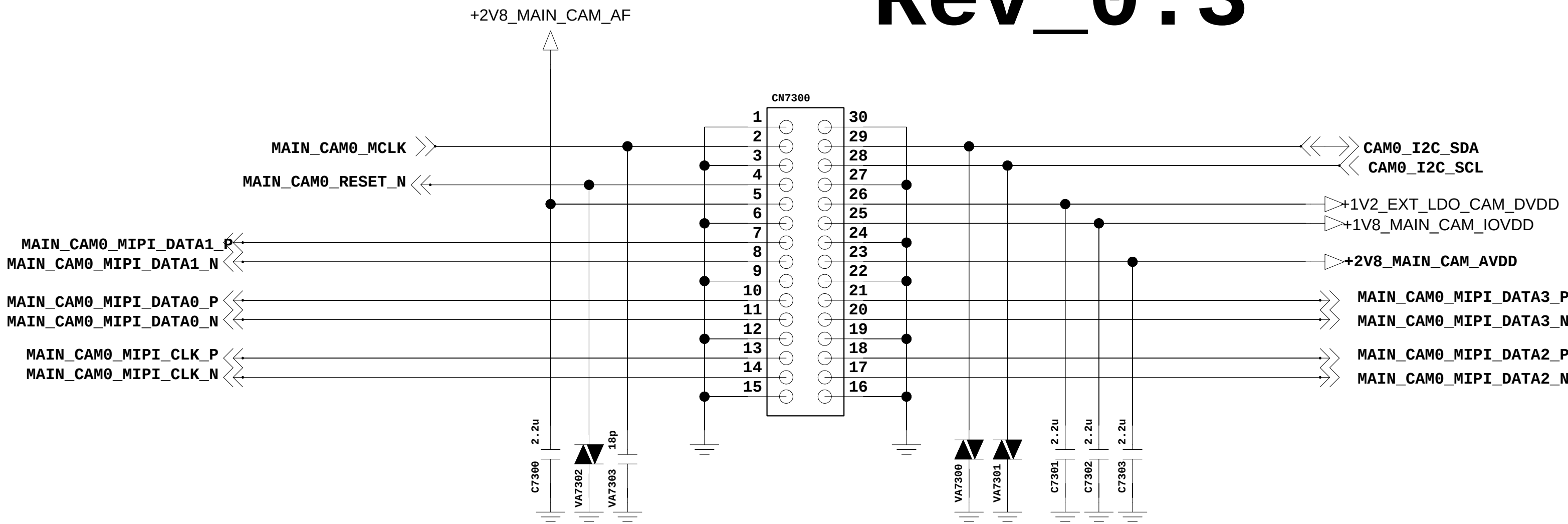
|      |                                            |       |                          |
|------|--------------------------------------------|-------|--------------------------|
| VSYS | 0.7-1.3V/1.35V (optional)<br>(6.25mV/step) | 2,800 | Processor / Digital Core |
| VSYS | 2.2                                        | 1,400 | System LDO input         |
| VPA  | 0.5V - 3.4V (0.1V/step)                    | 600   | 3G power amplifier       |

LDO types and brief specifications

| Type     | LDOname  | Vout (Volt)                        | I <sub>max</sub> (mA) | Application                    |
|----------|----------|------------------------------------|-----------------------|--------------------------------|
| ALDO     | VCN28    | 2.8                                | 30                    | RF chip                        |
| ALDO     | VTCXO    | 2.8                                | 40                    | 13/26MHz reference clock       |
| ALDO     | VA       | 2.8                                | 150                   | Analog baseband                |
| ALDO     | VCAMA    | 2.8                                | 150                   | Analog power for camera module |
| DLDO     | VCN_3V3  | 3.3/3.4/3.5/3.6                    | 240                   | WiFi                           |
| DLDO     | VIO28    | 2.8                                | 200                   | Digital IO                     |
| DLDO     | VSIM1    | 1.8/3.0                            | 50                    | 1st Sim card                   |
| DLDO     | VSIM2    | 1.8/3.0                            | 50                    | 2st SIM card                   |
| DLDO     | VSUB     | 3.3                                | 20                    | USB                            |
| DLDO     | VGP1     | 1.2/1.3/1.5/1.8<br>2.0/2.8/3.0/3.3 | 100                   | General purpose LDO            |
| DLDO     | VGP2     | 1.2/1.3/1.5/1.8<br>2.0/2.5/2.8/3.0 | 100                   | General purpose LDO            |
| DLDO     | VEMC_3V3 | 3.0/3.3                            | 400                   | 3.3V EMMC                      |
| DLDO     | VCAM_AF  | 1.2/1.3/1.5/1.8<br>2.0/2.8/3.0/3.3 | 100                   | AF application                 |
| ALDO     | VMC      | 1.8/3.3                            | 100                   | SD 2.0/3.0 memory card         |
| ALDO     | VMCH     | 3.0/3.3                            | 400                   | SD 3.0 memory card             |
| DLDO     | VIBR     | 1.2/1.3/1.5/1.8<br>2.0/2.8/3.0/3.3 | 100                   | Vibrator                       |
| PTCLDO   | VRTC     | 2.8                                | 2                     | Real-time clock                |
| VSYS LDO | VM       | 1.24/1.39/1.54/1.84                | 700                   | Memory power                   |
| VSYS LDO | VRF18    | 1.825                              | 200                   | RF application                 |
| VSYS LDO | VIO18    | 1.8                                | 300                   | IO pad power                   |
| VSYS LDO | VCAMD    | 1.2/1.3/1.5/1.8                    | 150                   | Camera application             |
| VSYS LDO | VCAM_IO  | 1.8                                | 100                   | Camera IO application          |
| VSYS LDO | VGP3     | 1.2/1.3/1.5/1.8                    | 200                   | General purpose LDO            |
| VSYS LDO | VCN18    | 1.8                                | 120                   | General purpose LDO            |

<7-3-5-1\_30pin\_5M\_8M\_13M\_AF> the pin map

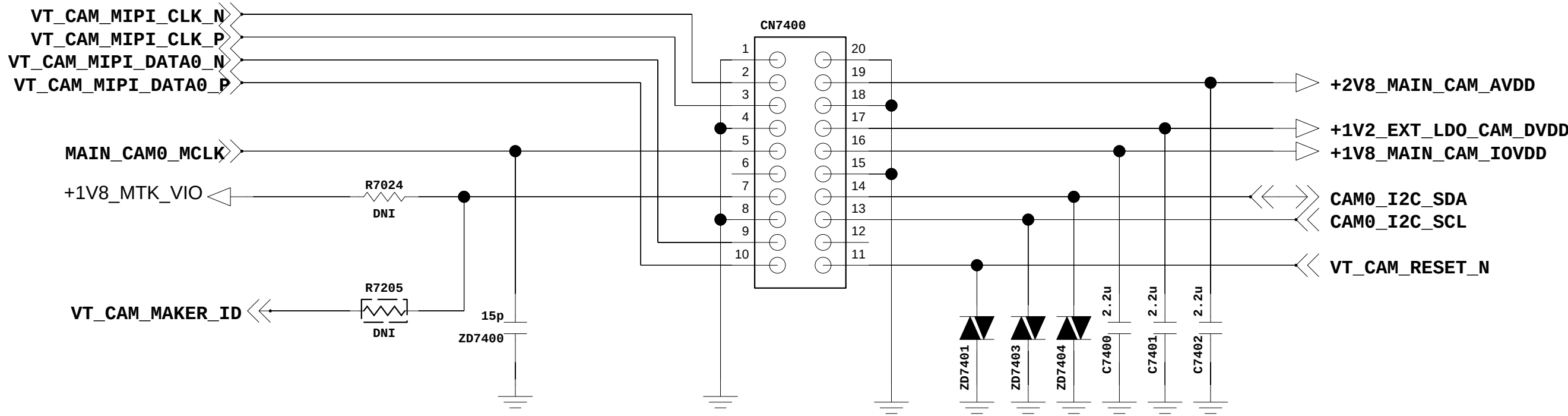
Rev\_0.3



< 7-4-1\_VT\_Camera\_20pin\_VGA\_1.26M\_1.3M\_2.4M>

Rev\_0.3

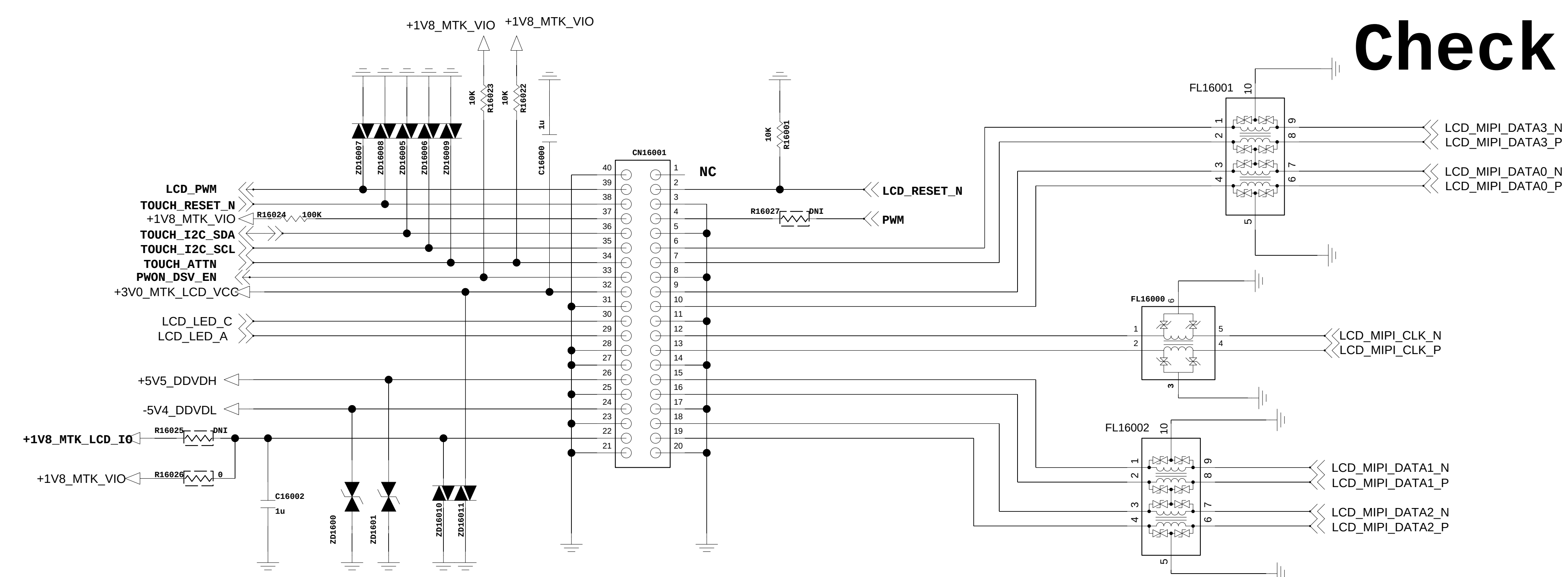
[ DGMS Guide ]  
MC-C00081-17: 1. Add ESD Protection curcuit for Camera signal lines  
2. Use under 100ohm/15pF ESD protection for over 2MP Camera



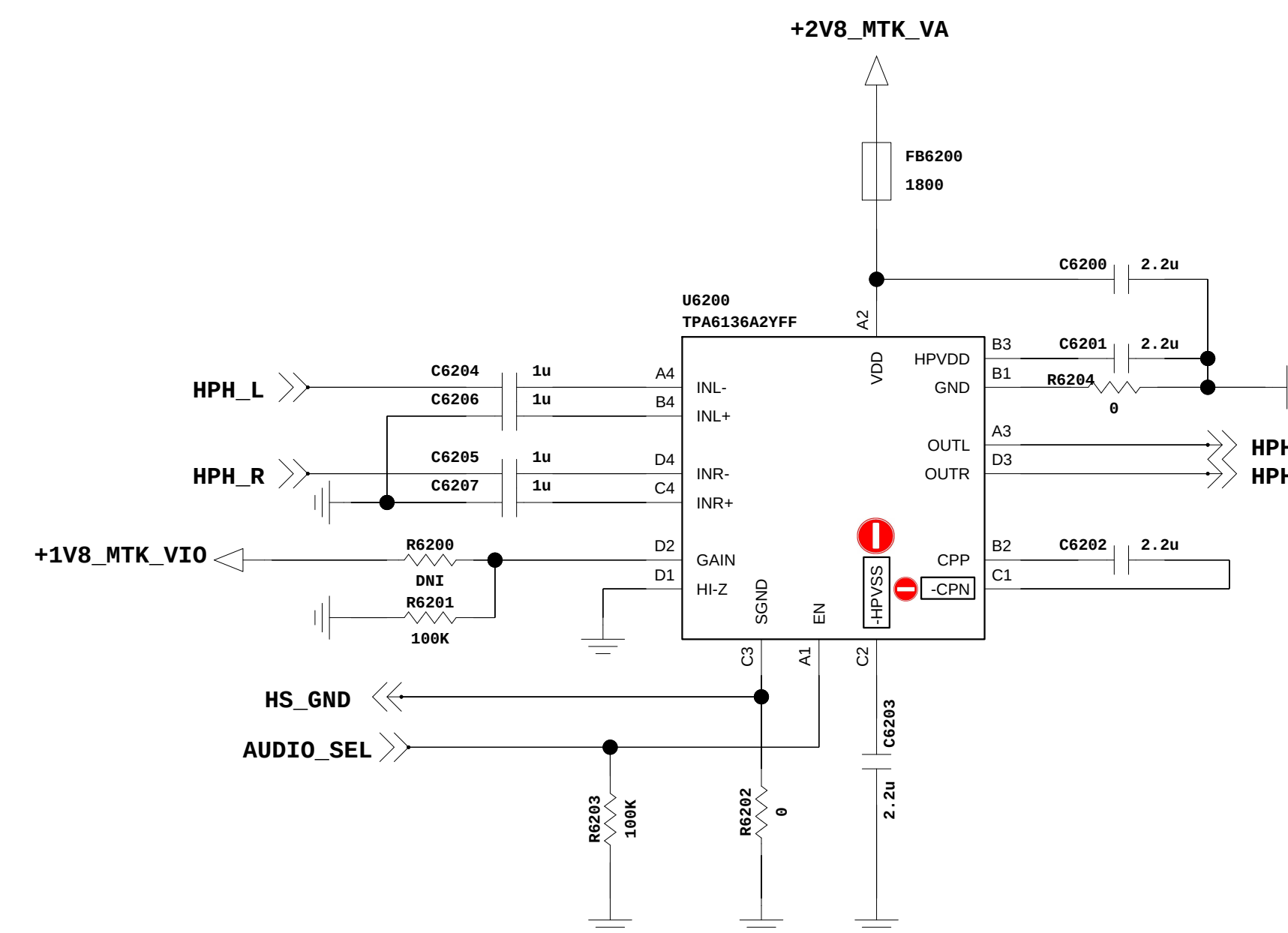
[ DGMS Guide ]  
MC-C01845-13 : Separate Analog Power and Digital Power

[ DGMS Guide ]  
MC-C00060-13 : Check the pull-up resistor PAD for I2C line

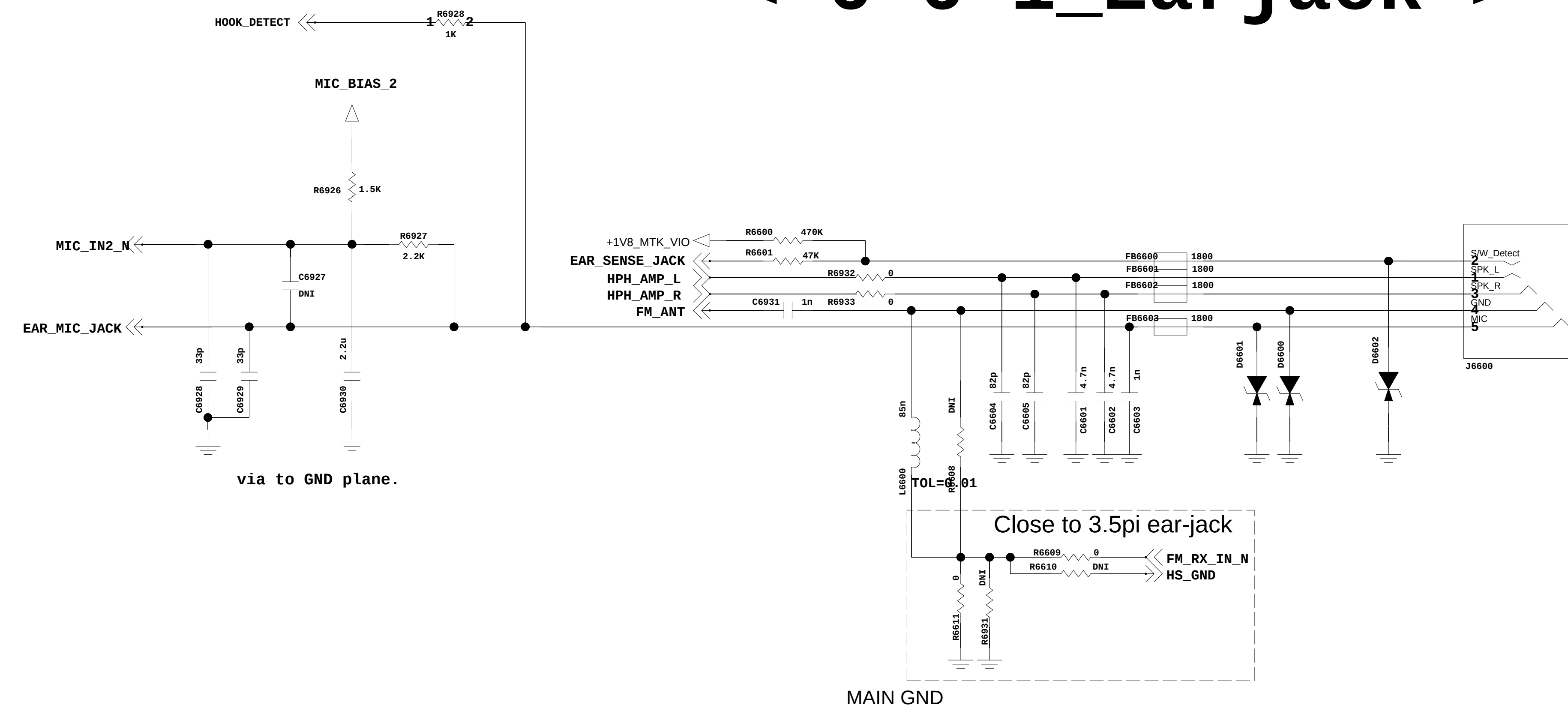
# LCD\_Touch Connector\_JDI Incell Touch



**< Headset Amp TAP6136>**

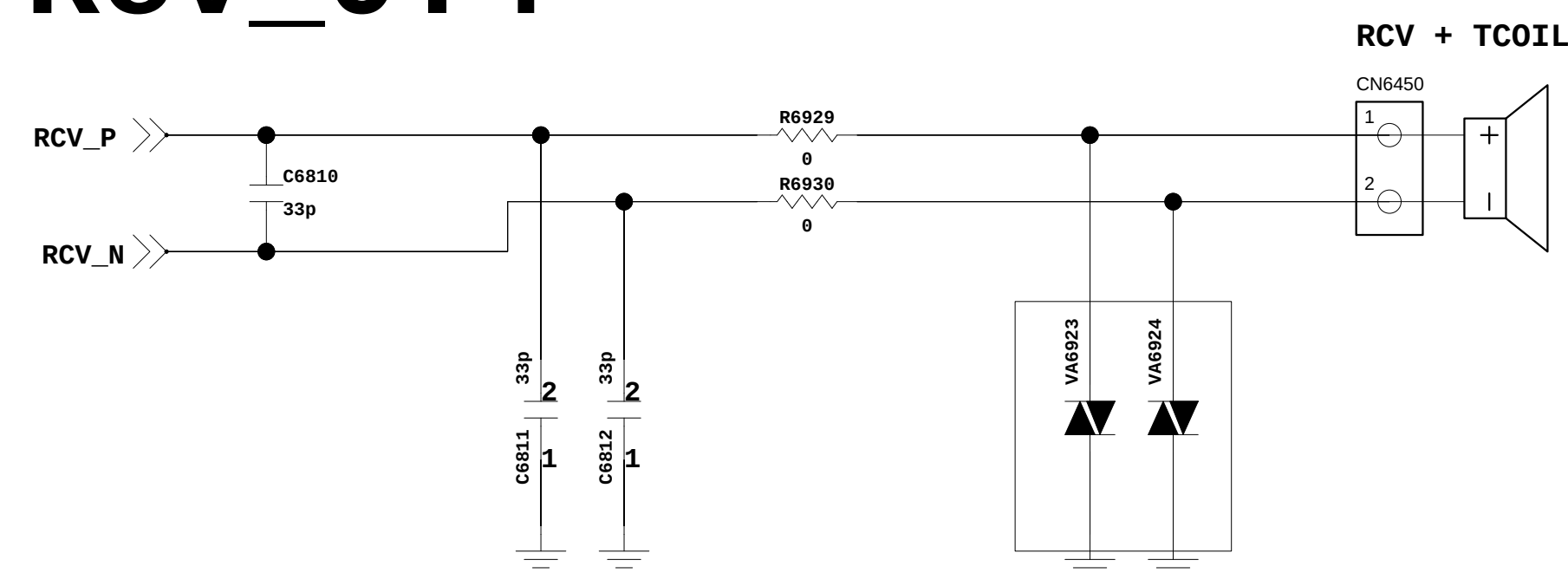


## < 6-6-1\_Earjack >



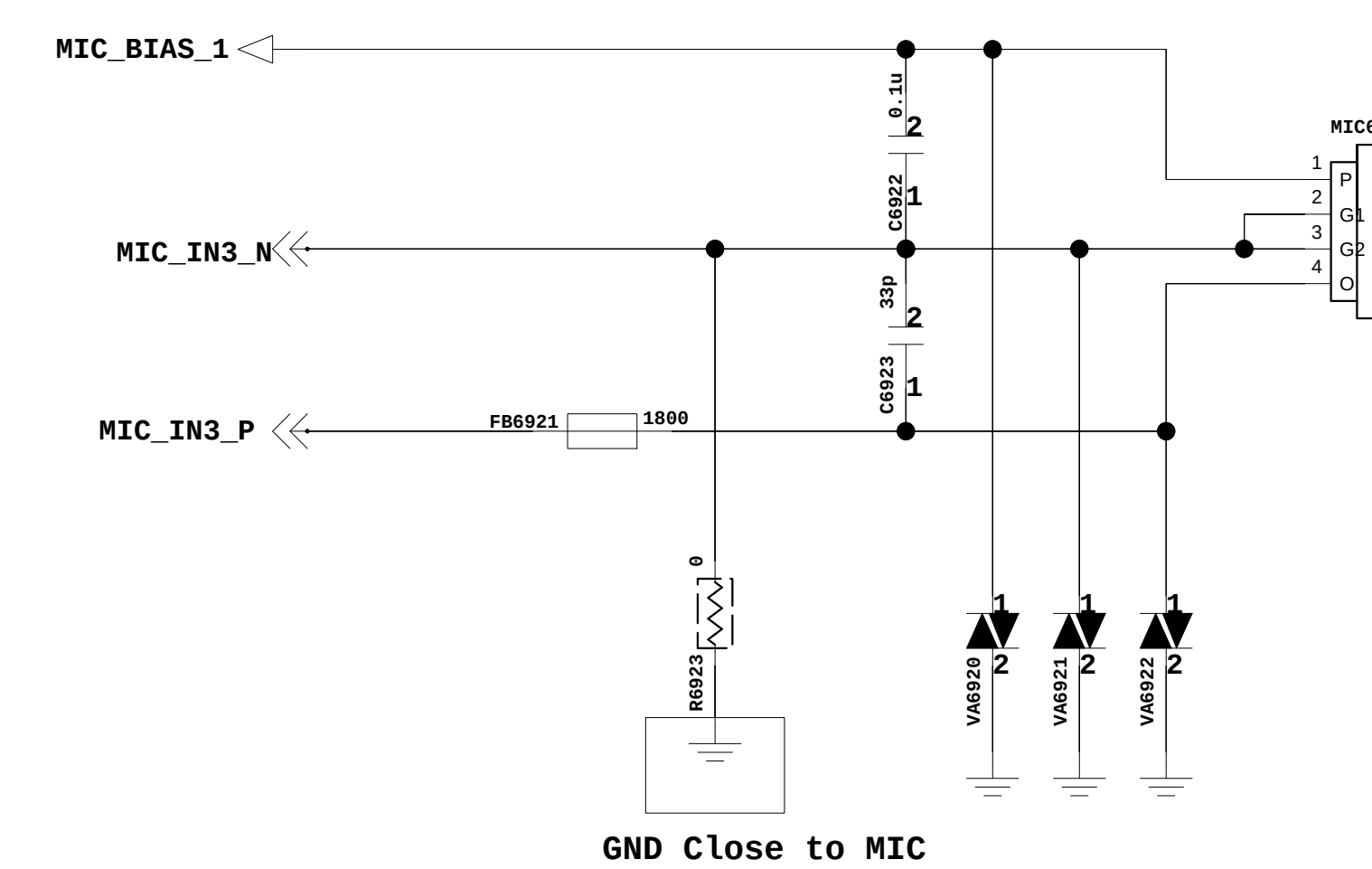
# < 6-8-2\_Receiver >

## Rev\_0.4



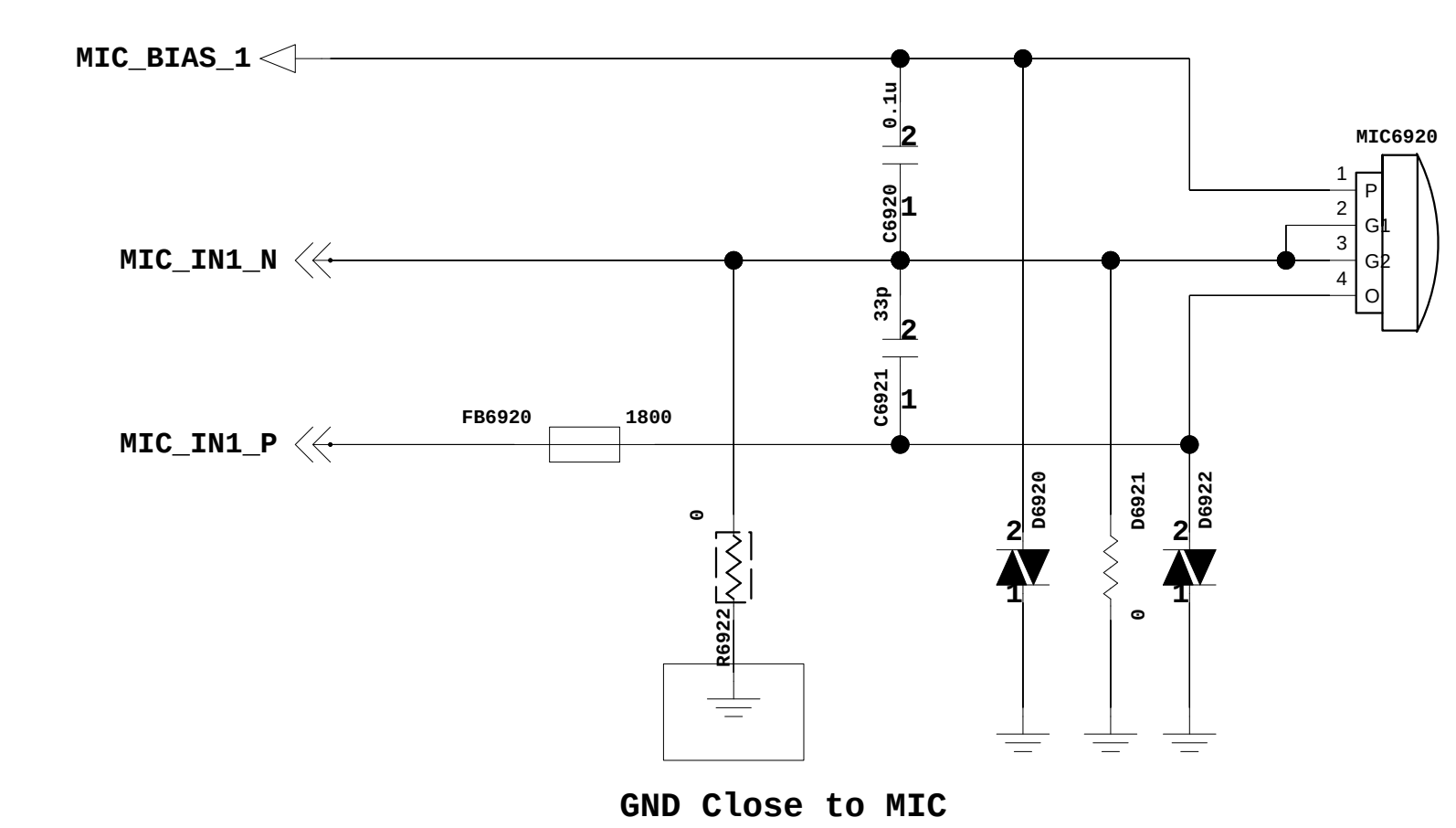
**< 6-9-2\_2MIC >**  
**Rev\_0.4**

**SUB\_MIC**



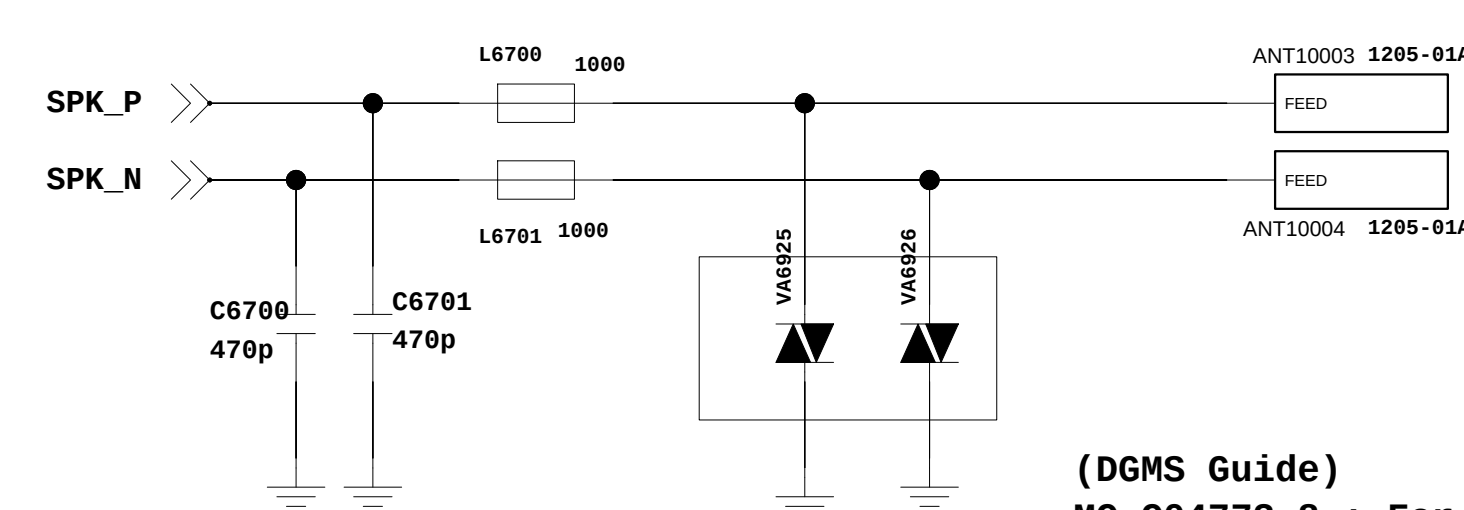
(DGMS Guide)  
MC-C00074-11 : For ESD Protection / Tuning points  
When USE) TVS devices's Working Voltage must be >

## MAIN\_MIC



(DGMS Guide)  
MC-C00074-11 : For ESD Protection / Tuning points  
When USE) TVS devices's Working Voltage must be greater than the Path voltage

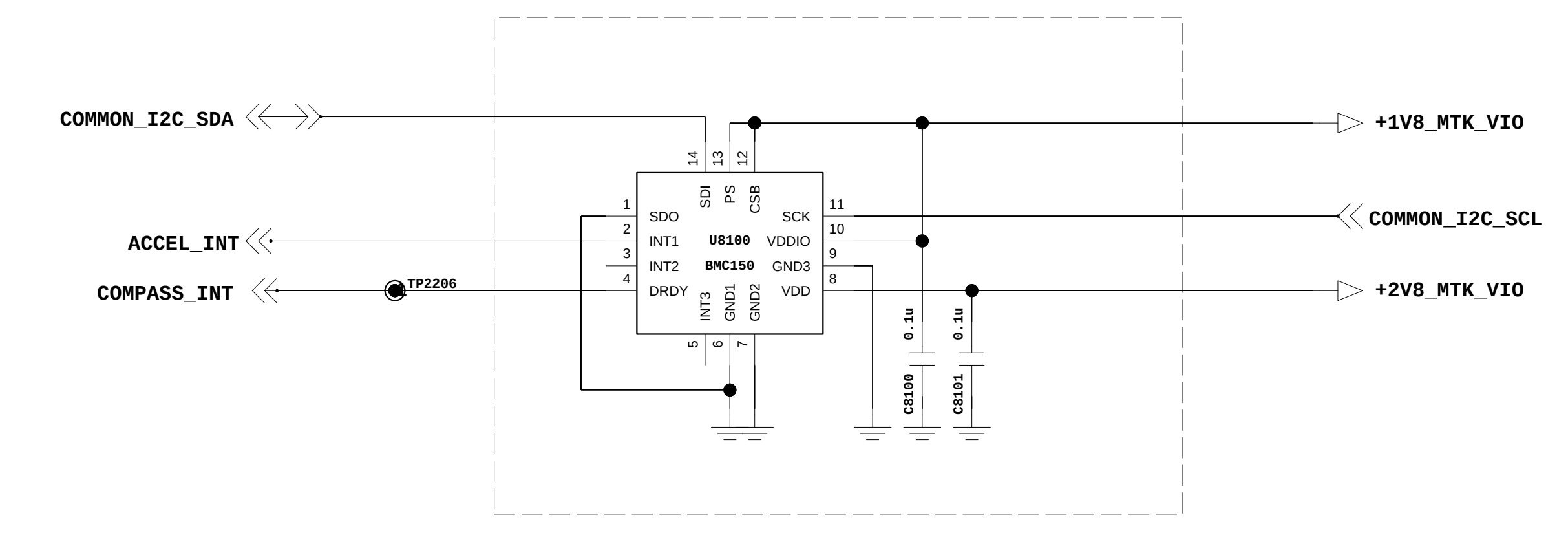
## < 6-7-1\_Speaker Rev\_0.4



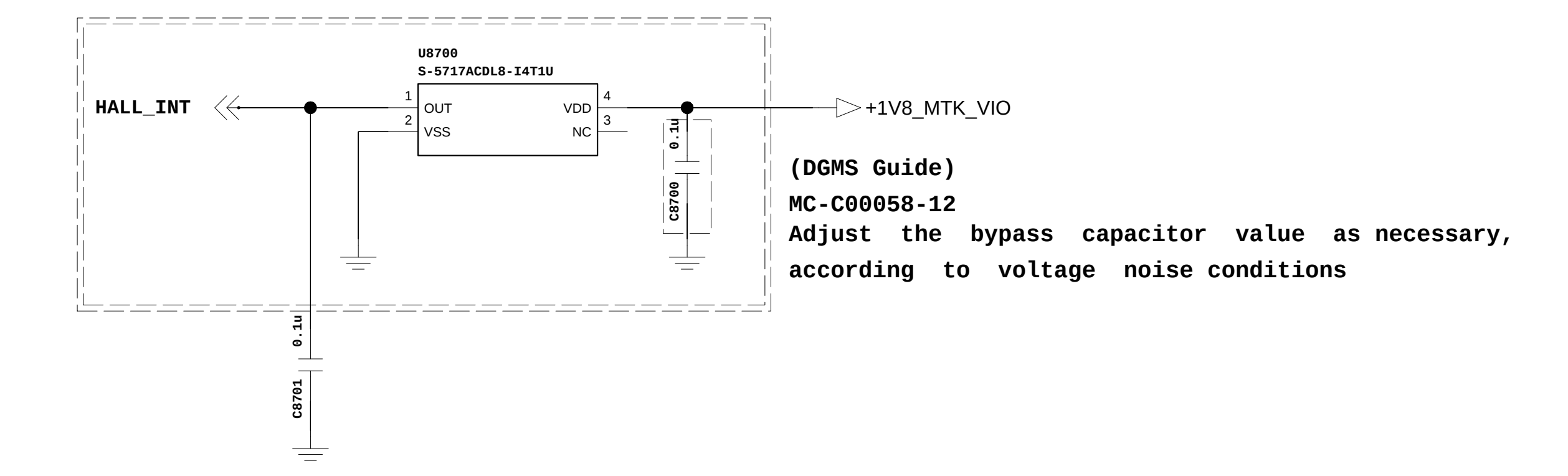
(DGMS Guide)  
MC-C04773-8 : For ESD Protection / Tuning points

When USE TVS devices's Working Voltage must be greater than the Path voltage should apply capacitor under 50pF value for ESD protection with d-class amp

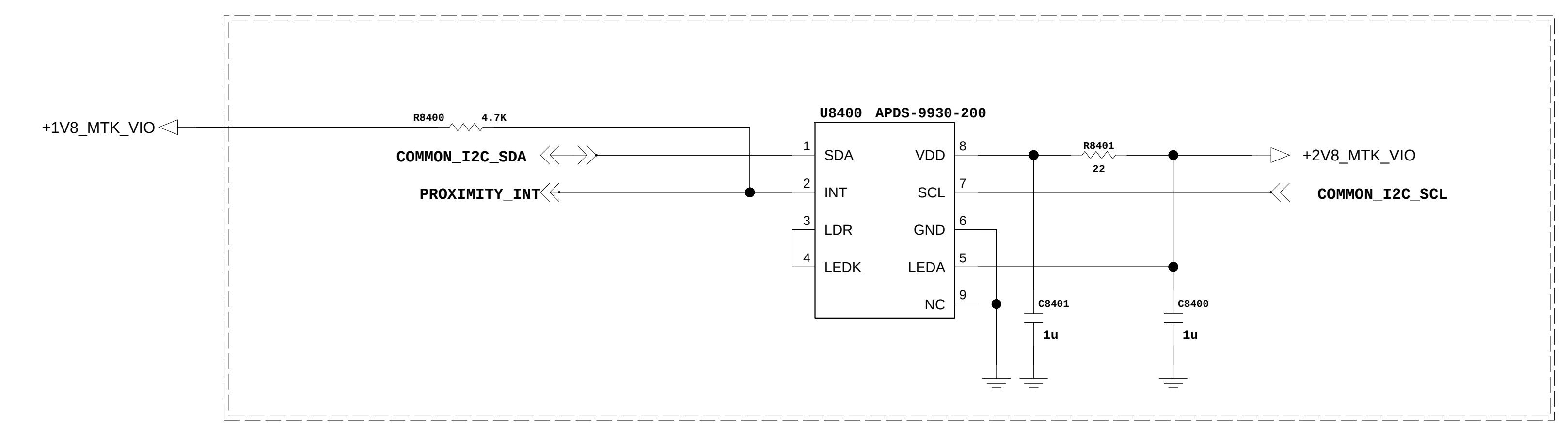
# <8-1-2-1\_Accel\_Compass\_BMC150>1.0



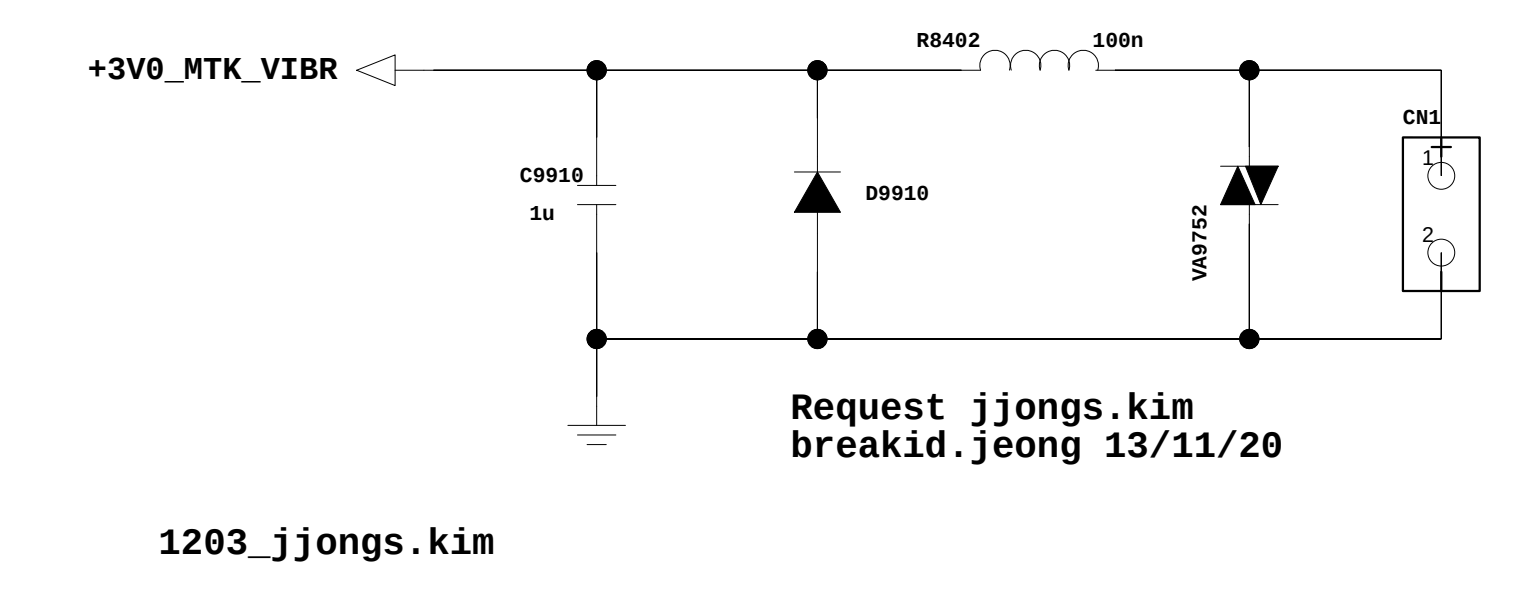
# <8-7-1-5\_Hall\_IC\_S5717>Rev\_1.0



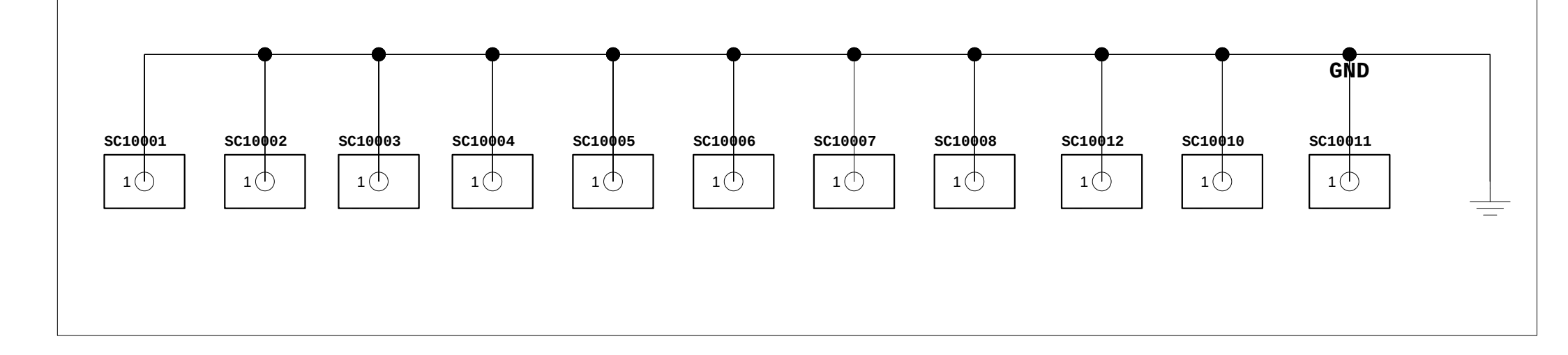
# <8-4-1-1\_Proximity\_APDS-9930>1.0



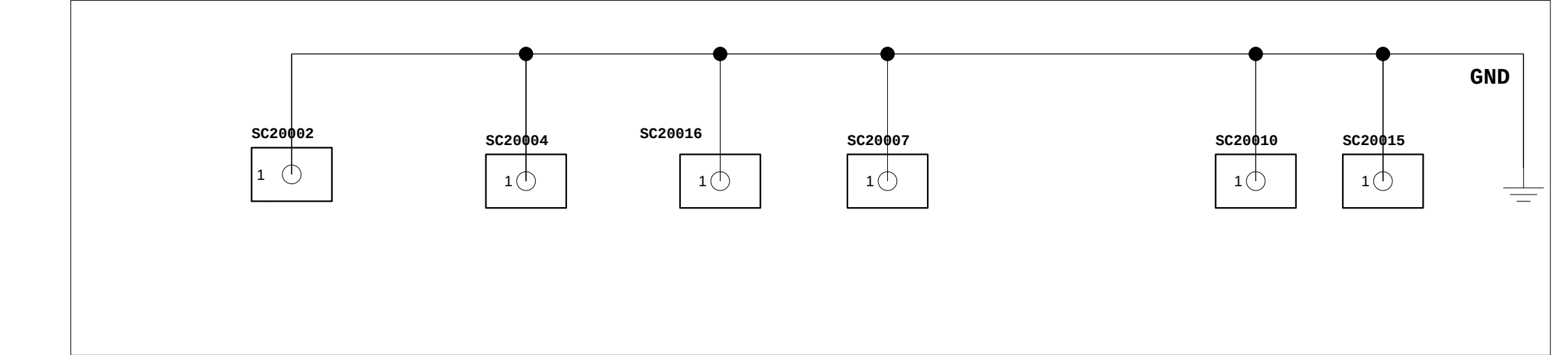
# < 9-9-1-1\_Motor > Rev\_0.3



## TOP SHIELD CAN CLIP



## BOTTOM SHIELD CAN CLIP

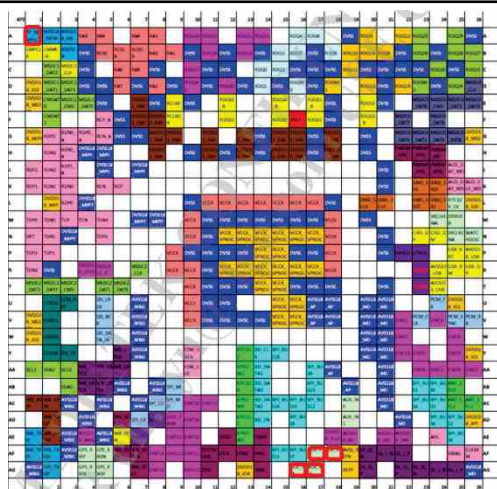






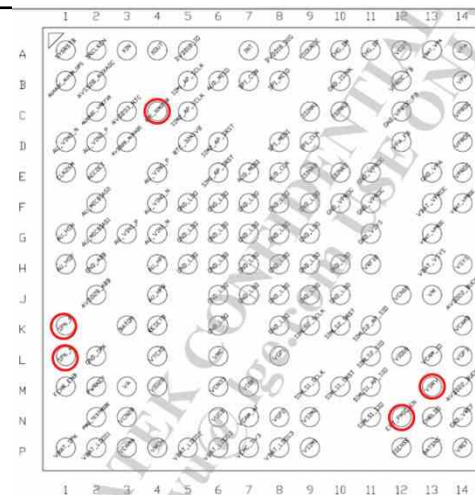
# 6. BGA PIN MAP

U2100\_MT6582\_LPDDR2



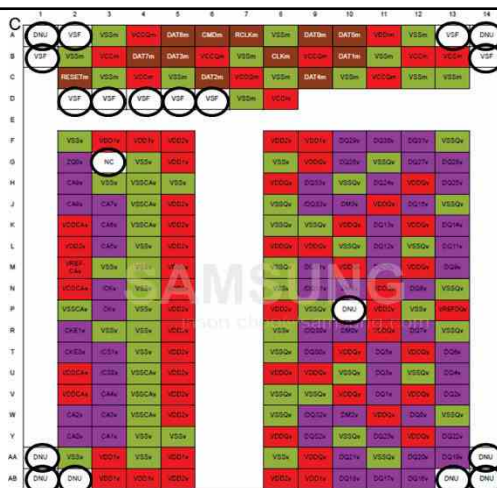
O Not used

U4100\_MT6323\_PMIC



O Not used

U3300\_eMMC



O Not used

Model name : MT6627 (U5100)  
Function : Connectivity IC  
– no BT/WiFi, no FM Radio, no GPS

Model name : OZ26000008 (X1500)  
Function : TCXO  
– no booting, no RF

Model name : MT6166 (U1500)  
Function : RF transmitter  
– no RF

Model name : SKY77761(U1600)  
Function : 3G PAM  
– no RF(WCDMA Band1/(2)/5/8)

Model name : BU52033NVX(U8700)  
Function : Hall IC  
no quick circle case

Model name : SKY77589-11(U1100)  
Function : RF FEM(Front End Module)  
– no RF

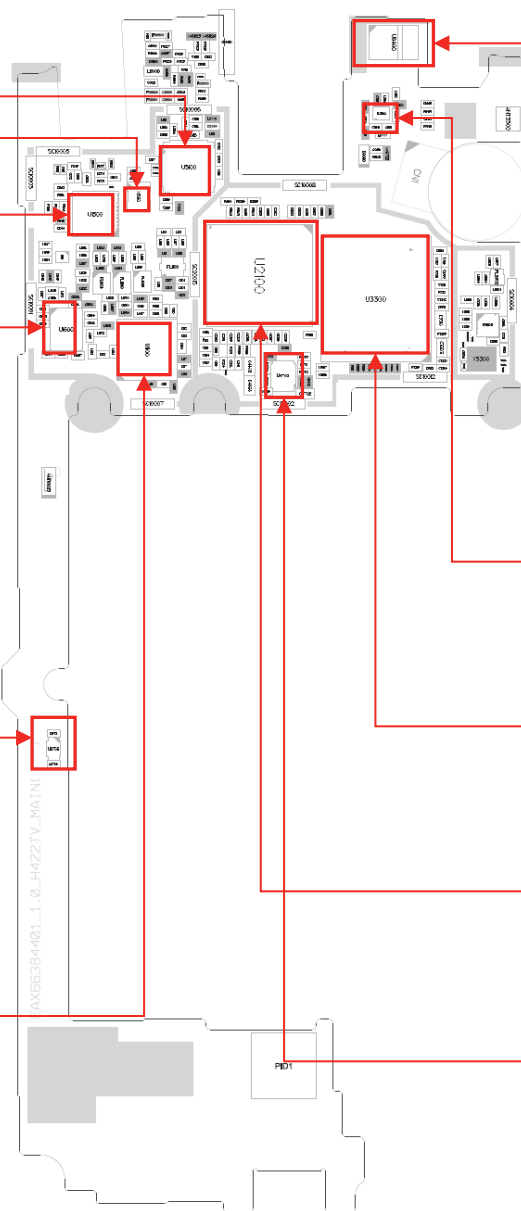
Model name : APDS-9130-200(U8400)  
Function : Proximity sensor  
– no proximity sensor

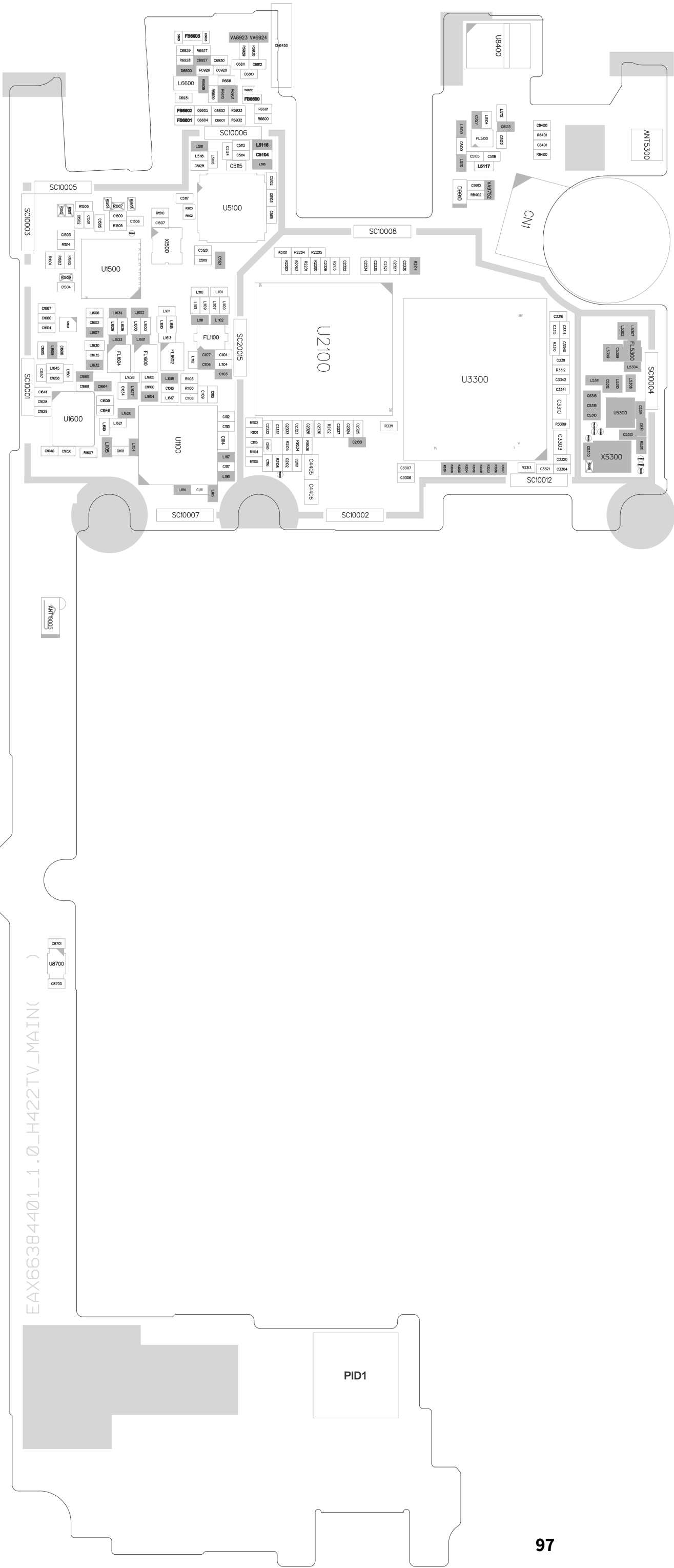
Model name : SFMG2XAB002 (FL5100)  
Function : GPS LNA  
– no GPS

Model name : KMQ7X000SA-B315(U3300)  
Function : Memory  
– no SIM detect, no SD card

Model name : MT6582(U2100)  
Function : Application processor  
– no booting, no function

Model name : RT9536GQW (U4700)  
Function : Linear Charger IC  
– no charge





Model name : GB042-20S-H10-E3000 (CN7400)  
Function : VT camera connector  
- no VT camera

Model name : GB042-30S-H10-E3000 (CN7300)  
Function : Main camera connector  
- no main camera

Model name : FC-135(X4100)  
Function : X-tal  
- no booting, no RF

Model name : MT6323(U4100)  
Function : PMIC  
- no booting, no touch, no LCD, no RF

Model name : GB042-40S-H10-E3000 (CN16001)  
Function : LCD B to B connector  
- no LCD

Model name : C-Clip (ANT1108,ANT1110)  
Function : Main antenna connection  
- no Service

Model name : 04-5151-005-104-883(CN10001)  
Function : USB connection  
- no charge, no USB connection

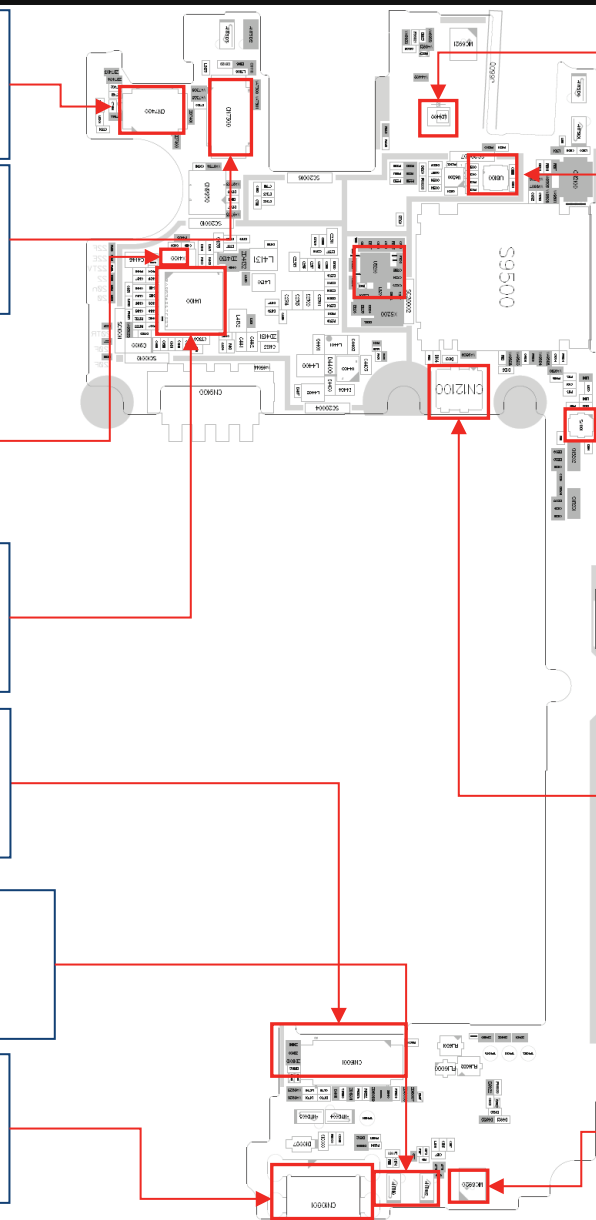
Model name : CL-773F-CW18C4-SDW-T (LD4400)  
Function : Fresh LED  
- no fresh LED

Model name : BMC150 (U8100)  
Function : Acceleration ,Compass  
- no Acceleration or compass sensor

Model name : MM8030 (SW1101)  
Function : RF switch  
- no RF

Model name : HSTC-04P20-1351 (CN12100)  
Function : Back key FPCB connector  
- no back key

Model name : S17OT423-001(MIC6920)  
Function : Microphone  
- no MIC.(Voice)



Model name : C-Clip(ANT1503)  
Function : GPS antenna connection  
- No GPS

Model name : BM10NB(0.8)-10DS (CN9950)  
Function : SIM2 FPCB connector  
- no SIM2 detect

Model name : KQ03L3-4R (CN9100)  
Function : Battery connector  
- no power, no booting

Model name : LM3632 (U4400)  
Function : DSV convertor  
- no LCD

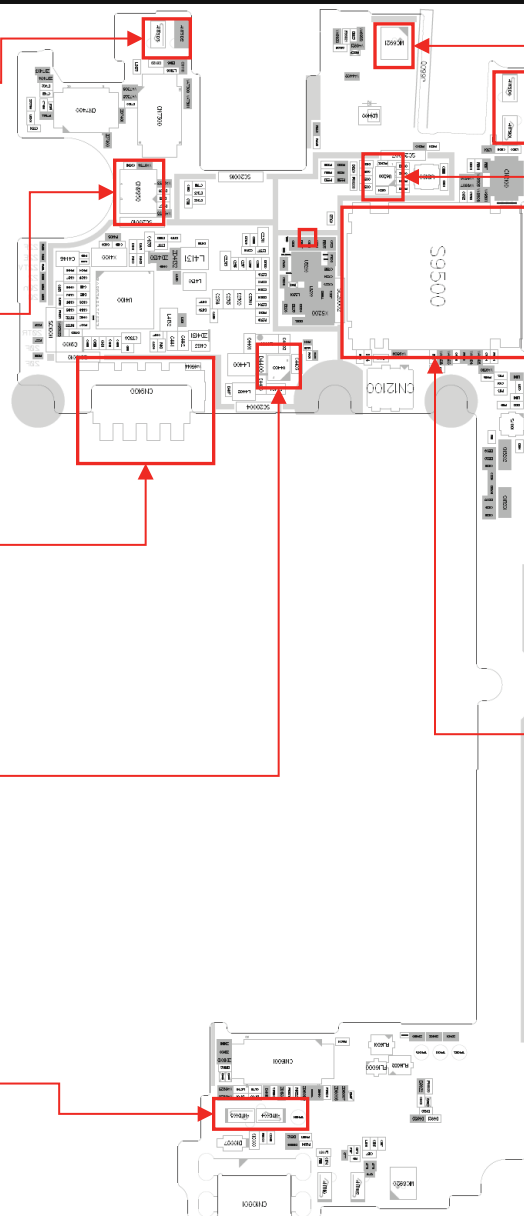
Model name : C-Clip(ANT10003,ANT10004)  
Function : Speaker contact  
- no speaker

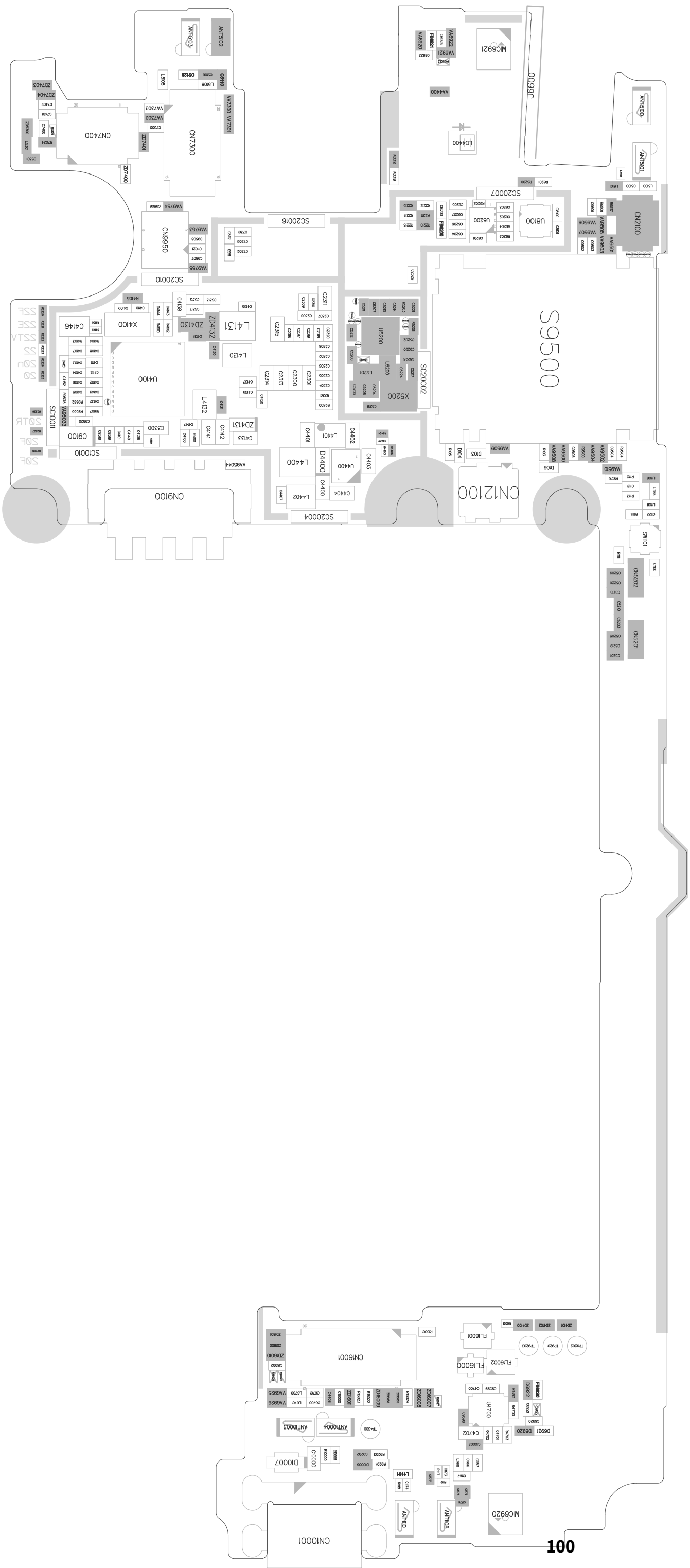
Model name : S17OT423-001(MIC6921)  
Function : Microphone  
- no MIC.(Voice)

Model name : C-Clip (ANT5100,ANT5101)  
Function : BT/WiFi antenna connection  
- No BT/WiFi

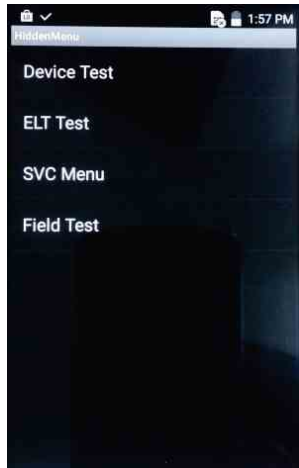
Model name : TPA6136A2YFF (U6200)  
Function : Headset Amp IC  
- no Headset

Model name : 1040681616 (S9500)  
Function : Combo socket  
- no SIM detect, no SD card



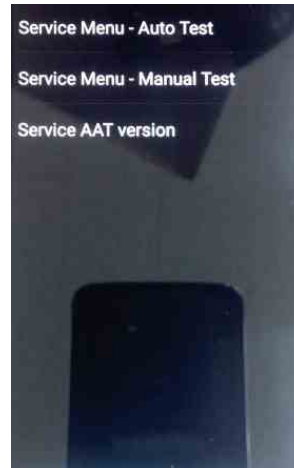


## 1. Hidden Menu Start



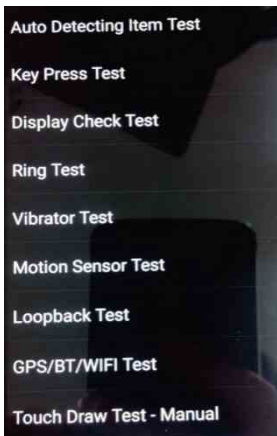
- Start shortcut key:  
\*#546368#\*422#
- Hidden Menu List  
: Start the desired menu, click

## 2. Device Test



- Service Menu - Auto test  
: All auto test

## 2. Device Test List



- Service Menu - Manual test  
: Start the desired item, click

|                                                    |                                              |                                              |                                            |
|----------------------------------------------------|----------------------------------------------|----------------------------------------------|--------------------------------------------|
| <div>1. Auto Detection Item Test</div> <div></div> | <div>2. Key Press Test</div> <div></div>     | <div>3. Display check test</div> <div></div> | <div>4. Ring test</div> <div></div>        |
| <div>5. Vibrator test</div> <div></div>            | <div>6. Motion Sensor test</div> <div></div> | <div>7. Loopback test</div> <div></div>      | <div>8. GPS/BT/WIFI test</div> <div></div> |

| 9. Touch Draw test-Manual                                                         | 10. Camera(Main) test                                                             | 11. Camcorder test                                                                  | 12. Camera(VT) test                                                                 |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |
|                                                                                   |                                                                                   |                                                                                     |                                                                                     |
|                                                                                   |                                                                                   |                                                                                     |                                                                                     |

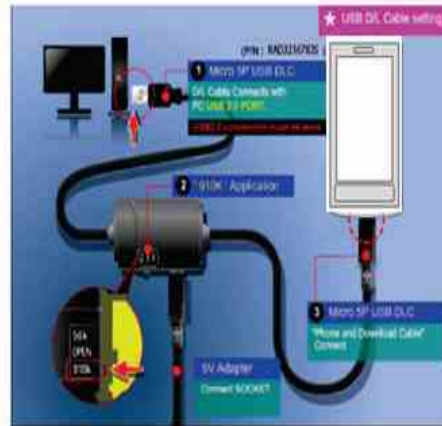
## 1. Summary

| Tool Version                      | DLL name                             | USB Driver                                            |     |
|-----------------------------------|--------------------------------------|-------------------------------------------------------|-----|
| LGFLAS Hv190                      | LGH422_20150310_LGFLASHv190_Download | LGUnitedMobileDriver_S50MAN311AP22_ML_WHQL_Ver_3.11.3 |     |
| Please Check the Version to “B2B” |                                      |                                                       |     |
| H/W                               |                                      |                                                       |     |
|                                   | Name                                 | Part No.                                              | SW  |
| D/L Cable                         | Micro 5P (56-open-910K) USB DLC      | RAD321 67835                                          | TOT |

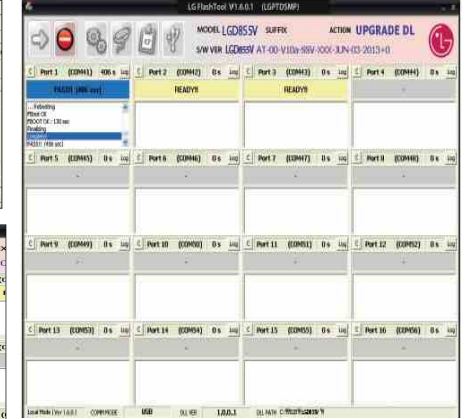
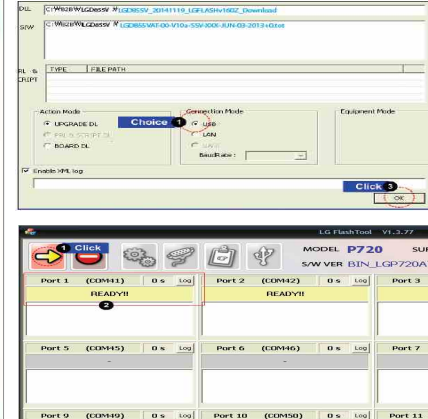
## 2. USB COM port Setting



## 3. USB D/L Cable setting



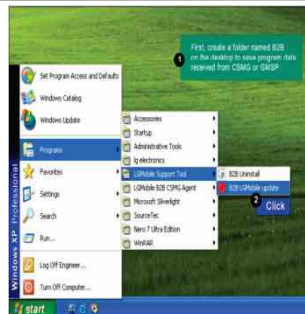
## 4. Flash tool D/L setting



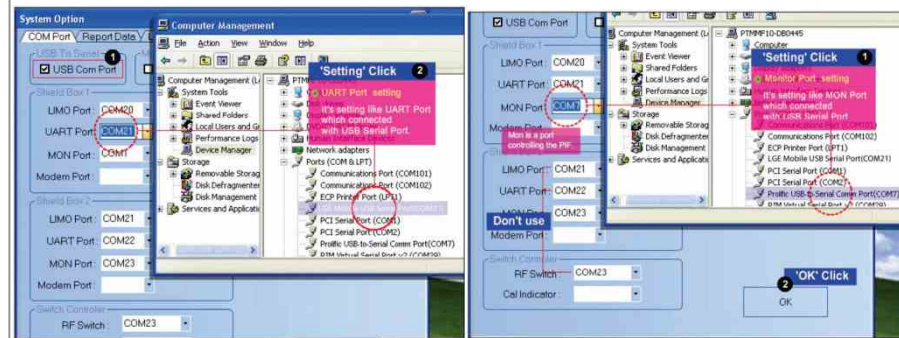
※ If you want more information, please refer B2B's Notification "Download User Guide".

## 1. Summary

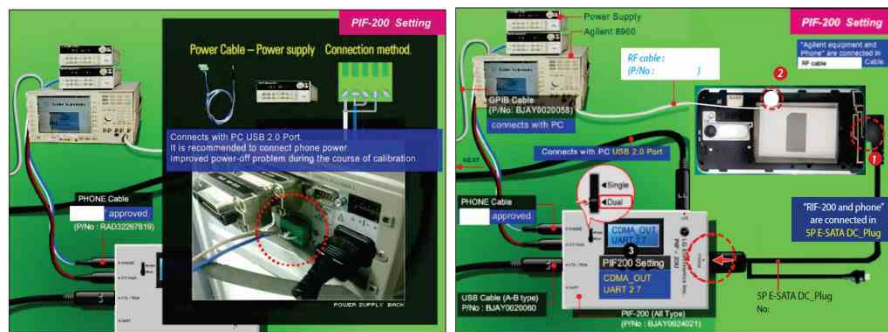
| CAL INFORMATION                                     |                                                                       |             |
|-----------------------------------------------------|-----------------------------------------------------------------------|-------------|
| S/W VERSION                                         |                                                                       |             |
| [TachyonV2013] LGH422 20150306 ALL File Calibration |                                                                       |             |
| Please Check the Version to "B2B"                   |                                                                       |             |
| H/W                                                 |                                                                       |             |
|                                                     | Name                                                                  | Part No.    |
| PIF                                                 | PIF200                                                                | BJAY0024021 |
| USB Cable                                           | USB Cable 2.0                                                         | RAD32247898 |
| Power Cable                                         | DC Power Cable                                                        | RAD32247878 |
| I/O Cable                                           | 5P E-SATA DC Plug                                                     | RAD32167861 |
| RF Cable Main                                       | MXFK1WT4000C2                                                         | RAD32667826 |
| Power Supply PIF                                    | Power supply 5.2V                                                     |             |
| Power Supply Phone                                  | Power supply 5.0V                                                     |             |
| PF Test Equipment                                   | Aqilent/CMW500                                                        |             |
| Notice                                              | 1. Use the Battery (Refer to Attached ppt)                            |             |
|                                                     | 1) Phone states: Power off<br>2) If do not use the battery, TX fails. |             |
| CMW 500 RF Cable connection                         | 2. Port Setting (Refer to Attached ppt)                               |             |
|                                                     | 1) Uart Port1 : Use the "LGE Mobile USB Serial Port"                  |             |



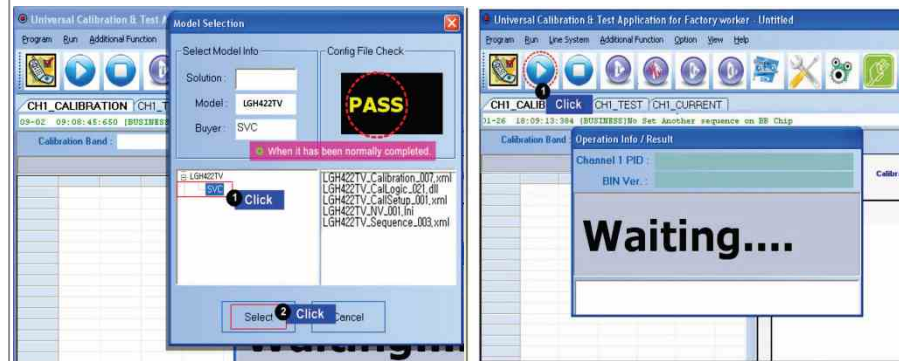
## 2. . USB COM port Setting



## 3. Calibration Cable setting

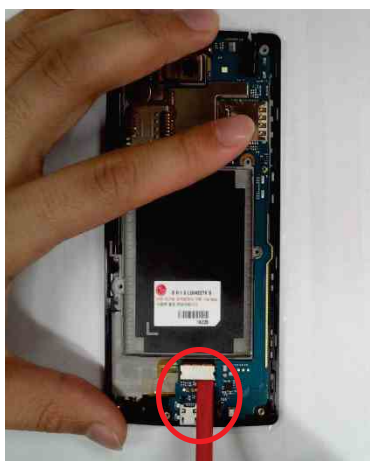


## 4. Flash tool D/L setting



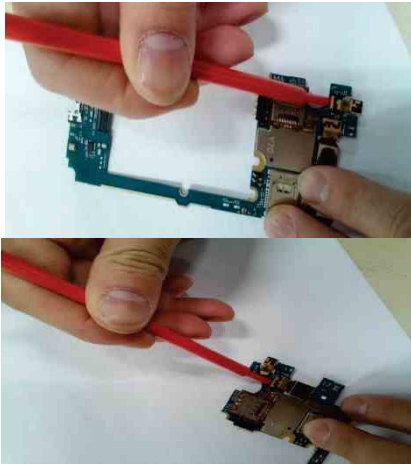
※ If you want more information, please refer B2B's Notification "RF Calibration User Guide".

# 11. DISASSEMBLE GUIDE

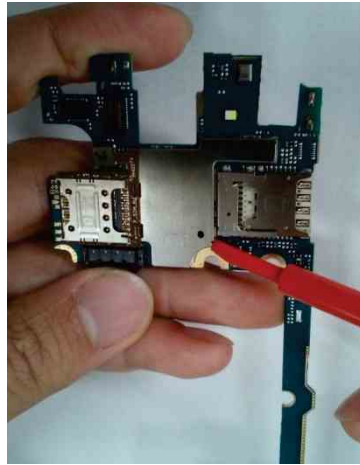
|                                                                                                                     |                                                                                                                        |                                                                                                                            |                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>1. Remove screws (10point)</b>  | <b>2. Disassemble Rear cover</b>      | <b>3. Disassemble Rear cover</b>        | <b>4. Disassemble Connector</b>        |
| <b>5. Disassemble Main PCB</b>     | <b>7. Disassemble HW parts(3ea)</b>  | <b>8. Take apart the Main antenna</b>  | <b>9. Take apart the Button assy</b>  |

# 11. DISASSEMBLE GUIDE

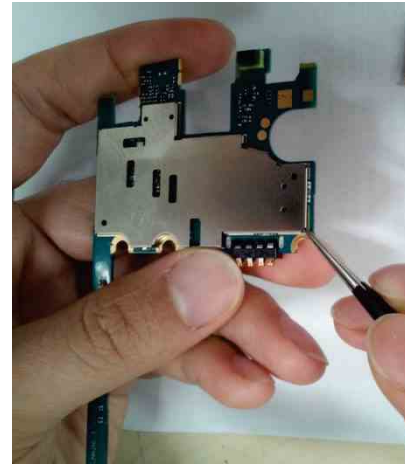
10. Take apart the connector



11. Take apart the Can shield(TOP)



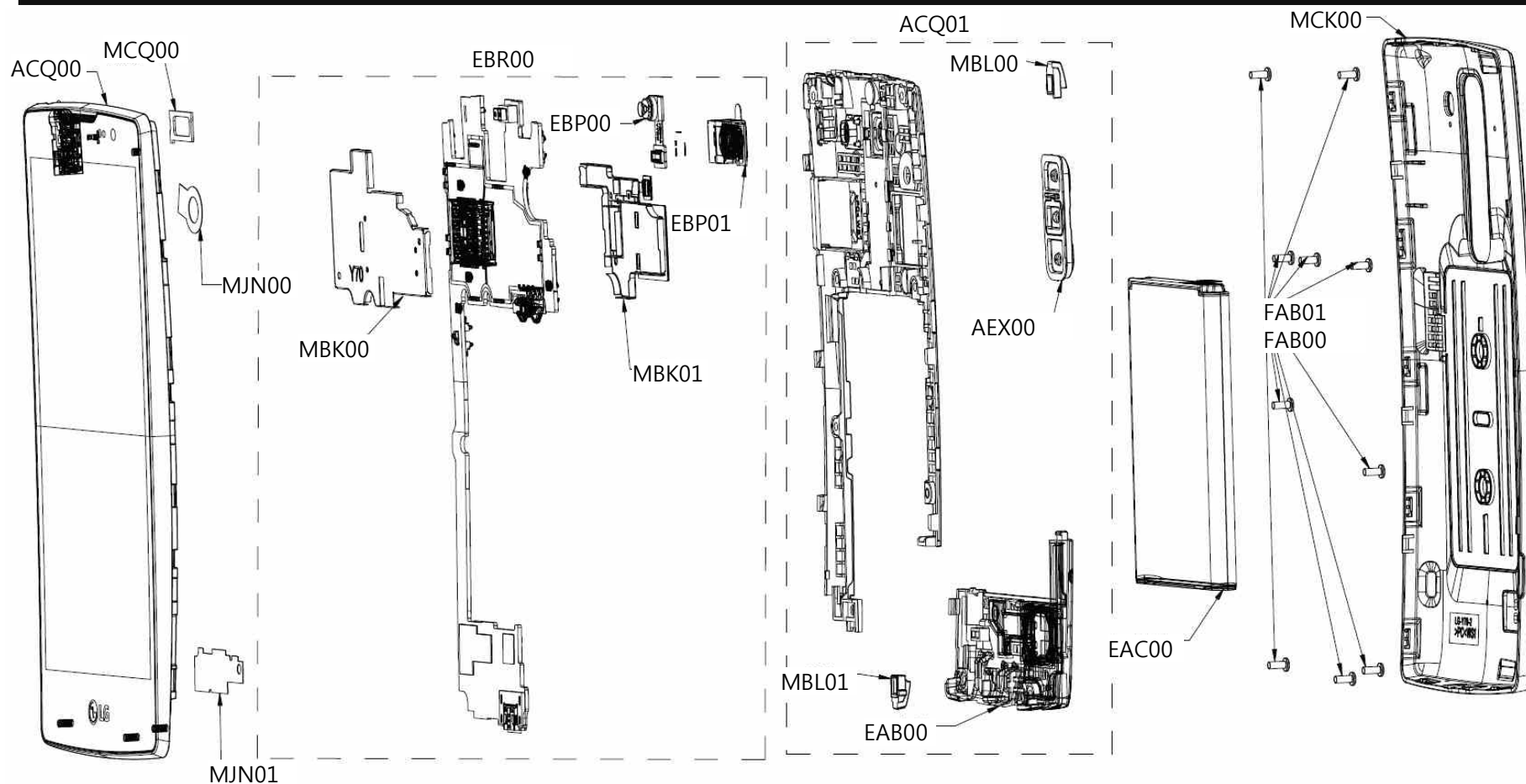
12. Take apart the Can shield(BOT)



13. Complete Disassembling



## 12. EXPLODED VIEW



|       |                      |       |                   |       |                                  |
|-------|----------------------|-------|-------------------|-------|----------------------------------|
| ACQ01 | Cover Assembly,Rear  | MJN01 | Tape              | EBP01 | Can,Shield                       |
| MBL00 | Cap                  | MCQ00 | Damper,Speaker    | MBK01 | Camera Module                    |
| AEX00 | Keypad Assembly,Main | FAB01 | Screw,Tapping     | EAC00 | Rechargeable Battery,Lithium Ion |
| MBL01 | Cap                  | FAB00 | Screw,Machine     | MCK00 | Cover,Battery                    |
| EAB00 | Speaker Module       | EBR00 | PCB Assembly,Main |       |                                  |
| ACQ00 | Cover Assembly       | MBK00 | Can,Shield        |       |                                  |
| MJN00 | Tape                 | EBP00 | Camera Module     |       |                                  |

# 13. REPLACEMENT PART LIST

| No | P/N         | Description               | Qty | Location no |
|----|-------------|---------------------------|-----|-------------|
| 1  | AGQ88333102 | Phone Assembly            | 1   | AGQ000000   |
| 2  | MEZ66175401 | Label,Approval            | 1   | MEZ002100   |
| 3  | ACQ88045302 | Cover Assembly,EMS        | 1   | ACQ100400   |
| 4  | ACQ88013602 | Cover Assembly,Rear       | 1   | ACQ01       |
| 5  | ACQ88074002 | Cover Assembly,Rear(SVC)  | 1   | ACQ105800   |
| 6  | MCR65966201 | Decor                     | 1   | MCR000000   |
| 7  | MBL66318201 | Cap                       | 1   | MBL00       |
| 8  | MKC65458801 | Window,Camera             | 1   | MKC009400   |
| 9  | MJN69547301 | Tape,Protect              | 1   | MJN061100   |
| 10 | MJN69427101 | Tape,Window               | 1   | MJN089300   |
| 11 | MFB63793201 | Lens,Flash                | 1   | MFB029600   |
| 12 | MCQ68265301 | Damper                    | 1   | MCQ000000   |
| 13 | MEZ64319901 | Label,After Service       | 1   | MEZ000900   |
| 14 | ACQ87937602 | Cover Assembly,Rear(Sub)  | 1   | ACQ063400   |
| 15 | MBF62844201 | Bush                      | 1   | MBF000000   |
| 16 | MCK68646202 | Cover,Rear                | 1   | MCK063300   |
| 17 | EAX66423901 | PCB,Keypad                | 1   | EAX010400   |
| 18 | MCQ68206201 | Damper,Camera             | 1   | MCQ009400   |
| 19 | MCQ68167701 | Damper,Speaker            | 1   | MCQ074200   |
| 20 | EAA63804901 | PIFA Antenna,WiFi         | 1   | EAA030100   |
| 21 | EAA63824901 | PIFA Antenna,GPS          | 1   | EAA030101   |
| 22 | AEX74357401 | Keypad Assembly,Main      | 1   | AEX00       |
| 23 | MBG65562601 | Button                    | 1   | MBG000001   |
| 24 | MBG65562501 | Button                    | 1   | MBG000000   |
| 25 | MEV65231701 | Insulator                 | 1   | MEV000000   |
| 26 | MBL66376901 | Cap                       | 1   | MBL01       |
| 27 | EAB63688701 | Speaker Module            | 1   | EAB00       |
| 28 | ACQ88035142 | Cover Assembly,Bar        | 1   | ACQ003400   |
| 29 | MJN69370201 | Tape,Protect              | 1   | MJN061100   |
| 30 | MJN69606901 | Tape,USP Film             | 1   | MJN107400   |
| 31 | MLAZ0038301 | Label                     | 1   | MEZ000000   |
| 32 | ACQ88311002 | Cover Assembly            | 1   | ACQ000000   |
| 33 | MCQ68207201 | Damper,Camera             | 1   | MCQ009400   |
| 34 | MEV65414201 | Insulator                 | 1   | MEV000000   |
| 35 | ACQ87773556 | Cover Assembly,Front      | 1   | ACQ00       |
| 36 | ACQ87800806 | Cover Assembly,Front(Sub) | 1   | ACQ033200   |
| 37 | MCK68547906 | Cover,Front               | 1   | MCK032700   |
| 38 | MDQ64257102 | Frame                     | 1   | MDQ000000   |
| 39 | MICE0016901 | Insert,Nut                | 1   | MET099501   |
| 40 | MICE0016907 | Insert,Nut                | 3   | MET099500   |
| 41 | MCQ68207101 | Damper                    | 1   | MCQ000000   |
| 42 | MCQ68245301 | Damper,LCD                | 1   | MCQ043300   |
| 43 | MCQ68265201 | Damper,Speaker            | 1   | MCQ00       |
| 44 | MJN69409201 | Tape                      | 1   | MJN00       |
| 45 | MDS65130901 | Gasket                    | 1   | MDS000002   |
| 46 | MHK64925201 | Sheet                     | 1   | MHK000000   |

# 13. REPLACEMENT PART LIST

|    |             |                                  |    |                                                                                                                                                        |
|----|-------------|----------------------------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 47 | MJN69370001 | Tape                             | 1  | MJN01                                                                                                                                                  |
| 48 | MJN69370101 | Tape                             | 1  | MJN000002                                                                                                                                              |
| 49 | MCR65967101 | Decor                            | 1  | MCR000000                                                                                                                                              |
| 50 | EAT62653201 | Module,Hybrid Touch LCD          | 1  | EAT130000                                                                                                                                              |
| 51 | MDS65091201 | Gasket                           | 1  | MDS000000                                                                                                                                              |
| 52 | EAG63550001 | Jack,Phone                       | 1  | EAG140300                                                                                                                                              |
| 53 | EAB63288401 | Receiver                         | 1  | EAB010400                                                                                                                                              |
| 54 | EAU62283801 | Motor,DC                         | 1  | EAU010000                                                                                                                                              |
| 55 | FAB32218901 | Screw,Tapping                    | 0  | FAB01                                                                                                                                                  |
| 56 | GMEY0014301 | Screw,Machine                    | 10 | FAB00                                                                                                                                                  |
| 57 | EBR80493302 | PCB Assembly,Main                | 1  | EBR00                                                                                                                                                  |
| 58 | EBR80563601 | PCB Assembly,Main,Insert         | 1  | EBR071500                                                                                                                                              |
| 59 | MBK64252501 | Can,Shield                       | 1  | MBK00                                                                                                                                                  |
| 60 | EBR80003501 | PCB Assembly,Flexible            | 1  | EBR070500                                                                                                                                              |
| 61 | EBR80043901 | PCB Assembly,Flexible,Insert     | 1  | EBR070100                                                                                                                                              |
| 62 | MJN69489501 | Tape                             | 1  | MJN000000                                                                                                                                              |
| 63 | EBR80043801 | PCB Assembly,Flexible,SMT        | 1  | EBR070400                                                                                                                                              |
| 64 | EBR80058701 | PCB Assembly,Flexible,SMT Bottom | 1  | EBR070200                                                                                                                                              |
| 65 | EAG64229801 | Socket,DIMM/SIMM                 | 1  | J100                                                                                                                                                   |
| 66 | EBR80078501 | PCB Assembly,Flexible,SMT Top    | 1  | EBR070300                                                                                                                                              |
| 67 | EAX66192001 | PCB,Sub                          | 1  | EAX010100                                                                                                                                              |
| 68 | EAG63393301 | Connector,BtoB                   | 1  | CN100                                                                                                                                                  |
| 69 | EBP62221801 | Camera Module                    | 1  | EBP00                                                                                                                                                  |
| 70 | MBK64272501 | Can,Shield                       | 1  | MBK01                                                                                                                                                  |
| 71 | MDS65211801 | Gasket                           | 1  | MDS000000                                                                                                                                              |
| 72 | BRAH0001305 | Resin,PC                         | 0  | RAA050100                                                                                                                                              |
| 73 | EBP62101706 | Camera Module                    | 1  | EBP01                                                                                                                                                  |
| 74 | EBR80514602 | PCB Assembly,Main,SMT            | 1  | EBR071800                                                                                                                                              |
| 75 | BRAH0001304 | Resin,PC                         | 0  | RAA050100                                                                                                                                              |
| 76 | EBR80493401 | PCB Assembly,Main,SMT Bottom     | 1  | EBR071600                                                                                                                                              |
| 77 | ECCH0000112 | Capacitor,Ceramic,Chip           | 1  | ZD7400                                                                                                                                                 |
| 78 | MBV62321701 | Clip                             | 7  | SC10001,SC10002,SC10003,SC10004,SC10005,SC10006,SC10007,SC10008,SC10012,SC20015                                                                        |
| 79 | EAN62888101 | IC,PMIC                          | 1  | U4100                                                                                                                                                  |
| 80 | ECCH0004904 | Capacitor,Ceramic,Chip           | 25 | C1500,C1507,C2321,C2336,C3306,C3342,C5116,C5119,C8400,C8401,C9910                                                                                      |
| 81 | ECCH0000110 | Capacitor,Ceramic,Chip           | 4  | C9504,C9505,C9506,C9507                                                                                                                                |
| 82 | ECCH0002001 | Capacitor,Ceramic,Chip           | 14 | C2322,C2323,C2324,C2325,C2326,C2327,C2328,C2330,C2331,C2332,C2333,C2334,C2335,C2337,C23307,C3311,C3314,C3315,C3316,C3321,C3340,C3341,C5102,C8700,C8701 |
| 83 | EAE62487001 | Capacitor,Low ESL                | 1  | C3301                                                                                                                                                  |
| 84 | ECZH0000813 | Capacitor,Ceramic,Chip           | 6  | C1113,C1634,C1640,C1641,C1656,C1660,C1667,C1668,C2101,C2102,C5103,C5114,C5118,C5120                                                                    |
| 85 | EAG64389801 | C-Clip                           | 7  | ANT10003,ANT10004,ANT1108,ANT1110,ANT5100,ANT5101,ANT5103                                                                                              |
| 86 | ECCH0005604 | Capacitor,Ceramic,Chip           | 7  | C2300,C2301,C2313,C2314,C2315,C4141,C4142                                                                                                              |
| 87 | ECZH0003103 | Capacitor,Ceramic,Chip           | 2  | C6920,C6922                                                                                                                                            |
| 88 | EAE62506501 | Capacitor,Ceramic,Chip           | 3  | C4137,C4139,C4150                                                                                                                                      |
| 89 | ERHZ0000404 | Resistor,Chip                    | 2  | R1101,R1102,R1104,R1105,R6928                                                                                                                          |
| 90 | EAN63646401 | IC,PMIC                          | 1  | U4400                                                                                                                                                  |
| 91 | ECCH0000198 | Capacitor,Ceramic,Chip           | 15 | C1609,C3320,C6930                                                                                                                                      |

# 13. REPLACEMENT PART LIST

|     |             |                                        |   |                                                       |
|-----|-------------|----------------------------------------|---|-------------------------------------------------------|
| 92  | EDNY0013602 | Diode,Zener                            | 1 | ZD4131                                                |
| 93  | EAE62883601 | Capacitor(High Frequency),Ceramic,Chip | 1 | C1167                                                 |
| 94  | ERHZ0000406 | Resistor,Chip                          | 7 | R2105                                                 |
| 95  | ERHZ0000402 | Resistor,Chip                          | 1 | R9504                                                 |
| 96  | ECZH0003503 | Capacitor,Ceramic,Chip                 | 2 | C10000,C4702                                          |
| 97  | ERHZ0000288 | Resistor,Chip                          | 1 | R9204                                                 |
| 98  | EAP62106301 | Inductor,Wire Wound,Chip               | 1 | L4400                                                 |
| 99  | EAE63284101 | Capacitor,TA,Polymer                   | 1 | C9100                                                 |
| 100 | ECCH0009103 | Capacitor,Ceramic,Chip                 | 3 | C1173,C1174,L5119                                     |
| 101 | ECZH0001215 | Capacitor,Ceramic,Chip                 | 5 | C4103,C4104,C4136,C4700,C9520                         |
| 102 | ERHZ0000465 | Resistor,Chip                          | 1 | R4101                                                 |
| 103 | EAV62991901 | LED,Flash                              | 1 | LD4400                                                |
| 104 | EXXY0018701 | Crystal                                | 1 | X4100                                                 |
| 105 | ERHY0003301 | Resistor,Chip                          | 2 | R9534                                                 |
| 106 | EAG63772101 | Connector,RF                           | 1 | SW1101                                                |
| 107 | ENBY0036001 | Connector,BtoB                         | 1 | CN16001                                               |
| 108 | EAN62677301 | IC,Charger                             | 1 | U4700                                                 |
| 109 | EAP63106901 | Inductor,Wire Wound,Chip               | 2 | L4130,L4132                                           |
| 110 | EAH61992701 | Diode,Schottky                         | 1 | D4400                                                 |
| 111 | EAE62962301 | Capacitor,Ceramic,Chip                 | 1 | C1629,C5113                                           |
| 112 | ERHZ0000295 | Resistor,Chip                          | 1 | R4102                                                 |
| 113 | EAP62187501 | Inductor,Multilayer,Chip               | 1 | L4402                                                 |
| 114 | ERHZ0000405 | Resistor,Chip                          | 5 | R1100                                                 |
| 115 | SAFP0000401 | Wire Pad,Short                         | 4 | R1507,R1509,R1512,R5310                               |
| 116 | ERHY0009501 | Resistor,Chip                          | 5 | R5102,R5103                                           |
| 117 | ENBY0039601 | Connector,BtoB                         | 1 | CN7400                                                |
| 118 | ECCH0007802 | Capacitor,Ceramic,Chip                 | 1 | C4138                                                 |
| 119 | EAG64370001 | Connector,Terminal Block               | 1 | CN12100                                               |
| 120 | ERHZ0000401 | Resistor,Chip                          | 7 | R1505,R1506,R2101,R6609,R6611,R6929,R6930,R6932,R6933 |
| 121 | EAE62969601 | Capacitor,Ceramic,Chip                 | 1 | C4133                                                 |
| 122 | EAG63530401 | Connector,Terminal Block               | 1 | CN9100                                                |
| 123 | ENBY0040701 | Connector,BtoB                         | 1 | CN7300                                                |
| 124 | ERHZ0000204 | Resistor,Chip                          | 2 | R1510                                                 |
| 125 | ERHY0000298 | Resistor,Chip                          | 1 | R1113                                                 |
| 126 | SFBH0008102 | Filter,Bead                            | 1 | FB6921                                                |
| 127 | ELCH0003835 | Inductor,Multilayer,Chip               | 1 | C5128                                                 |
| 128 | EAG63393201 | Connector,BtoB                         | 1 | CN9950                                                |
| 129 | ELCH0003815 | Inductor,Multilayer,Chip               | 1 | C1616                                                 |
| 130 | ELCH0003841 | Inductor,Multilayer,Chip               | 1 | L1108                                                 |
| 131 | EAE62843001 | Capacitor,Ceramic,Chip                 | 2 | C4405,C4406                                           |
| 132 | ERHZ0000486 | Resistor,Chip                          | 2 | R6601                                                 |
| 133 | SEVY0005201 | Varistor                               | 3 | D103,D104,D106                                        |
| 134 | EAM62770301 | Filter,EMI/Power                       | 2 | FL16001,FL16002                                       |
| 135 | ERHY0009507 | Resistor,Chip                          | 1 | R1117                                                 |
| 136 | ECZH0004402 | Capacitor,Ceramic,Chip                 | 3 | C3304                                                 |
| 137 | EAE62767801 | Capacitor,Ceramic,Chip                 | 1 | C4146                                                 |
| 138 | ECCH0000182 | Capacitor,Ceramic,Chip                 | 3 | C4105,C4106,C9518                                     |

# 13. REPLACEMENT PART LIST

|     |             |                                        |    |                                                                         |
|-----|-------------|----------------------------------------|----|-------------------------------------------------------------------------|
| 139 | EUSY0412501 | IC,Headphone Amplifier                 | 1  | U6200                                                                   |
| 140 | ECZH0000830 | Capacitor,Ceramic,Chip                 | 3  | C1111,C1117,C1600,C1606,C5109,C6810,C6811,C6812,C6928,C6929,L1610,L1615 |
| 141 | EAN62637201 | IC,Acceleration Sensor                 | 1  | U8100                                                                   |
| 142 | EAB62911301 | Microphone,Condenser                   | 2  | MIC6920,MIC6921                                                         |
| 143 | EAG64249801 | Socket,DIMM/SIMM                       | 1  | S9500                                                                   |
| 144 | EAG63510401 | Connector,I/O                          | 1  | CN10001                                                                 |
| 145 | ERHZ0000308 | Resistor,Chip                          | 1  | R9532                                                                   |
| 146 | ECCH0006201 | Capacitor,Ceramic,Chip                 | 2  | C3310                                                                   |
| 147 | ERHZ0000538 | Resistor,Chip                          | 1  | R4100                                                                   |
| 148 | ECCH0007803 | Capacitor,Ceramic,Chip                 | 2  | C4401,C4402                                                             |
| 149 | EAP63185901 | Inductor,Multilayer,Chip               | 1  | L4401                                                                   |
| 150 | ECCH0000117 | Capacitor,Ceramic,Chip                 | 1  | C4109                                                                   |
| 151 | SFBH0008105 | Filter,Bead                            | 1  | FB6600                                                                  |
| 152 | EBC62235701 | Resistor,Chip                          | 1  | R4130                                                                   |
| 153 | ERHY0035501 | Resistor,Chip                          | 1  | R4702                                                                   |
| 154 | ECZH0001216 | Capacitor,Ceramic,Chip                 | 1  | C5111                                                                   |
| 155 | SAFO0000401 | Wire Pad,Open                          | 3  | R1504,R1508,R1511                                                       |
| 156 | ECCH0000113 | Capacitor,Ceramic,Chip                 | 2  | L5105,VA7303                                                            |
| 157 | ERHY0009316 | Resistor,Chip                          | 1  | R1118                                                                   |
| 158 | ECCH0009101 | Capacitor,Ceramic,Chip                 | 1  | C4145                                                                   |
| 159 | EAH61872601 | Diode,TVS                              | 1  | D10007                                                                  |
| 160 | EAM62710701 | Filter,Bead                            | 1  | FB6601,FB6602,FB6603                                                    |
| 161 | EAE62883901 | Capacitor(High Frequency),Ceramic,Chip | 1  | C1157                                                                   |
| 162 | SEVY0005402 | Varistor                               | 1  | VA95044                                                                 |
| 163 | EBC62036001 | Resistor,Chip                          | 1  | R10001                                                                  |
| 164 | EAM62770201 | Filter,EMI/Power                       | 1  | FL16000                                                                 |
| 165 | ECCH0000115 | Capacitor,Ceramic,Chip                 | 2  | C1109,C1110,C1115,C1116,C1646                                           |
| 166 | ECCH0007804 | Capacitor,Ceramic,Chip                 | 1  | C4101                                                                   |
| 167 | EAP62666001 | Inductor,Wire Wound,Chip               | 1  | L4131                                                                   |
| 168 | EAE62505701 | Capacitor,Ceramic,Chip                 | 1  | C4400                                                                   |
| 169 | ERHZ0000316 | Resistor,Chip                          | 1  | R1112                                                                   |
| 170 | ERHZ0000407 | Resistor,Chip                          | 1  | R9516                                                                   |
| 171 | EAE62885201 | Capacitor,Ceramic,Chip                 | 2  | C6700,C6701                                                             |
| 172 | ELCH0003819 | Inductor,Multilayer,Chip               | 1  | L1629,L1638                                                             |
| 173 | ELCH0003828 | Inductor,Multilayer,Chip               | 1  | L1161                                                                   |
| 174 | ECCH0000129 | Capacitor,Ceramic,Chip                 | 1  | C4407                                                                   |
| 175 | ERHZ0000299 | Resistor,Chip                          | 1  | R4703                                                                   |
| 176 | EAM62630901 | Filter,Bead                            | 2  | L6700,L6701                                                             |
| 177 | EAE62882501 | Capacitor(High Frequency),Ceramic,Chip | 1  | C1166                                                                   |
| 178 | ELCH0004717 | Inductor,Multilayer,Chip               | 1  | L5106                                                                   |
| 179 | ELCH0003832 | Inductor,Multilayer,Chip               | 1  | L1100,L1107                                                             |
| 180 | EBR80493501 | PCB Assembly,Main,SMT Top              | 1  | EBR071700                                                               |
| 181 | EAX66384401 | PCB,Main                               | 1  | EAX010000                                                               |
| 182 | ECZH0000830 | Capacitor,Ceramic,Chip                 | 12 | C1111,C1117,C1600,C1606,C5109,C6810,C6811,C6812,C6928,C6929,L1610,L1615 |
| 183 | ERHZ0000401 | Resistor,Chip                          | 9  | R1505,R1506,R2101,R6609,R6611,R6929,R6930,R6932,R6933                   |
| 184 | ECCH0000198 | Capacitor,Ceramic,Chip                 | 3  | C1609,C3320,C6930                                                       |
| 185 | ELCH0003820 | Inductor,Multilayer,Chip               | 1  | L1621                                                                   |

# 13. REPLACEMENT PART LIST

|     |             |                                        |    |                                                                                                                                                       |
|-----|-------------|----------------------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 186 | ECCH0002001 | Capacitor,Ceramic,Chip                 | 25 | C2322,C2323,C2324,C2325,C2326,C2327,C2328,C2330,C2331,C2332,C2333,C2334,C2335,C2337,C3307,C3311,C3314,C3315,C3316,C3321,C3340,C3341,C5102,C8700,C8701 |
| 187 | EAE62843001 | Capacitor,Ceramic,Chip                 | 2  | C4405,C4406                                                                                                                                           |
| 188 | ERHZ0000405 | Resistor,Chip                          | 1  | R1100                                                                                                                                                 |
| 189 | ERHZ0000404 | Resistor,Chip                          | 5  | R1101,R1102,R1104,R1105,R6928                                                                                                                         |
| 190 | ERHZ0000204 | Resistor,Chip                          | 1  | R1510                                                                                                                                                 |
| 191 | ELCH0005001 | Inductor,Multilayer,Chip               | 3  | C1635,L1104,L1112                                                                                                                                     |
| 192 | ECCH0000115 | Capacitor,Ceramic,Chip                 | 5  | C1109,C1110,C1115,C1116,C1646                                                                                                                         |
| 193 | ERHY0000254 | Resistor,Chip                          | 5  | R2202,R2203,R3313,R8400,R9536                                                                                                                         |
| 194 | EAN62969801 | IC,Hall Effect Switch                  | 1  | U8700                                                                                                                                                 |
| 195 | ECCH0000143 | Capacitor,Ceramic,Chip                 | 3  | C1604,C1628,C6931                                                                                                                                     |
| 196 | ERHY0000193 | Resistor,Chip                          | 3  | R1601,R1602,R3311                                                                                                                                     |
| 197 | MBV62321701 | Clip                                   | 10 | SC10001,SC10002,SC10003,SC10004,SC10005,SC10006,SC10007,SC10008,SC10012,SC20015                                                                       |
| 198 | ELCH0003826 | Inductor,Multilayer,Chip               | 4  | C1602,C1657,C1658,L1101                                                                                                                               |
| 199 | ECZH0001421 | Capacitor,Ceramic,Chip                 | 1  | C3303                                                                                                                                                 |
| 200 | ECZH0001217 | Capacitor,Ceramic,Chip                 | 6  | C1501,C1502,C1503,C1504,C1505,C1506                                                                                                                   |
| 201 | EAN62568201 | IC,Proximity                           | 1  | U8400                                                                                                                                                 |
| 202 | EAM62513501 | Filter,Duplexer                        | 1  | FL1600                                                                                                                                                |
| 203 | EAM62710701 | Filter,Bead                            | 3  | FB6601,FB6602,FB6603                                                                                                                                  |
| 204 | ERHZ0000250 | Resistor,Chip                          | 1  | R3309                                                                                                                                                 |
| 205 | EAN63366601 | IC,MCP,eMMC                            | 1  | U3300                                                                                                                                                 |
| 206 | ERHZ0000236 | Resistor,Chip                          | 1  | R1514                                                                                                                                                 |
| 207 | ECCH0004904 | Capacitor,Ceramic,Chip                 | 11 | C1500,C1507,C2321,C2336,C3306,C3342,C5116,C5119,C8400,C8401,C9910                                                                                     |
| 208 | ECCH0000122 | Capacitor,Ceramic,Chip                 | 1  | C1112                                                                                                                                                 |
| 209 | ECZH0000813 | Capacitor,Ceramic,Chip                 | 14 | C1113,C1634,C1640,C1641,C1656,C1660,C1667,C1668,C2101,C2102,C5103,C5114,C5118,C5120                                                                   |
| 210 | ELCH0001049 | Inductor,Multilayer,Chip               | 1  | L1611                                                                                                                                                 |
| 211 | EAE62882801 | Capacitor(High Frequency),Ceramic,Chip | 1  | L1617                                                                                                                                                 |
| 212 | EAM62090901 | Filter,Duplexer,IMT                    | 1  | FL1602                                                                                                                                                |
| 213 | ELCH0003816 | Inductor,Multilayer,Chip               | 1  | L1613                                                                                                                                                 |
| 214 | ERHZ0000441 | Resistor,Chip                          | 1  | R8401                                                                                                                                                 |
| 215 | ERHZ0000406 | Resistor,Chip                          | 1  | R2105                                                                                                                                                 |
| 216 | ECCH0000151 | Capacitor,Ceramic,Chip                 | 3  | C5117,C6601,C6602                                                                                                                                     |
| 217 | ELCH0001056 | Inductor,Multilayer,Chip               | 1  | C5104                                                                                                                                                 |
| 218 | SFBH0008105 | Filter,Bead                            | 1  | FB6600                                                                                                                                                |
| 219 | EAN62887801 | IC,RF Transceiver,3G                   | 1  | U1500                                                                                                                                                 |
| 220 | EBC62463701 | Resistor,Chip                          | 2  | R3310,R3312                                                                                                                                           |
| 221 | ECCH0000155 | Capacitor,Ceramic,Chip                 | 1  | C5124                                                                                                                                                 |
| 222 | EAE62945401 | Capacitor(High Frequency),Ceramic,Chip | 1  | L1645                                                                                                                                                 |
| 223 | EAE62282201 | Capacitor,Ceramic,Chip                 | 1  | C6603                                                                                                                                                 |
| 224 | ELCH0001425 | Inductor,Multilayer,Chip               | 1  | L5108                                                                                                                                                 |
| 225 | ELCH0004720 | Inductor,Multilayer,Chip               | 1  | L5112                                                                                                                                                 |
| 226 | ERHZ0000487 | Resistor,Chip                          | 1  | R6600                                                                                                                                                 |
| 227 | EAW62283501 | Crystal                                | 1  | X1500                                                                                                                                                 |
| 228 | EAN63106301 | IC,RF Amplifier                        | 1  | FL5100                                                                                                                                                |
| 229 | ELCH0003838 | Inductor,Multilayer,Chip               | 1  | L5104                                                                                                                                                 |
| 230 | EAT61914601 | Module,Tx Module                       | 1  | U1100                                                                                                                                                 |

# 13. REPLACEMENT PART LIST

|     |             |                                        |   |                               |
|-----|-------------|----------------------------------------|---|-------------------------------|
| 231 | ELCH0001031 | Inductor,Multilayer,Chip               | 3 | L1600,L1603,L1606             |
| 232 | EAN62909001 | IC,WiFi                                | 1 | U5100                         |
| 233 | EAE62945301 | Capacitor(High Frequency),Ceramic,Chip | 1 | L1619                         |
| 234 | EAE62882601 | Capacitor(High Frequency),Ceramic,Chip | 1 | L5118                         |
| 235 | ERHZ0000443 | Resistor,Chip                          | 5 | R2200,R2201,R2204,R2205,R6927 |
| 236 | ELCH0004726 | Inductor,Multilayer,Chip               | 1 | C1101                         |
| 237 | ERHZ0000220 | Resistor,Chip                          | 1 | R2103                         |
| 238 | ELCH0003835 | Inductor,Multilayer,Chip               | 1 | C5128                         |
| 239 | ECCH0000133 | Capacitor,Ceramic,Chip                 | 1 | C1108                         |
| 240 | EAE62962301 | Capacitor,Ceramic,Chip                 | 2 | C1629,C5113                   |
| 241 | EAM63050001 | Filter,Separator,Switch                | 1 | U1601                         |
| 242 | ELCH0003839 | Inductor,Multilayer,Chip               | 1 | L1110                         |
| 243 | ELCH0003819 | Inductor,Multilayer,Chip               | 2 | L1629,L1638                   |
| 244 | ELCH0001054 | Inductor,Multilayer,Chip               | 1 | C5122                         |
| 245 | EAH62033201 | Diode,TVS                              | 2 | D6601,D6602                   |
| 246 | EAN63388401 | IC,Digital Baseband Processor,3G       | 1 | U2100                         |
| 247 | MEZ65049701 | Label                                  | 1 | PID1                          |
| 248 | ERHZ0000449 | Resistor,Chip                          | 1 | R1103                         |
| 249 | EAG64409801 | C-Clip                                 | 1 | ANT10005                      |
| 250 | ELCH0003847 | Inductor,Multilayer,Chip               | 1 | C1605                         |
| 251 | ECCH0017601 | Capacitor,Ceramic,Chip                 | 1 | C5115                         |
| 252 | EAM62531301 | Filter,Saw,Dual                        | 1 | FL1100                        |
| 253 | EAN62796101 | IC,Power Amplifier                     | 1 | U1600                         |
| 254 | ERHZ0000486 | Resistor,Chip                          | 1 | R6601                         |
| 255 | ECZH0004402 | Capacitor,Ceramic,Chip                 | 1 | C3304                         |
| 256 | ELCH0003815 | Inductor,Multilayer,Chip               | 1 | C1616                         |
| 257 | EAE62502901 | Capacitor,Ceramic,Chip                 | 1 | C1114                         |
| 258 | ELCH0003832 | Inductor,Multilayer,Chip               | 2 | L1100,L1107                   |
| 259 | ELCH0001430 | Inductor,Multilayer,Chip               | 1 | R8402                         |
| 260 | EAE62882701 | Capacitor(High Frequency),Ceramic,Chip | 1 | C1104                         |
| 261 | EAE62685701 | Capacitor,Ceramic,Chip                 | 1 | C5105                         |
| 262 | EAP62109801 | Inductor,Wire Wound,Chip               | 1 | L6600                         |
| 263 | ERHZ0000469 | Resistor,Chip                          | 1 | R1603                         |
| 264 | EAM62514001 | Filter,Duplexer                        | 1 | FL1604                        |
| 265 | ELCH0001048 | Inductor,Multilayer,Chip               | 1 | L5117                         |
| 266 | ELCH0003814 | Inductor,Multilayer,Chip               | 1 | L1605                         |
| 267 | ERHY0003301 | Resistor,Chip                          | 1 | R9534                         |
| 268 | ERHY0000289 | Resistor,Chip                          | 1 | R2106                         |
| 269 | ERHY0009501 | Resistor,Chip                          | 2 | R5102,R5103                   |
| 270 | ELCH0005003 | Inductor,Multilayer,Chip               | 2 | L1109,L1113                   |
| 271 | ELCH0012508 | Inductor,Multilayer,Chip               | 1 | L1501                         |
| 272 | ELCH0003824 | Inductor,Multilayer,Chip               | 1 | L1630                         |
| 273 | SAFP0000401 | Wire Pad,Short                         | 4 | R1507,R1509,R1512,R5310       |
| 274 | ERHZ0000291 | Resistor,Chip                          | 1 | R1607                         |
| 275 | ERHZ0000529 | Resistor,Chip                          | 1 | R6926                         |
| 276 | ELCH0003834 | Inductor,Multilayer,Chip               | 1 | L1628                         |
| 277 | ERHZ0000294 | Resistor,Chip                          | 1 | R2102                         |

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|     |             |                                  |   |                   |
|-----|-------------|----------------------------------|---|-------------------|
| 278 | EAH61532901 | Diode,Switching                  | 1 | D9910             |
| 279 | ECCH0006201 | Capacitor,Ceramic,Chip           | 1 | C3310             |
| 280 | EAE62286801 | Capacitor,Ceramic,Chip           | 1 | C2103             |
| 281 | SAFO0000401 | Wire Pad,Open                    | 3 | R1504,R1508,R1511 |
| 282 | ECCH0000127 | Capacitor,Ceramic,Chip           | 2 | C6604,C6605       |
| 283 | SAD35252001 | Software,Mobile                  | 1 | SAD010000         |
| 284 | SAF30980201 | Software Assembly,Common         | 1 | SAF010100         |
| 285 | SAD35384601 | Software,USB Driver              | 1 | SAD010500         |
| 286 | SAD35404301 | Software,OSP                     | 1 | SAD010600         |
| 287 | SAD35344501 | Software,Application             | 1 | SAD010100         |
| 288 | AGF78058402 | Package Assembly                 | 1 | AGF000000         |
| 289 | MLAJ0004402 | Label,Master Box                 | 0 | MEZ047200         |
| 290 | AGJ74178402 | Pallet Assembly                  | 1 | AGJ000000         |
| 291 | MAY66471717 | Box,Pallet Sleeve                | 0 | MAY115200         |
| 292 | MGA62599507 | Pallet                           | 0 | MGA000001         |
| 293 | MLAJ0004402 | Label,Master Box                 | 0 | MEZ047200         |
| 294 | MLAZ0050901 | Label                            | 0 | MEZ000000         |
| 295 | MEZ64598733 | Label,Unit Box                   | 1 | MEZ084100         |
| 296 | MAY67169303 | Box,Unit                         | 1 | MAY084000         |
| 297 | MAY66470404 | Box,Master                       | 0 | MAY047100         |
| 298 | MAF63267402 | Bag,Vinyl                        | 1 | MAF086500         |
| 299 | MEZ64578901 | Label,Unit Box                   | 1 | MEZ084102         |
| 300 | AAD87393202 | Addition Assembly                | 1 | AAD000000         |
| 301 | MFL69080501 | Manual,Operation                 | 0 | MFL053800         |
| 302 | MCK68746002 | Cover,Battery                    | 1 | MCK00             |
| 303 | EAY62989701 | Adapters                         | 1 | EAY060001         |
| 304 | EAC62798201 | Rechargeable Battery,Lithium Ion | 1 | EAC00             |
| 305 | EAB62209304 | Earphone,Stereo                  | 1 | EAB010200         |
| 306 | AFN77132701 | Manual Assembly,Operation        | 1 | AFN053800         |
| 307 | MBM65097101 | Card,Quick Reference             | 1 | MBM062600         |
| 308 | MBM64995932 | Card,Warranty                    | 1 | MBM087200         |
| 309 | EAD62377902 | Cable,Assembly                   | 1 | EAD010000         |

